Flows of geometric structures III: stability, continuation and solitons

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Abstract

We study Ricci–harmonic flows of tensorial H-structures with fixed natural quadratic torsion terms. Torsion-free G_2- and $\mathrm{Spin}(7)$-structures of full holonomy are stable under all such flows on closed manifolds. For general tensorial H, we prove local derivative estimates retaining the initial regularity and pointed compactness up to the initial boundary under finite initial derivative and injectivity bounds. We prove continuation from a positive continuous metric endpoint with time-integrable squared torsion norm, or under integrability of the full velocity. The estimates are uniform over bounded families of fixed coefficients. A weighted metric-error criterion gives complete ancient singularity models.

For the modified Ricci–harmonic (MHR) family, volume dissipation identifies compact torsion-free limits of noncollapsed immortal solutions under a uniform curvature–torsion bound in the coercive parameter ranges; a full-holonomy limit gives convergence of the whole flow. We classify closed steady G_2 solitons at the critical parameter, give a bracket-flow reduction and a finite homogeneous catalogue, and construct complete cylindrical solitons. An explicit $\mathrm{SU}(3)$ dictionary determines the Type IIA and product comparisons. Compact quotients also give collapsing Type I singularities. We prove variational obstructions for G_2 and $\mathrm{Spin}(7)$ within the specified class of invariant quadratic functionals and pairings, including fixed-measure weighted functionals. Finally, we establish finite-time perturbation estimates in a parabolic gauge and determine the tangent correction of the literal Taylor-selected $\mathrm{Spin}(7)$ expression.

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1 Introduction

A tensorial H-structure consists of a tensor, or a compatible collection of tensors, whose stabiliser is a subgroup $\mathrm{H} \subset \mathrm{SO}(n)$. Its intrinsic torsion measures the failure of the defining tensor to be parallel. For $\mathrm{H} = \mathrm{G}_2$ and $\mathrm{Spin}(7)$, vanishing torsion gives a Ricci-flat metric with holonomy contained in H. Geometric flows provide a way to seek such structures from initial data with nonzero torsion. When the metric evolves as well, curvature and intrinsic torsion must be controlled together. Their interaction governs both stability near a torsion-free structure and the formation of singularities.

For a fixed metric, Loubeau–Sá Earp formulate the harmonic section flow and relate it to equivariant harmonic map heat flow [LSE23, Theorem 1 and Proposition 17]. Paper I develops the harmonic flow within a fixed isometric class and proves an almost-monotonicity formula for it [FLMSE24, Theorem 2.5]. Paper II introduces the moving-metric Ricci–harmonic flow and proves short-time existence, derivative estimates and a continuation criterion [FFL$^+$26, Theorem 4.2, Theorem 5.2, Corollary 5.3 and Theorem 5.4]. The purpose of the present paper is to study the behaviour beyond short-time existence: stability near torsion-free structures, continuation under weaker bounds, and explicit soliton and singularity models. We also examine the extent to which these equations admit a variational interpretation.

Let ξ be an H-structure, $g = g_\xi$ its induced metric, and T its intrinsic torsion. We use the diamond action of endomorphisms on tensors, so that $\nabla_X \xi = T_X \diamond \xi$. Our main example is the two-parameter modified Ricci–harmonic flow

$$\partial_t \xi = \mathcal{F}_{a,b}(\xi) = \big[-\mathrm{Ric} + a(T \star T) + b|T|^2 g + \mathrm{div}\, T\big] \diamond \xi, \qquad a, b \in \mathbb{R},$$

abbreviated to $\mathrm{MHR}_{a,b}$. Here $(T \star T)_{ij} = T_{i,pq} T_{j,pq}$ and $(\mathrm{div}\, T)_{ij} = \nabla_p T_{p,ij}$ in an orthonormal frame. The metric therefore evolves by

$$\partial_t g = -2\mathrm{Ric} + 2a(T \star T) + 2b|T|^2 g.$$

The quadratic terms leave the principal symbol unchanged but alter the stationary and self-similar solutions. This distinction determines the organisation of the results: the analytic theory applies to the larger fixed quadratic class, while the soliton and dissipation identities retain the individual MHR coefficients.

Stability near torsion-free structures has been established for several other flows. Lotay–Wei treat the Laplacian flow of closed G_2 forms in a fixed cohomology class [LW19, Theorem 1.3], and Weiß–Witt treat the Dirichlet energy flow for arbitrary nearby positive 3-forms [WW12, Theorem 8.1 and Corollary 8.2]. Our first result concerns unrestricted nearby G_2- and $\mathrm{Spin}(7)$-structures at a full-holonomy background. We allow any fixed equation of the form

$$\partial_t \xi = [-\mathrm{Ric} + \mathrm{div}\, T + Q(T)] \diamond \xi,$$

where Q is any fixed H-equivariant quadratic map from the intrinsic-torsion module to symmetric endomorphisms and the orthogonal complement of $\mathrm{Lie}(\mathrm{H})$.

Theorem A (Theorem 3.1)**.** Let M be connected and closed, and let $\bar{\xi}$ be a torsion-free H-structure with full holonomy $\mathrm{H} = \mathrm{G}_2$ or $\mathrm{Spin}(7)$. For every fixed flow in the quadratic Ricci–harmonic class, sufficiently small smooth perturbations of $\bar{\xi}$ have solutions for all positive time which converge smoothly to a torsion-free structure. Smallness may be measured in $W^{2-2/p,p}$, with $p > \dim M + 2$. In the DeTurck gauge the convergence is exponential in this norm.

The first variation of $Q(T)$ vanishes at $T = 0$. In the standard DeTurck gauge, the remaining linearisation is the direct sum of the Lichnerowicz Laplacian on symmetric tensors and the rough Laplacian in the vertical directions. The parallel-spinor argument of [DWW05, Theorem 2.5] controls its spectrum, and the torsion-free deformation theory [Nor13, Proposition 3.1 and Lemma 3.13] identifies its kernel with the tangent space to the

equilibrium manifold.[1] We then apply the normal stability theorem of [PSZ09, Theorem 2.1]. Thus stability means decay towards the local manifold of torsion-free structures; the limiting structure may differ from the initial background.

Dwivedi's Ricci–harmonic G_2 equation belongs to the same quadratic class [Dwi26, Definition 3.2]. Thus Theorem A answers the full-holonomy case of his stability question [Dwi26, Question 8.4]. Its identification with a particular MHR equation requires a qualification: the two equations agree on the normal-torsion locus, but differ elsewhere by the commutator displayed in Proposition 2.2. The stability argument does not require this locus to be preserved.

For the fixed quadratic class, Proposition 4.6 gives uniform doubling and continuation under a curvature–torsion bound. The pointed compactness and weaker metric-endpoint criterion below also hold in this class, for arbitrary tensorial H. The relevant scale is

$$\Lambda_{\mathrm{H}} = \big(|\mathrm{Rm}|^2 + |\nabla T|^2 + |T|^4\big)^{1/2}.$$

Paper II shows that $\sup_M \Lambda_{\mathrm{H}}$ diverges at a finite maximal time [FFL$^+$26, Theorem 5.4]. Our continuation result uses weaker information about the metric and torsion. Its metric-endpoint hypothesis follows the continuation results of Lotay–Wei for the Laplacian flow [LW17, Theorem 8.1] and Duthie for reasonable $\mathrm{Spin}(7)$-flows [Dut26, Theorem 7.1]. We replace their uniform torsion bound by a time-integrability condition for every fixed quadratic Ricci–harmonic equation and arbitrary tensorial H. A second criterion uses the full velocity of the defining tensor.

Theorem B (Theorem 4.15 and Corollaries 4.16, 4.19)**.** Let a solution of one fixed equation in the quadratic Ricci–harmonic class on a closed manifold be defined on $[0, T_0)$, where $T_0 < \infty$. If $g(t)$ converges uniformly to a continuous positive definite metric and

$$\int_0^{T_0} \sup_M |T|^2 \, dt < \infty,$$

then the solution extends smoothly through T_0. Alternatively, the condition

$$\int_0^{T_0} \sup_M |\partial_t \xi|_{g(t)} \, dt < \infty$$

alone gives continuation. Consequently every finite maximal time satisfies

$$\int_0^{T_0} \sup_M \big(|\mathrm{Ric}| + |T|^2\big) \, dt = \infty.$$

The Ricci–torsion divergence criterion is already known for reasonable G_2-flows [Che18, Theorem 5.2, (74)] and reasonable $\mathrm{Spin}(7)$-flows [Dut26, Theorem 7.3, (7.16)]. Here it is obtained for general tensorial H in the fixed quadratic class. Dwivedi's bounded-velocity criterion for his G_2 equation [Dwi26, Theorem 5.9, (5.30)] is the corresponding predecessor of the full-velocity assertion. Our integral hypothesis controls both the metric and the vertical motion of the structure.

Both continuation arguments proceed by contradiction through a smooth pointed blow-up. The compactness theorem allows a bounded family of fixed quadratic models; after extraction, the models converge and determine the limiting equation. In particular, a sequence solving one fixed equation has a limit solving that equation.

At a hypothetical singular time, the continuous metric endpoint makes the rescaled metric spaces Euclidean and supplies the injectivity bound. There are then two ways to eliminate torsion from the smooth limit. In the first argument, the torsion integral makes its torsion vanish. In the second, integrability of the full velocity supplies a continuous endpoint for the defining tensor, which makes the limiting tensor parallel. The latter argument compares the endpoint charts with the smooth convergence charts using only first derivatives of their transition maps. In either case, the limiting curvature and torsion vanish, contradicting the blow-up normalisation $\Lambda_{\mathrm{H}}(p, 0) = 1$. The same compactness theorem gives complete ancient singularity models whenever a rescaled injectivity bound is available. One sufficient condition for this bound is

$$\int_0^{T_0} (T_0 - t) \sup_M |T(t)|^4 \, dt < \infty.$$

[1] Locators for [Nor13] refer to arXiv:1008.0663v1.

Indeed, Chen's relative noncollapse theorem [Che18, Definition 4.1 and Theorem 4.2, (62)] and our singularity lower rate give the required rescaled volume and injectivity bounds; see Corollary 4.12. The same weighted torsion condition appears in Chen's analysis of reasonable G_2-flows [Che18, Theorems 5.3–5.4, (75) and (89)] and in Duthie's analysis of reasonable $\mathrm{Spin}(7)$-flows [Dut26, Theorems 7.6–7.7], where it gives noncollapse on the scale $\sqrt{T_0 - t}$ and, under an additional growth bound, torsion-free blow-up limits of maximal volume growth. Here it is used for every fixed equation of the quadratic class and arbitrary tensorial H, and yields an ancient limit of the same equation without that growth bound. The hypothesis cannot simply be omitted: Corollary 5.23 gives a compact Heisenberg solution whose diameter and volume tend to zero even after normalisation at its Type I singularity scale.

We also prove local derivative estimates which retain the regularity of the initial data. Here the equation may be any fixed flow in the quadratic Ricci–harmonic class.

Theorem C (Theorem 4.4)**.** Let $\xi(t)$ be a smooth solution on a closed manifold for $0 \le t \le \tau \le \tau_0$. On an initial-metric ball $B_{g(0)}(p, r)$, suppose $\Lambda_{\mathrm{H}} \le K$ throughout this interval and

$$|\nabla^j \mathrm{Rm}(0)| + |\nabla^{j+1} T(0)| \le K_\ell, \qquad 0 \le j \le \ell.$$

Then, on $\overline{B_{g(0)}(p, r/2)}$ and for $0 < t \le \tau$,

$$|\nabla^m \mathrm{Rm}(t)| + |\nabla^{m+1} T(t)| \le C t^{-(m-\ell)_+/2}, \qquad m \ge 0.$$

The constants are independent of higher initial derivatives and of the geometry outside the control ball. They are uniform over bounded families of fixed quadratic terms.

Dwivedi proves this retained exponent for his G_2 flow [Dwi26, Theorem 4.7, (4.50)–(4.53)]; his product Bernstein argument [Dwi26, Lemma 4.10 and (4.57)] also underlies our localisation. Duthie's local estimate for reasonable $\mathrm{Spin}(7)$-flows has the ordinary $t^{-m/2}$ loss [Dut26, Theorem 4.2]. We extend the retained estimate to arbitrary tensorial H in the fixed quadratic class, uniformly over bounded coefficient families. A global estimate first establishes the derivative hierarchy. A moving-distance cutoff then localises it without an injectivity-radius assumption.

If the initial derivative bounds hold globally, they also control the initial boundary of pointed limits. For sequences with a uniform global Λ_{H} bound, bounded fixed coefficient models and a uniform initial base-point injectivity bound, Theorem 4.9 gives convergence of the metrics and defining tensors in

$$C\big([0, \tau]; C^{\ell+1}_{\mathrm{loc}}\big),$$

and smooth convergence for positive time, in the same time-independent comparison maps. The proof uses the integrable $t^{-1/2}$ bound at the first unretained derivative order. It gives a finite-regularity initial trace, without constructing a solution from nonsmooth initial data.

Appendix A gives a complementary estimate for approximate solutions on a fixed finite interval. In a fixed parabolic gauge, the residual measures the failure of a path to satisfy the flow equation. A small L^p residual and a small initial discrepancy in $W^{2-2/p,p}$ give a nearby exact solution in $W^{1,p}_t L^p_x \cap L^p_t W^{2,p}_x$, for $p > n + 2$. The proof combines maximal regularity for the principal part with a Volterra–Gronwall estimate for the rough lower-order coefficients.

The dependence on (a, b) becomes particularly explicit for solitons. We use the convention

$$\mathcal{F}_{a,b}(\xi) = \rho\, g \diamond \xi + \mathcal{L}_Y \xi,$$

so that the metric dilation is $1 + 2\rho t$: negative, zero and positive ρ correspond to shrinking, steady and expanding solutions, respectively. The trace equation depends on $\mu = a + nb$. On closed manifolds, scalar-torsion identities then give sign restrictions and volume dissipation. At the critical G_2 parameter $\mu = 9$, we obtain a classification: the steady solitons are precisely the nearly parallel structures, including the torsion-free case, with an infinitesimal automorphism as soliton field (Proposition 5.16).

The same volume identity gives a convergence criterion for immortal solutions. In the ranges $\mu < -3/2$ for G_2 and $\mu < -2$ for $\mathrm{Spin}(7)$, assume a uniform bound for Λ_{H} and a positive basepoint injectivity-radius bound along some sequence of times tending to infinity. Every pointed limit is then compact and torsion-free. If one limit has full holonomy, the whole flow converges smoothly and exponentially on the original manifold

(Corollary 5.13). The volume bound first rules out a noncompact Ricci-flat limit. Once a subsequence enters the stability neighbourhood, Theorem A controls the entire remaining flow.

There are also complete gradient $\mathrm{Spin}(7)$ solitons for every MHR parameter pair. If $\mathcal{T} = cg_N$, $c \neq 0$, is a complete nearly parallel G_2 seed, its product with a line carries

$$\Phi = ds \wedge \varphi + \psi, \qquad \rho = \tfrac{1}{2}(a + 7b - 12)c^2, \qquad F(s) = \tfrac{1}{4}(12 - a)c^2 s^2.$$

These give shrinkers, steady solitons and expanders as the sign of $a+7b-12$ varies (Proposition 5.17). On a circle product the soliton equation holds exactly when $a = 12$. In particular, $(a, b) = (12, 0)$ gives a compact stationary structure with nonzero torsion, proving sharpness of the upper $\mathrm{Spin}(7)$ rigidity endpoint. The construction uses the harmonicity of locally conformally parallel $\mathrm{Spin}(7)$-structures [FSE24, arXiv:2301.12494v2, Corollary 3.4].

For complete noncompact examples, let G be the simply connected seven-dimensional Heisenberg group. Choose a left-invariant coframe $(e^1, \dots, e^6, \eta)$ with

$$de^j = 0, \qquad d\eta = \omega := e^{12} + e^{34} + e^{56}, \qquad \Upsilon = (e^1 + \mathrm{i}e^2) \wedge (e^3 + \mathrm{i}e^4) \wedge (e^5 + \mathrm{i}e^6).$$

Theorem D (Proposition 5.22)**.** For every $(a, b) \in \mathbb{R}^2$ and $c_0 > 0$, the left-invariant G_2-structure

$$\varphi_{c_0} = c_0\eta \wedge \omega + \mathrm{Re}\,\Upsilon, \qquad g_{c_0} = c_0^2\eta^2 + \sum_{j=1}^{6}(e^j)^2$$

is a complete $\mathrm{MHR}_{a,b}$ soliton with rate

$$\rho = c_0^2\left(\frac{5}{2} - \frac{7a}{6} + \frac{5b}{2}\right).$$

It admits no gradient soliton representation when $a \neq 3/2$.

The construction continues the contact Calabi–Yau examples for the Laplacian flow and coflow of [LSES22, Theorems A–C], the modified-coflow solutions of [SESS25, Theorem 4.2 and Corollary 4.3], the Ricci–harmonic solutions of [DS26, Theorems 3.1 and 4.6], and the Heisenberg-group coflow evolutions of [BFF20, Corollary 5 and Theorem 7]. Our formula treats every MHR parameter pair and recovers specific trajectories of these earlier flows, as detailed in Section 5.5. It reduces the flow to two scalar equations, which can be integrated explicitly. On the Heisenberg group the resulting solution is self-similar under its grading automorphisms. The non-gradient assertion uses both soliton equations. The skew equation and invertibility of the torsion force any gradient potential to be constant. The metric equation then fails when $a \neq 3/2$, because the metric velocity is not a scalar multiple of the metric. The evolving structures also descend to compact quotients; the soliton vector field need not descend.

We finally ask whether MHR admits a gradient interpretation. Gianniotis–Zacharopoulos give a principal-order obstruction for their standard product L^2 pairing [GZ26, (3.1) and Proposition 3.3]. We ask whether this obstruction persists for arbitrary nondegenerate invariant block weights, within a specified class of curvature and quadratic-torsion functionals. Write $\mathcal{T} = \mathcal{T}_1 + \mathcal{T}_7 + \mathcal{T}_{14} + \mathcal{T}_{27}$ for the irreducible decomposition of the full torsion tensor, defined by $\nabla_i\varphi_{jkl} = \mathcal{T}_{im}\psi_{mjkl}$, where $\psi = *_\varphi\varphi$. Consider the quadratic functionals

$$\mathcal{W}(\varphi) = \int_M \left(\alpha R + \sum_{k\in\{1,7,14,27\}} \beta_k|\mathcal{T}_k|^2\right) dV,$$

where all coefficients are constant. The allowed metrics on the space of G_2-structures are the nondegenerate invariant L^2 metrics with constant weights on the 1-, 7- and 27-dimensional variation summands. A natural vector field linear in torsion is a constant multiple of $(V_\mathcal{T})_k = \mathcal{T}_{ij}\varphi_{ijk}$.

Theorem E (Theorem 6.1)**.** For no MHR parameter pair (a, b) is there a functional $\mathcal{W}$ and a nondegenerate invariant L^2 metric G in the preceding classes, together with a constant c, such that

$$\mathcal{F}_{a,b}(\varphi) = -\operatorname{grad}_\mathsf{G}\mathcal{W}(\varphi) + \mathcal{L}_{cV_\mathcal{T}}\varphi$$

for every smooth G_2-structure on a closed manifold.

The first-variation formulae of [DGK25, Corollary 5.36] reduce the question to an algebraic identity in the torsion. Three tests, on the two skew summands and on symmetric torsion, force the pairing to vanish on the vector variation block, contrary to nondegeneracy. Evaluation at a constant weight gives the same obstruction for the fixed-measure weighted functionals in Corollary 6.2. For $\mathrm{Spin}(7)$, the same conclusion holds for $\int(\alpha R+\beta_8|T_8|^2+\beta_{48}|T_{48}|^2)\,dV$, every nondegenerate invariant pairing, and the unique torsion-linear generator cV, where $V_k=T_{i,ik}$ (Proposition 6.3). Five exact variations on one compact nilmanifold force the skew pairing to vanish. The same tests prove the corresponding fixed-measure weighted obstruction in Corollary 6.4. The assertion is restricted to these functional, metric and gauge classes; in particular, it does not preclude other monotonicity formulae or a construction using an optimising weight.

The homogeneous and product constructions are developed further in Theorem 5.27, Proposition 5.30, and Remark 5.34. The bracket formulation is adapted from Lauret's bracket flow for the Laplacian flow [Lau17]. It retains the skew torsion term and yields exact soliton certificates for a finite catalogue of left-invariant closed and co-closed structures from [CF11, BFF18, Nic18]. For $\mathrm{SU}(3)$, the defining 2- and 3-forms have different scaling weights. We compute the torsion dictionary and the MHR velocity for the compatible tuple, identify the precise Type IIA comparison, and determine the correction required by a product with a circle (Propositions 5.37, 5.43 and 5.44). These calculations also give compact steady examples and sharp parameter exclusions.

Broader extensions and the unresolved monotonicity and classification questions are discussed in the afterword.

The paper is organised as follows. Section 2 describes the equations and compares MHR with Dwivedi's G_2 flow and Duthie's reasonable $\mathrm{Spin}(7)$ framework. Section 3 proves Theorem A. Section 4 proves the retained estimates of Theorem C, pointed compactness including its finite-regularity initial trace, the continuation criteria of Theorem B and weighted-error ancient limits. Section 5 sets up the gauged soliton system, develops the soliton and volume identities and the stationarity of immortal limits, then constructs the explicit models of Theorem D and the collapsing singularity example, before treating bracket flow, the finite homogeneous catalogue, the $\mathrm{SU}(3)$ comparisons and compact endpoint sharpness. Section 6 proves Theorem E. Appendix A gives finite-time perturbation estimates in a parabolic gauge. Appendix B examines the literal Taylor-selected $\mathrm{Spin}(7)$ expression: its quadratic term can fail to be tangent to the Cayley orbit, and its orthogonal tangent correction gives $\mathrm{MHR}_{0,-1/7}$ when the divergence coefficient is one.

Notation and conventions. Let M^n be a connected oriented manifold without boundary; closed means compact without boundary. Fix a tensor, or a compatible tuple of tensors, $\xi_\circ$ on $\mathbb{R}^n$ whose full stabiliser in $\mathrm{GL}(n,\mathbb{R})$ is a closed subgroup $\mathrm{H}\subset\mathrm{SO}(n)$. An H-structure is a section of the associated $\mathrm{GL}^+(n,\mathbb{R})$-orbit bundle; its metric is denoted by $g=g_\xi$. All contractions, orthogonal projections and covariant derivatives refer to this metric unless another metric is specified.

We use the right action of $\mathrm{GL}(n,\mathbb{R})$ on tensors and denote its infinitesimal action by $\diamond$. Symmetric endomorphisms are identified with symmetric covariant 2-tensors using g; a variation $(h+A)\diamond\xi$, with h symmetric and A skew, induces $\partial_t g=2h$. The intrinsic torsion is denoted by T and is defined by $\nabla_X\xi=T_X\diamond\xi$. For G_2-structures, $\mathcal{T}$ denotes the full torsion tensor defined by $\nabla_i\varphi_{jkl}=\mathcal{T}_{im}\psi_{mjkl}$, where $\psi=*_\varphi\varphi$; it is distinct from T.

Repeated indices are summed in an orthonormal frame. The norm of a d-form has the usual factor $1/d!$; in particular, the squared endomorphism norm of a skew 2-tensor is twice its squared form norm. We write $\Delta=\nabla^i\nabla_i=-\nabla^*\nabla$ for the rough Laplacian. The notation $A*B$ denotes a finite sum of contractions with the fixed algebraic tensors of the structure, with coefficients controlled by the fixed equation.

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2 Geometric structures and Ricci–harmonic flows

We recall the tensorial description of an H-structure and fix the normalisations used in the flow equations. The distinction between intrinsic torsion and the full G_2 torsion tensor will be needed both for comparison with other flows and for the explicit examples. For the exceptional groups, the metric and torsion variation formulae were developed by Karigiannis [Kar09, Corollary 3.2 and Theorem 3.8] [Kar08, Proposition 3.1 and Theorem 3.4]. Paper I expresses both decompositions in terms of the tensor orbit and its stabiliser. We use this common description throughout.

Intrinsic torsion and the diamond action. For a tensor u of type (p, q) and $A \in \mathrm{GL}(n, \mathbb{R})$, we use the right action

$$u \cdot A = \big((A^{-1})^{\otimes p} \otimes (A^*)^{\otimes q}\big) u.$$

The *diamond action* is its infinitesimal action,

$$A \diamond \xi := \frac{d}{ds}\Big|_{s=0} \xi \cdot \exp(sA),$$

where now A is an endomorphism. For a covariant tensor this reads

$$(A \diamond u)(X_1, \dots, X_q) = \sum_{j=1}^{q} u(X_1, \dots, AX_j, \dots, X_q).$$

Write $\mathfrak{h} = \mathrm{Lie}(\mathrm{H})$ and $\mathfrak{m} = \mathfrak{h}^{\perp} \subset \mathfrak{so}(n)$. The tangent space to the orbit is identified with $\Sigma^2 \oplus \Lambda^2_{\mathfrak{m}}$ by the diamond action, where Σ^2 denotes symmetric endomorphisms. Thus a variation has a unique representative $(S + C) \diamond \xi$, with S symmetric and $C \in \mathfrak{m}$, and its metric variation is $2S$. The intrinsic torsion is the unique $\mathfrak{m}$-valued one-form characterised by

$$\nabla_X \xi = T_X \diamond \xi, \qquad T \in \Omega^1(M, \mathfrak{m}). \tag{2.1}$$

These conventions are those of [FLMSE24, Sections 1.2–1.4, Lemma 1.4 and Lemma 1.17]. Thus $T = 0$ precisely when the defining tensor is parallel. Since the action is a right action,

$$A \diamond (B \diamond \xi) - B \diamond (A \diamond \xi) = -[A, B] \diamond \xi.$$

A skew endomorphism acts skew-adjointly on forms. For a tensor of type (p, q), the identity endomorphism acts by multiplication by its net degree $w = q - p$; in particular, $g \diamond \varphi = 3\varphi$ for a G_2 form and $g \diamond \Phi = 4\Phi$ for a Cayley form.

We identify an endomorphism E with the covariant 2-tensor $A(x, y) = g(Ex, y)$. In an orthonormal frame the matrix of E is A^{t}; this transpose is important for the signs of skew tensors. We use the full tensor norm on intrinsic torsion:

$$(T \star T)_{ij} = T_{i,pq} T_{j,pq}, \qquad |T|^2 = T_{i,pq} T_{i,pq}, \qquad (\operatorname{div} T)_{ij} = \nabla_p T_{p,ij}. \tag{2.2}$$

The flow equations.

Definition 2.1 (Quadratic Ricci–harmonic flow)**.** Fix an H-equivariant quadratic map from the intrinsic-torsion module $(\mathbb{R}^n)^* \otimes \mathfrak{m}$ to $\Sigma^2 \oplus \mathfrak{m}$. Its associated tensorial expression is denoted by $Q(T)$. The corresponding quadratic Ricci–harmonic flow is

$$\partial_t \xi = \big[-\mathrm{Ric} + \operatorname{div} T + Q(T)\big] \diamond \xi. \tag{2.3}$$

The coefficients of Q are fixed; there are no prescribed external fields in this class.

The leading terms in (2.3) couple Ricci flow of the metric to harmonic motion in its evolving isometric class. A related coupling occurs in Müller's metric–map flow [Mül12, Section 1, $(\mathrm{RH})_\alpha$]. There the harmonic variable

is a map into a fixed target; here it is a reduction of the tangent bundle, whose connection varies with the induced metric. We focus on the two-parameter modified Ricci–harmonic flow

$$\partial_t \xi = \mathcal{F}_{a,b}(\xi) := \big[-\mathrm{Ric} + a(T \star T) + b|T|^2 g + \operatorname{div} T\big] \diamond \xi, \qquad a, b \in \mathbb{R}. \tag{2.4}$$

We abbreviate it to $\mathrm{MHR}_{a,b}$. The relation with the parameters in [FFL+26, (3.27), Section 3.3.2] is

$$a = \lambda - n\hat{\lambda}, \qquad b = \hat{\lambda}, \qquad \mu := a + nb = \lambda. \tag{2.5}$$

Its metric and volume equations are

$$\partial_t g = -2\mathrm{Ric} + 2a(T \star T) + 2b|T|^2 g, \tag{2.6}$$

$$\partial_t \operatorname{dvol}_g = (-R + \mu|T|^2)\, \operatorname{dvol}_g . \tag{2.7}$$

In particular, a stationary MHR structure satisfies

$$R = \mu|T|^2.$$

Paper II proves short-time existence and uniqueness for the whole class (2.3), whereas its doubling, Shi and continuation estimates are stated for MHR [FFL+26, Theorem 4.2, Proposition 5.1, Theorem 5.2, Corollary 5.3 and Theorem 5.4]. We record those MHR estimates in theorem 4.1, then prove their extensions to (2.3) before using them for compactness and continuation. The stability theorem also applies to this larger class, since every quadratic torsion term has zero first variation at a torsion-free structure. The soliton and volume identities retain the exact MHR coefficients.

Gauge and scaling. For a fixed background metric $\bar{g}$, let

$$W^k = g^{ij}\big(\Gamma(g)^k_{ij} - \Gamma(\bar{g})^k_{ij}\big)$$

be the standard DeTurck vector field. Diffeomorphism invariance makes the original equation only weakly parabolic. The gauged equation is obtained by adding $\mathcal{L}_W \xi$ to (2.3); this is the field $2X_{\mathrm{DT}}$ in [FFL+26, Proposition 4.1 and proof of Theorem 4.2]. The identity

$$\mathcal{L}_Y \xi = \big[T_Y + \tfrac{1}{2}\mathcal{L}_Y g + \tfrac{1}{2} dY^\flat\big] \diamond \xi \tag{2.8}$$

fixes its normalisation. In the orbit chart $u = S + C \mapsto \bar{\xi} \cdot \exp(u)$, this modification has scalar heat principal symbol and is strictly parabolic. Pullback by the diffeomorphisms generated by $-W$ recovers the geometric equation.

The scaling is determined by the metric. For $s > 0$, let $\delta_s \xi$ multiply each tensor component of net degree w by $s^{w/2}$. Then $g_{\delta_s \xi} = s g_\xi$ and the parabolic rescaling

$$\xi_s(t) = \delta_s \xi(t/s)$$

preserves every fixed equation (2.3). In particular, $\varphi_s(t) = s^{3/2}\varphi(t/s)$ and $\Phi_s(t) = s^2 \Phi(t/s)$. The control quantity

$$\Lambda_{\mathrm{H}}(x,t) = \big(|\mathrm{Rm}|^2 + |\nabla T|^2 + |T|^4\big)^{1/2}, \qquad \Lambda_{\mathrm{H}}(t) = \sup_M \Lambda_{\mathrm{H}}(\cdot, t) \tag{2.9}$$

scales by s^{-1}. For a finite maximal time T_0, we call the singularity Type I if

$$\sup_{t<T_0} (T_0 - t)\Lambda_{\mathrm{H}}(t) < \infty,$$

and Type IIa otherwise. An immortal solution is Type III if $\sup_{t \geq 0} t\Lambda_{\mathrm{H}}(t) < \infty$, and Type IIb otherwise. These terms describe rates; a soliton conclusion requires a separate argument.

G_2 torsion conventions and Dwivedi's flow. We use the orientation convention of [FLMSE24, (1.10)] and [Dwi26, (2.2), (2.4)]:

$$(u\lrcorner\varphi)\wedge(v\lrcorner\varphi)\wedge\varphi=-6g(u,v)\,dV,\qquad \psi=*\varphi.$$

The Hodge star below uses this orientation. Reversing it reverses ψ and the full torsion $\mathcal{T}$, but leaves the intrinsic torsion and MHR unchanged. Define $\mathcal{T}$ by

$$\nabla_i\varphi_{jkl}=\mathcal{T}_{im}\psi_{mjkl}. \tag{2.10}$$

In this comparison, write T^{H} for the intrinsic torsion used in Papers I–II, to distinguish it from $\mathcal{T}$. Dwivedi's equation is [Dwi26, Definition 3.2, (RHF)]

$$\partial_t\varphi=[-\mathrm{Ric}+3\mathcal{T}^{\mathsf{t}}\mathcal{T}-|\mathcal{T}|^2g]\diamond\varphi+(\operatorname{div}\mathcal{T})\lrcorner\psi. \tag{2.11}$$

Proposition 2.2 (MHR versus Dwivedi in the G_2 case)**.** *The two torsion conventions satisfy*

$$T^{\mathrm{H}}_{i;pq}=-\tfrac{1}{3}\mathcal{T}_{im}\varphi_{mpq},\quad |T^{\mathrm{H}}|^2=\tfrac{2}{3}|\mathcal{T}|^2,\quad (T^{\mathrm{H}}\star T^{\mathrm{H}})=\tfrac{2}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}}, \tag{2.12}$$

and

$$(\operatorname{div}T^{\mathrm{H}})\diamond\varphi=(\operatorname{div}\mathcal{T})\lrcorner\psi. \tag{2.13}$$

Consequently, MHR$_{a,b}$ becomes

$$\partial_t\varphi=\left[-\mathrm{Ric}+\tfrac{2a}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}}+\tfrac{2b}{3}|\mathcal{T}|^2g\right]\diamond\varphi+(\operatorname{div}\mathcal{T})\lrcorner\psi. \tag{2.14}$$

For

$$(a,b)=\left(\tfrac{9}{2},-\tfrac{3}{2}\right),\qquad (\lambda,\hat{\lambda})=\left(-6,-\tfrac{3}{2}\right), \tag{2.15}$$

the MHR quadratic term is $3\mathcal{T}\mathcal{T}^{\mathsf{t}}-|\mathcal{T}|^2g$, whereas Dwivedi's is $3\mathcal{T}^{\mathsf{t}}\mathcal{T}-|\mathcal{T}|^2g$. Their discrepancy is

$$3\big(\mathcal{T}\mathcal{T}^{\mathsf{t}}-\mathcal{T}^{\mathsf{t}}\mathcal{T}\big). \tag{2.16}$$

Thus the equations agree exactly wherever $\mathcal{T}$ is normal. This includes symmetric torsion (in particular co-closed and nearly parallel structures) and skew torsion (in particular closed structures).

Proof. The algebraic identity $(-\frac{1}{3}v\lrcorner\varphi)\diamond\varphi=v\lrcorner\psi$, together with (2.10) and injectivity of the diamond action on $\mathfrak{m}$, proves the first identity in (2.12). Contracting and using $\varphi_{mpq}\varphi_{npq}=6g_{mn}$ gives the next two identities. Also,

$$(\operatorname{div}T^{\mathrm{H}})_{ij}=-\tfrac{1}{3}(\operatorname{div}\mathcal{T})_m\varphi_{mij}-\tfrac{1}{3}\mathcal{T}_{pm}\mathcal{T}_{pa}\psi_{amij}.$$

The last term vanishes because $\mathcal{T}_{pm}\mathcal{T}_{pa}$ is symmetric in m,a. Applying the same algebraic identity proves (2.13). Substitution gives (2.14), and coefficient matching gives (2.15)–(2.16). □

Remark 2.3. No preservation theorem for the normal-torsion locus is being asserted. Proposition 2.2 gives an exact identification for solutions known to remain there and a lower-order comparison in general. Dwivedi's equation is nevertheless an exact member of the broader class (2.3).

MHR and reasonable $\mathrm{Spin}(7)$-flows. A flow $\partial_t\Phi=(h+X)\diamond\Phi$ is *reasonable* in Duthie's sense [Dut26, Definition 3.1 and (3.2)–(3.4)] if it has short-time existence and uniqueness for its lifespan and, schematically,

$$2h=-2\mathrm{Ric}+L(T)+T*T+C, \tag{2.17}$$

$$X=L(\nabla T)+L(\mathrm{Rm})+L(T)+T*T+C, \tag{2.18}$$

$$\partial_tT=\Delta T+L(\nabla T)+L(T)+\mathrm{Rm}*T+\nabla T*T+T*T*T+T*T. \tag{2.19}$$

Here L denotes controlled linear tensorial maps and C is prescribed with all derivatives bounded.

Proposition 2.4 ($\mathrm{Spin}(7)$ MHR is a reasonable flow)**.** *For every fixed (a,b), MHR$_{a,b}$ with $\mathrm{H}=\mathrm{Spin}(7)$ is reasonable in Duthie's sense.*

Proof. Short-time existence and uniqueness on a closed manifold follow from [FFL+26, Theorem 4.2 and its proof]; the added quadratic terms are of lower order for every (a, b). For uniqueness throughout the common lifespan, let t^* be the supremum of the times up to which two solutions with the same initial data agree. Continuity gives agreement at t^*, and short-time uniqueness restarted there extends it if t^* is interior to both lifespans. By (2.6),

$$2h = -2\mathrm{Ric} + 2a(T \star T) + 2b|T|^2 g,$$

which has the form (2.17). The skew velocity is $X = \operatorname{div} T = L(\nabla T)$, already valued in Ω^2_7 [Dut26, Definition 2.15], giving (2.18). Finally, Paper II's torsion equation [FFL+26, Corollary 3.22 and (5.2)], whose quadratic-term clause applies to every (a, b), is a special case of (2.19), with $C = 0$. These are precisely the defining requirements. □

Consequently Duthie's estimates and compactness theorems for a fixed reasonable $\mathrm{Spin}(7)$ equation apply to MHR, with their stated hypotheses [Dut26, Theorems 4.2, 4.5, 5.1 and 6.1–6.2]. This application is independent of the Taylor expression considered in Appendix B.

3 Stability near torsion-free structures

We ask whether a small perturbation of a torsion-free structure exists for all time and converges to a nearby torsion-free structure. The possible limits form a local moduli space, whose tangent directions are neutral modes of the linearised flow. We must therefore prove decay transverse to this space and control the motion along it.

With the metric fixed, the isometric G_2- and $\mathrm{Spin}(7)$-flows admit long-time existence and harmonic limits under small-torsion or small-entropy hypotheses [DGK21, Theorems 5.13 and 5.15] [DLSE24, Theorems 5.9 and 5.10]. Such limits satisfy $\operatorname{div} T = 0$ and need not be torsion-free. For the moving metric alone, Haslhofer–Müller establish dynamical stability of compact Ricci-flat metrics that locally maximise Perelman's λ-functional, without assuming integrability [HM14, Theorem 1.1]. The argument below instead uses the integrable torsion-free moduli space to control the coupled metric and structure.

At a torsion-free structure, $T = 0$, so quadratic torsion terms have zero first variation. Every flow in the class below consequently has the same gauged linearisation: a Lichnerowicz Laplacian in the metric directions and a rough Laplacian in the vertical directions. Full exceptional holonomy removes the vertical kernel, and deformation theory identifies the metric kernel with the tangent space of the torsion-free moduli. Throughout this section M is connected and closed, and all norms and Sobolev spaces are defined using the background structure.

Theorem 3.1 (Stability of torsion-free structures)**.** *Let* $\mathrm{H} \in \{\mathrm{G}_2, \mathrm{Spin}(7)\}$ *and let* $\bar{\xi}$ *be a torsion-free* H*-structure with* $\mathrm{Hol}(\bar{g}) = \mathrm{H}$*. Let* $Q(T)$ *be a smooth autonomous natural quadratic torsion term, and consider*

$$\partial_t \xi = \big[-\mathrm{Ric} + \operatorname{div} T + Q(T)\big] \diamond \xi. \tag{3.1}$$

Let $W = W(g, \bar{g})$ *be the standard DeTurck vector field*

$$W^k = g^{ij}\big(\Gamma(g)^k_{ij} - \Gamma(\bar{g})^k_{ij}\big), \tag{3.2}$$

and add $\mathcal{L}_W \xi$ *to the right-hand side of* (3.1)*. In a smooth admissible-orbit chart at* $\bar{\xi}$*, modelled on* $E = \Sigma^2 \oplus \Lambda^2_{\mathfrak{m}}$*, fix* $p > n + 2$ *and put*

$$X_0 = L^p(E), \qquad X_1 = W^{2,p}(E), \qquad X_\gamma = W^{2-2/p,p}(E).$$

Then $\bar{\xi}$ *is normally stable for this gauged equation. Every sufficiently small* X_γ *perturbation has a global gauged solution converging exponentially in* X_γ *to a nearby torsion-free structure. For smooth initial data the corresponding ungauged solution exists for all time and converges smoothly, after the DeTurck reconstruction, to a torsion-free structure.*

In particular, the conclusion holds for $\mathrm{MHR}_{a,b}$ *with arbitrary parameters* a, b*, and for Dwivedi's* G_2 *Ricci–harmonic flow.*

Remark 3.2. For Dwivedi's equation, Theorem 3.1 settles the full-holonomy case of [Dwi26, Question 8.4]. Its hypotheses do not include reduced-holonomy backgrounds. In that setting the vertical rough Laplacian can have parallel sections and harmonic vector fields affect the gauge slice; those kernel directions require a separate analysis.

3.1 Deformation and parabolic inputs

We use the nonpositive Laplacians

$$\Delta = \nabla^i \nabla_i = -\nabla^* \nabla, \qquad \Delta_L h = \Delta h + 2\mathring{R} h$$

at a Ricci-flat metric. The parallel-spinor estimate of [DWW05, Lemma 2.4 and Theorem 2.5] states that $-\Delta_L$ is nonnegative on all symmetric 2-tensors at a torsion-free exceptional structure; their L_g is our $-\Delta_L$. The following specialisation of the deformation theory identifies the metric kernel explicitly in the $\mathrm{Spin}(7)$ case.

Lemma 3.3 (The Lichnerowicz kernel for $\mathrm{Spin}(7)$)**.** *Let $\bar{\Phi}$ be a torsion-free $\mathrm{Spin}(7)$-structure with $\mathrm{Hol}(\bar{g}) = \mathrm{Spin}(7)$, oriented so that $\bar{\Phi}$ is self-dual. On trace-free symmetric 2-tensors set*

$$\Psi(h) = h_{ij}\, e^i \wedge (e_j \lrcorner \bar{\Phi}).$$

Then

$$\Delta_d \Psi(h) = -\Psi(\Delta_L h). \tag{3.3}$$

In particular, Ψ identifies the trace-free divergence-free kernel of Δ_L with $\mathcal{H}^4_{35}$, and

$$\dim \ker \Delta_L = 1 + b^4_-(M).$$

This kernel is the metric tangent space, in Bianchi gauge, of the local moduli space of torsion-free $\mathrm{Spin}(7)$-structures.

Proof. The parallel $\mathrm{Spin}(7)$-equivariant map Ψ identifies Σ^2_0 with the anti-self-dual bundle Λ^4_{35}, and its extension to Σ^2 satisfies $\Psi(\bar{g}) = 4\bar{\Phi}$; see [Kar08, Proposition 2.4, Corollary 2.6 and Proposition 2.8]. Parallel equivariant maps intertwine the corresponding standard Laplacians [Nor13, Lemmas 2.5–2.6]. On forms this operator is Δ_d, and on symmetric 2-tensors at a Ricci-flat metric it is $-\Delta_L$. This proves (3.3).

Full holonomy and the Bochner formula give $b^1(M) = 0$. If $\Delta_L h = 0$, taking the trace shows that $\mathrm{tr}\, h$ is constant. Moreover, the Ricci-flat commutation identity $\Delta_L \mathcal{L}_Y \bar{g} = \mathcal{L}_{\Delta Y} \bar{g}$, together with self-adjointness, shows that $\mathrm{div}\, h$ is a harmonic 1-form. It therefore vanishes. Consequently

$$\ker \Delta_L = \mathbb{R}\bar{g} \oplus \{h : \mathrm{tr}\, h = 0,\ \mathrm{div}\, h = 0,\ \Delta_L h = 0\},$$

and the intertwining identifies the second summand with all harmonic anti-self-dual 4-forms.

To identify the full moduli tangent space, we also need $\mathcal{H}^4_7 = 0$. A unit parallel positive spinor σ gives the parallel splitting $S^+ = \mathbb{R}\sigma \oplus (\Lambda^2_7 \cdot \sigma)$. For harmonic $\alpha \in \Omega^2_7$, the Clifford identity $D(\alpha \cdot \sigma) = (d\alpha + \delta\alpha) \cdot \sigma = 0$ and the scalar-flat Lichnerowicz formula show that $\alpha \cdot \sigma$ is parallel. Full $\mathrm{Spin}(7)$ holonomy has only the invariant line $\mathbb{R}\sigma$, whereas $\alpha \cdot \sigma \perp \sigma$. Thus $\alpha = 0$. The parallel equivariant isomorphism $\Lambda^2_7 \to \Lambda^4_7$, $\alpha \mapsto \alpha \diamond \bar{\Phi}$, again intertwines the standard Laplacians, so $\mathcal{H}^4_7 = 0$.

Finally, the torsion-free pre-moduli space is smooth with tangent space the harmonic sections of $\Lambda^4_1 \oplus \Lambda^4_7 \oplus \Lambda^4_{35}$ [Nor13, Proposition 3.1]. Its metric derivative maps onto the infinitesimal Ricci-flat deformations [Nor13, Lemma 3.13]. The preceding identifications give precisely the stated kernel, with one scaling direction and b^4_- trace-free directions. □

This identifies the neutral directions. The following parabolic theorem converts decay in the complementary directions into convergence to an equilibrium.

Established theorem 3.4 (Normal stability in maximal regularity)**.** [PSZ09, §2, (2.1)–(2.5), Theorem 2.1].

Let $X_1 \hookrightarrow X_0$ be densely and continuously embedded Banach spaces, $1 < p < \infty$, and $X_\gamma = (X_0, X_1)_{1-1/p,p}$. Consider

$$\partial_t u + A(u)u = f(u), \qquad (A, f) \in C^1\big(V; \mathcal{L}(X_1, X_0) \times X_0\big)$$

on an open subset $V \subset X_\gamma$. Suppose that $\bar{u} \in V \cap X_1$ is an equilibrium and $A(\bar{u})$ has maximal L^p-regularity. Write L for the linearisation of $-A(u)u + f(u)$ at $\bar{u}$. Assume that the nearby equilibria form a finite-dimensional C^1 manifold in X_1, tangent to $\ker L$ at $\bar{u}$, that $X_0 = \ker L \oplus \mathrm{ran}\, L$, and that $\sigma(L) \setminus \{0\} \subset \{\mathrm{Re}\, z < 0\}$. Then sufficiently small X_γ perturbations exist for all positive time and converge exponentially in X_γ to a nearby equilibrium.

The next proof checks these hypotheses in geometric terms. Its three main inputs are the block form of the gauged linearisation, the identification of all nearby equilibria with torsion-free structures, and maximal regularity. The final step removes the gauge.

3.2 Proof of stability

Proof of Theorem 3.1. We identify the linearisation and the local equilibrium manifold, then apply theorem 3.4 and reconstruct the geometric flow.

The gauged linearisation. Write a variation as $(S + C) \diamond \bar{\xi}$, so that $\delta g = 2S$. All operators in the linearisation below are computed with $\bar{g}$. At $\bar{\xi}$ both T and W vanish. Since Q is quadratic, its first variation vanishes as well. Thus the linearisation is independent of Q.

The metric equation of the gauged flow has linearisation $\partial_t(2S) = \Delta_L(2S)$, since its linear part is the Ricci–DeTurck equation. The intrinsic torsion variation is [FFL+26, Proposition 3.10, (3.15)]

$$\delta T_l = \nabla_l C - \pi_{\mathfrak{m}}(\Lambda \nabla S_l), \qquad (\Lambda \nabla S_l)_{ab} = \nabla_a S_{bl} - \nabla_b S_{al}.$$

No connection or projection variation remains in $\delta(\operatorname{div} T)$, because each such term multiplies $T = 0$. Also

$$\delta W = 2 \operatorname{div} S - d \operatorname{tr} S.$$

The exact Lie derivative decomposition is

$$\mathcal{L}_W \xi = \left(\tfrac{1}{2}\mathcal{L}_W g + \tfrac{1}{2} dW^\flat + T_W\right) \diamond \xi,$$

where only the $\mathfrak{m}$ component of the skew term acts on ξ. The term T_W has zero linearisation at $\bar{\xi}$. Combining the remaining skew variations gives

$$\Delta C + \pi_{\mathfrak{m}}\big([\nabla_a, \nabla^l] S_{bl} - [\nabla_b, \nabla^l] S_{al}\big)_{ab}.$$

The displayed commutator term is zero. Indeed, its curvature contractions $R_{albp} S_{lp}$ are symmetric in a, b, by pair symmetry of the curvature tensor and symmetry of S; the other contractions contain $\mathrm{Ric} = 0$. The linearisation is therefore exactly

$$L(S, C) = (\Delta_L S, \Delta C). \tag{3.4}$$

Spectrum and kernel. The parallel-spinor estimate recalled above gives $\Delta_L \leq 0$ on all symmetric 2-tensors. The rough Laplacian on $\Lambda^2_{\mathfrak{m}}$ is nonpositive and has kernel the parallel sections. This bundle is the nontrivial irreducible 7-dimensional holonomy module in both cases, so full holonomy makes that kernel zero. Both diagonal operators are self-adjoint on L^2, elliptic, and have compact resolvent. Their L^p spectra and kernels agree with their L^2 counterparts by elliptic regularity. Hence

$$\ker L = \ker \Delta_L \oplus \{0\}, \qquad X_0 = \ker L \oplus \operatorname{ran} L, \tag{3.5}$$

and the nonzero spectrum of L lies in $(-\infty, -\omega]$ for some $\omega > 0$.

The equilibrium manifold. Torsion-free H-structures have Ricci-flat metrics. Their local pre-moduli space is smooth, its tangent space consists of harmonic orbit-tangent forms, and its metric derivative maps onto the entire space of infinitesimal Ricci-flat deformations [Nor13, Proposition 3.1 and Lemma 3.13]. These facts apply to both groups. In the $\mathrm{Spin}(7)$ case the metric kernel is described explicitly by Lemma 3.3; in the G_2 case it has dimension $b^3(M)$, including scaling [DWW05, §3, (3.20)–(3.22)].

We choose representatives whose metrics satisfy $W(g, \bar{g}) = 0$. This choice is possible and smooth: the derivative at the identity of $\phi \mapsto W(\phi^* \bar{g}, \bar{g})$ is $Y \mapsto \Delta Y$. By the Bochner formula and full holonomy, $b^1(M) = 0$, so this operator is invertible. The implicit function theorem, applied in $C^{k+2,\alpha}$ diffeomorphisms and $C^{k,\alpha}$ vector fields, therefore supplies a unique small gauge correction for each member of the smooth finite-dimensional pre-moduli family. Denote the resulting family of structures by $\mathcal{E}$. Every member of $\mathcal{E}$ is a gauged equilibrium.

The metric derivative of $\mathcal{E}$ at $\bar{\xi}$ is precisely $\ker \Delta_L$. To see that the gauge correction does not remove any such direction, a kernel tensor has constant trace and zero divergence, by the trace and divergence commutation identities used above. It therefore already satisfies the linearised equation $\delta W = 0$. Conversely, every metric derivative in this slice is an infinitesimal Ricci-flat deformation. Furthermore, the metric map is injective on the tangent of the pre-moduli family: if a torsion-free variation has $S = 0$, differentiating $T = 0$ gives $\nabla C = 0$; full holonomy then gives $C = 0$. It follows that

$$T_{\bar{\xi}} \mathcal{E} = \ker L. \tag{3.6}$$

The family is consequently an embedded finite-dimensional manifold near $\bar{\xi}$.

There are no additional equilibria sufficiently near $\bar{\xi}$. Indeed, the gauged operator $P : X_1 \supset U \to X_0$ is smooth. With the splitting in (3.5), the range component of $P = 0$ is solved uniquely as a graph over $\ker L$ by the implicit function theorem. By (3.6), the projection of $\mathcal{E}$ onto $\ker L$ contains a neighbourhood of zero. Thus its graph is the entire local solution graph of the range component, and the remaining finite-dimensional equation vanishes on that neighbourhood. All nearby equilibria therefore belong to $\mathcal{E}$.

Global existence and convergence in the gauge. In the orbit chart the gauged equation is a smooth second-order quasilinear system with scalar heat symbol [FFL+26, Theorem 4.2 and its proof]. Its principal coefficients depend smoothly on the structure, and its remaining terms depend smoothly on the first jet. Since $p > n + 2$, the embedding $X_\gamma \hookrightarrow C^{1,\alpha}$ holds for some $\alpha > 0$. Consequently it has the form $\partial_t u + A(u)u = f(u)$ with $(A, f) \in C^1(V; \mathcal{L}(X_1, X_0) \times X_0)$. The smooth strongly elliptic background operator has maximal L^p-regularity. Localise in finitely many trivialising balls and apply [PSZ09, Proposition 3.2, (3.8)] to the local Dirichlet problems. Their scalar strongly elliptic principal symbols satisfy (E) and (LS), and the cut-off sections vanish near the artificial boundaries. A partition of unity introduces only first-order commutators; interpolation absorbs these after a spectral shift and time subdivision. This gives the a-priori estimate on a short interval $[0, a]$. For existence, let χ_j be cut-offs supported in the balls and equal to one on the supports of the partition ψ_j, and let S_j be the solution operators of the local Dirichlet problems with zero initial and boundary data. Then $u = \sum_j \chi_j S_j(\psi_j f)$ solves $\partial_t u + A(\bar{u})u = f + \sum_j [A(\bar{u}), \chi_j] S_j(\psi_j f)$. The error is of first order, so its norm in $L^p(0, a; L^p)$ is at most $Ca^\theta \|f\|$ for some $\theta > 0$, and a Neumann series gives a solution when a is small. With the estimate and time subdivision, this is maximal L^p-regularity on every finite interval. Together with (3.5)–(3.6), this verifies every hypothesis of theorem 3.4 and proves the asserted global existence and exponential X_γ convergence.

Reconstruction of the geometric flow. For smooth initial data, local parabolic regularity on time strips and the uniform small X_γ bound give uniform bounds in every higher Sobolev norm for $t \geq 1$. Interpolation with exponential X_γ convergence gives exponential convergence in every C^k norm, with a possibly smaller exponent. The limiting equilibrium has $W = 0$, so $W(t)$ also decays exponentially in every C^k norm. Let F_t solve $\partial_t F_t = -W(t) \circ F_t$, $F_0 = \mathrm{Id}$. These diffeomorphisms exist for all time on the closed manifold. Integrability of every C^k norm of $W(t)$ for $t \geq 1$, and the corresponding flow equations for derivatives and inverses, show that F_t converges smoothly to a diffeomorphism F_∞. Integrating from t to infinity in the same flow equations shows that this convergence is exponential in every C^k norm. Naturality gives an ungauged solution $F_t^* \xi(t)$, which converges smoothly to the torsion-free structure $F_\infty^* \xi_\infty$. □

4 Compactness and continuation

We proceed from derivative estimates to pointed limits, and then to continuation. A bound for Λ_{H} controls all higher derivatives away from the initial time. Together with an injectivity bound, these estimates give smooth pointed limits of fixed quadratic Ricci–harmonic flows. At a putative singular time, the resulting blow-ups allow us to test whether weaker information about the metric and torsion already forces continuation.

The compactness argument combines Hamilton's pointed Riemannian compactness theorem [Ham95, Theorem 2.3] with the time-independent comparison maps used for Laplacian flow by Lotay–Wei [LW17, Theorem 7.2]. Analogous dynamic results hold for Dwivedi's Ricci–harmonic G_2 flow [Dwi26, Theorems 5.5–5.6] and for a given reasonable $\mathrm{Spin}(7)$ flow [Dut26, Theorem 6.2]. Here we carry this framework to arbitrary tensorial H in the fixed quadratic class. MHR is the principal application; its relation to Dwivedi's equation is the one specified in Proposition 2.2.

Throughout this section, $\mathrm{H} \subset \mathrm{SO}(n)$ is a closed tensorial subgroup. Retained regularity, doubling and continuation from bounded Λ_{H} apply to the fixed quadratic class (2.3). The compactness theorem permits a bounded sequence of fixed coefficient models, with a subsequence converging to a fixed limiting model. Each individual operator is autonomous. The continuation and blow-up results concern one fixed quadratic equation. The source manifolds are closed, whereas the pointed limits may be noncompact. All norms and covariant derivatives use the induced metric, and Λ_{H} denotes the curvature–torsion quantity in (2.9).

Analytic input. We collect the estimates from Paper II used below. The doubling estimate provides a common time interval after rescaling, the Shi estimates supply smooth compactness, and the continuation criterion detects finite-time singularities. The principal symbol will also be used in Appendix A.

Established theorem 4.1 (MHR estimates from Paper II)**.** [FFL+26, Theorem 4.2; Corollary 3.22; (5.2), (5.6); Proposition 5.1 and Theorems 5.2 and 5.4]

Smooth initial data on a closed manifold admit a unique smooth MHR solution for a short time. The DeTurck modification has scalar heat principal symbol, with the source's symbol convention,

$$\sigma(D\mathcal{P})(x,\chi)(S+C) = |\chi|^2(S+C), \qquad S \in \Sigma^2, \quad C \in \Lambda^2_{\mathfrak{m}}. \tag{4.1}$$

Its torsion and curvature satisfy the schematic equations

$$\partial_t T = \Delta T + \mathrm{Rm} * T * \Xi + \nabla T * T * \Xi + T * T * T, \tag{4.2}$$

$$\partial_t \mathrm{Rm} = \Delta \mathrm{Rm} + \mathrm{Rm} * (\mathrm{Rm} + T * T) + \nabla^2 T * T + \nabla T * \nabla T, \tag{4.3}$$

where Ξ is the algebraic tensor defining the $\mathfrak{m}$-projection and $*$ denotes universal contractions with coefficients depending on the flow. If the solution exists on $[0,\delta)$, then

$$\Lambda_{\mathrm{H}}(t) \le 2\Lambda_{\mathrm{H}}(0) \quad \text{for } 0 \le t < \min\{\delta, (C\Lambda_{\mathrm{H}}(0))^{-1}\}. \tag{4.4}$$

If $K > 0$ and $\Lambda_{\mathrm{H}} \le K$ on $M \times [0, K^{-1}]$, then

$$|\nabla^m \mathrm{Rm}| + |\nabla^{m+1} T| \le C_m K t^{-m/2}, \qquad 0 < t \le K^{-1}, \quad m \ge 0. \tag{4.5}$$

At every finite maximal time T_0,

$$\lim_{t \uparrow T_0} \Lambda_{\mathrm{H}}(t) = \infty, \qquad \Lambda_{\mathrm{H}}(t) \ge \frac{c}{T_0 - t}. \tag{4.6}$$

4.1 Retained regularity and continuation

The ordinary Shi estimate allows derivatives to grow as the initial time is approached, even when those derivatives are initially bounded. The retained estimate removes this loss through each controlled initial order; smoothing is needed only above it. Dwivedi proves the corresponding local estimate for his particular G_2 equation [Dwi26, Theorem 4.7, (4.50)–(4.53), and Lemma 4.10]. We first establish a global estimate for the whole quadratic class.

Give the finite-dimensional coefficient space in Definition 2.1 a fixed norm. The constants may depend on the tensor model and a bound for this norm, but not on higher initial derivatives. The exponent in the theorem has a direct interpretation: orders $m \le \ell$ stay uniformly bounded, and the first unretained order has at most a $t^{-1/2}$ singularity.

Theorem 4.2 (Retained initial regularity)**.** *Let M be closed, let $\mathcal{K}$ be a compact family of the quadratic maps in Definition 2.1, and let $\ell \ge 0$ be an integer. Suppose a smooth solution of (2.3), for one fixed $Q \in \mathcal{K}$, exists on $M \times [0,\tau]$, where $0 < \tau \le \tau_0$, and satisfies*

$$\Lambda_{\mathrm{H}} \le K \quad \text{on } M \times [0,\tau], \qquad \sup_M \big(|\nabla^j \mathrm{Rm}| + |\nabla^{j+1} T|\big)(0) \le K_\ell \quad (0 \le j \le \ell).$$

For every integer $m \ge 0$ there is a constant $C = C(n, \mathrm{H}, \xi_\circ, \sup_{\mathcal{K}} |Q|, \ell, m, K, K_\ell, \tau_0)$ such that

$$|\nabla^m \mathrm{Rm}| + |\nabla^{m+1} T| \le C t^{-\frac{1}{2}(m-\ell)_+}, \qquad 0 < t \le \tau. \tag{4.7}$$

For $m \le \ell$ the bound includes $t = 0$. The constant is independent of M and of higher initial derivatives; smoothness through $t = 0$ is used only qualitatively.

Proof. We use a curvature–torsion energy hierarchy. The main point is to count derivatives of the algebraic coefficients as well as derivatives of curvature and torsion.

Algebraic coefficients and evolution. Call a tensor algebraic if its components in adapted frames are a fixed H-invariant model. This includes all coefficient and projection tensors in the flow. The metric connection $\nabla^{\mathrm{H}} = \nabla - T\diamond$ preserves ξ, hence transports adapted frames to adapted frames. Since H is the full stabiliser of $\xi_\circ$, it preserves every such tensor $\mathcal{A}$. Therefore

$$\nabla \mathcal{A} = T \diamond \mathcal{A}, \qquad \nabla^s \mathcal{A} = \sum \mathcal{C} * \nabla^{a_1} T * \cdots * \nabla^{a_q} T, \quad \sum_i (a_i + 1) = s, \tag{4.8}$$

where the $\mathcal{C}$ are algebraic with bounded models.

The torsion first variation is linear in the symmetric and skew velocity components [FFL+26, Proposition 3.10, (3.15)]. For $-\mathrm{Ric} + \operatorname{div} T$, the differentiated Ricci term cancels and leaves the torsion heat equation [FFL+26, Theorem 3.20, (3.34)]. Adding $Q(T)$ contributes $\nabla Q + Q * T$, which by (4.8) has types $T * \nabla T$ and T^3. This supplies the general-quadratic clause of [FFL+26, Corollary 3.22, (3.35)] explicitly. The metric variation is $\partial_t g = -2\mathrm{Ric} + 2Q_\Sigma(T)$. The curvature variation, linear in this metric velocity, and (4.8) give

$$(\partial_t - \Delta)T = \mathrm{Rm} * T + T * \nabla T + T^3, \tag{4.9}$$

$$(\partial_t - \Delta)\mathrm{Rm} = \mathrm{Rm} * \mathrm{Rm} + \mathrm{Rm} * T * T + T * \nabla^2 T + \nabla T * \nabla T + T * T * \nabla T + T^4. \tag{4.10}$$

These are schematic equalities with algebraic coefficients. The last two curvature types are needed for general Q; the metric contractions of MHR do not produce them. Compare [FFL+26, §5.1 and (5.6)].

Differentiated equations. Assign weight $a + 2$ to $\nabla^a \mathrm{Rm}$, weight $b + 1$ to $\nabla^b T$, and weight zero to algebraic coefficients. The right sides of (4.9)–(4.10) have weights 3 and 4, respectively, and every term has at least two factors. Spatial differentiation preserves this property by (4.8). The commutator identities

$$\partial_t \Gamma = g^{-1} * \nabla \partial_t g, \qquad [\nabla, \Delta]A = \mathrm{Rm} * \nabla A + \nabla \mathrm{Rm} * A$$

then show inductively that $(\partial_t - \Delta)\nabla^k \mathrm{Rm}$ and $(\partial_t - \Delta)\nabla^{k+1} T$ are sums of terms of weight $k + 4$, each with at least two factors. In particular the metric-connection commutator includes the derivatives of Q_Σ.

Give $\nabla^a \mathrm{Rm}$ level a and $\nabla^b T$, $b \geq 1$, level $b - 1$; bare T has weight one and is treated separately. A factor of level L has weight $L + 2$. In a product of weight $k + 4$ with at least two factors:

- every level is at most $k + 1$; a level-$k + 1$ factor occurs only with one bare T;
- for $k \geq 1$, a level-k factor is unique and its remaining factors are Rm, ∇T, or two copies of T;
- if all levels are below k, then $\sum_i (L_i - \ell)_+ \leq (k - \ell)_+$.

For the last assertion, if only one $L_i > \ell$, use $L_i \leq k - 1$. If at least two exceed ℓ, write p for the number of factors with a level and q for the number of bare factors. Then $\sum_i L_i = k+4-2p-q$, so $\sum_i (L_i - \ell)_+ \leq k - 2\ell \leq (k-\ell)_+$. If none exceeds ℓ the assertion is immediate. For $k = 0$, products without level one contain only Rm, ∇T and T.

The energy inequality. Put $E_k = |\nabla^k \mathrm{Rm}|^2 + |\nabla^{k+1} T|^2$. Suppose inductively that the claimed bounds hold below level k. Evolving norms introduces only $\partial_t g * \nabla^k \mathrm{Rm} * \nabla^k \mathrm{Rm}$ and the analogous torsion term, bounded by CKE_k. The differentiated equations supply the good term $-2E_{k+1}$. Their level-$k + 1$ factors contribute at most $CK^{1/2} E_k^{1/2} E_{k+1}^{1/2}$; Young's inequality absorbs these into E_{k+1}, leaving CKE_k. The level-k factors are bounded by CKE_k. Finally the lower products and the preceding level count give $Ct^{-(k-\ell)_+/2} E_k^{1/2}$, bounded by $E_k + Ct^{-(k-\ell)_+}$ with a fresh constant. At $k = 0$ the remaining products are bounded by CK^2, giving the same inequality. Consequently

$$(\partial_t - \Delta)E_k \leq -E_{k+1} + C_1 E_k + C_2 t^{-(k-\ell)_+}. \tag{4.11}$$

Here C_1 depends only on the fixed model, k, K and the coefficient bound; C_2 may also depend on the already established lower-level bounds, ℓ and τ_0. No higher initial derivative enters.

Retained orders and smoothing. The case $m = 0$ follows from $\Lambda_{\mathrm{H}} \leq K$. For $1 \leq m \leq \ell$, discard the negative term in (4.11) and apply the maximum principle to $e^{-C_1 t} E_m$. This gives the uniform bound from the initial $E_m(0) \leq K_\ell^2$ and the lower-level induction.

Let $m = \ell + j$, $j \geq 1$, and take common constants C_1, C_2 for levels $\ell, \dots, \ell + j$. Enlarge K_ℓ to $\max(K_\ell, 2K)$ if needed. Choose

$$A_{j-1} = j + C_1 \tau_0, \qquad A_{r-1} = A_r(r + C_1 \tau_0) \quad (1 \leq r < j), \qquad F = t^j E_{\ell+j} + \sum_{r=0}^{j-1} A_r t^r E_{\ell+r}.$$

Using (4.11), the coefficient of $E_{\ell+j}$ in $(\partial_t - \Delta)F$ is at most $t^{j-1}(j + C_1 t - A_{j-1}) \leq 0$. For $1 \leq r < j$ the corresponding coefficient is $t^{r-1}(A_r(r + C_1 t) - A_{r-1}) \leq 0$. The singular forcing is cancelled by its time weight, and the only remaining energy is $A_0 C_1 E_\ell$, already uniformly bounded. Thus $(\partial_t - \Delta)F \leq C_*$. Smoothness through $t = 0$ makes F continuous there, with $F(0) = A_0 E_\ell(0) \leq A_0 K_\ell^2$; the positive-time weights remove all higher initial energies. The maximum principle yields $t^j E_{\ell+j} \leq F \leq A_0 K_\ell^2 + C_* \tau_0$, proving the induction. All coefficient models are polynomial in the fixed quadratic map, so every constant is uniform over $\mathcal{K}$. □

Corollary 4.3 (Scale-invariant retained estimate)**.** *Under the hypotheses of Theorem 4.2, assume instead that* $\Lambda_{\mathrm{H}} \le K$, $0 < \tau \le \alpha/K$, *and initially*

$$|\nabla^j \mathrm{Rm}| + |\nabla^{j+1} T| \le \beta K^{1+j/2}, \qquad 0 \le j \le \ell.$$

Then

$$|\nabla^m \mathrm{Rm}| + |\nabla^{m+1} T| \le C K^{1+m/2} (Kt)^{-\frac{1}{2}(m-\ell)_+}, \qquad 0 < t \le \tau,$$

where C *depends only on the model, the coefficient bound,* ℓ, m, α, β. *For* $\ell = 0$, *one may take* $\beta = 2$ *and this is the global Shi estimate* $CKt^{-m/2}$ *for the quadratic class.*

Proof. Under $\widetilde{g}(s) = Kg(s/K)$, the norms of $\nabla^a \mathrm{Rm}$ and $\nabla^b T$ scale by $K^{-1-a/2}$ and $K^{-(b+1)/2}$. The fixed quadratic equation is preserved by Section 2. Apply Theorem 4.2 with bounds $1, \beta$ and time interval at most α, then scale back. □

The next estimate uses bounds only on an initial-metric ball. Its proof applies the same derivative hierarchy with a moving cutoff and shrinks the ball at successive derivative orders. The geometry outside the control ball never enters the constants.

Theorem 4.4 (Local retained regularity for a fixed quadratic flow)**.** *Let* $K, K_\ell, \tau_0 > 0$. *Let* M *be a closed* n*-manifold, let* H *be a closed tensorial subgroup of* $\mathrm{SO}(n)$ *with fixed model* $\xi_\circ$, *and let* $\mathcal{K}$ *be a bounded set in the finite-dimensional space of natural quadratic maps of Definition 2.1. Fix* $Q \in \mathcal{K}$, *and let* $\xi(t)$ *be a smooth solution of*

$$\partial_t \xi = (-\mathrm{Ric} + \operatorname{div} T + Q(T)) \diamond \xi, \qquad \partial_t g = -2\mathrm{Ric} + 2Q_\Sigma(T)$$

on $M \times [0, \tau]$, *where* $0 < \tau \le \tau_0$. *For* $p \in M$ *and* $r > 0$, *put* $B_0(s) = B_{g(0)}(p, s)$. *Suppose, for an integer* $\ell \ge 0$,

$$\Lambda_{\mathrm{H}} \le K \quad \text{on } B_0(r) \times [0, \tau], \qquad |\nabla^j \mathrm{Rm}(0)| + |\nabla^{j+1} T(0)| \le K_\ell \quad \text{on } B_0(r), \quad 0 \le j \le \ell.$$

For every integer $m \ge 0$ *there is a constant*

$$C = C(n, \mathrm{H}, \xi_\circ, \sup_{\mathcal{K}} |Q|, K, K_\ell, \tau_0, r, \ell, m)$$

such that

$$|\nabla^m \mathrm{Rm}| + |\nabla^{m+1} T| \le C t^{-(m-\ell)_+/2} \quad \text{on } \overline{B_0(r/2)} \times (0, \tau].$$

For $m \le \ell$, *the bound holds through* $t = 0$. *The constant is independent of the geometry outside* $B_0(r)$, *of the choice of* M, *and of higher initial derivatives.*

The product quantity used below is the retained-regularity quantity of [Dwi26, (4.55) and Lemma 4.10], with indices shifted by one. That paper proves the local estimate for its particular G_2 equation [Dwi26, Theorem 4.7]. Duthie's reasonable-$\mathrm{Spin}(7)$ estimate [Dut26, Theorem 4.2] gives the ordinary $t^{-m/2}$ loss. Here the contribution is the local retained estimate for every closed tensorial H and fixed natural quadratic map, with explicit uniformity over bounded coefficient families.

Proof. Write $L = \partial_t - \Delta_{g(t)}$ and $E_j = |\nabla^j \mathrm{Rm}|^2 + |\nabla^{j+1} T|^2$. The pointwise argument proving (4.11) uses only the bound for Λ_{H} and lower derivatives at the point in question. In particular, on any spatial subdomain on which the lower-order inductive bounds hold,

$$LE_j \le -E_{j+1} + C_1 E_j + C_2 t^{-(j-\ell)_+}. \tag{4.12}$$

This is a pointwise consequence of the already proved coefficient and commutator calculus, rather than an application of a global estimate. The constants have the dependence in Theorem 4.2. Also, by the metric compatibility of ∇ and Cauchy–Schwarz,

$$|\nabla E_j| \le 2(E_j E_{j+1})^{1/2}. \tag{4.13}$$

A moving-distance cutoff. We first record the cutoff needed to apply a local maximum principle. On $B_0(r)$, the metric equation and $\Lambda_{\mathrm{H}} \le K$ give

$$-2Dg(t) \le \partial_t g(t) \le 2Dg(t), \qquad |\mathrm{Rm}| \le K, \qquad D = C(n, \sup_{\mathcal{K}} |Q|)K.$$

Consequently, on this domain,

$$e^{-2D\tau_0}g(0) \le g(t) \le e^{2D\tau_0}g(0). \tag{4.14}$$

Let $0 < s' < s \le r$ and $x \in \overline{B_0(s')}$. Put $d = s - s' > 0$ and $R = e^{-D\tau_0}d/4$. Every curve from x to the complement of $B_0(s)$ has, up to its first exit, initial length at least d, and therefore $g(t)$-length at least $e^{-D\tau_0}d = 4R$. Thus every $g(t)$-minimising geodesic from x of length at most $2R$ remains in $B_0(s)$. In particular, these geodesics meet only the region on which the metric and curvature bounds are assumed. Their endpoints are contained in the fixed closed initial ball $\overline{B_{g(0)}(x, d/2)}$ when their length is at most $2R$.

Choose a smooth nonincreasing $\chi : [0,\infty) \to [0,1]$ that is one on $[0,1/2]$ and zero on $[1,\infty)$, with $|\chi'|^2 \le C\chi$ and $|\chi''| \le C$. Set

$$\eta(y,t) = \chi\big(d_{g(t)}(x,y)/R\big).$$

It is supported in a fixed compact subset of $B_0(s)$, satisfies $\eta(x,t) = 1$, and, wherever smooth and positive,

$$L\eta \le C_\eta, \qquad \frac{|\nabla\eta|^2}{\eta} \le C_\eta, \qquad C_\eta = C(n, D, K, R^{-1}). \tag{4.15}$$

Indeed, variation of length gives $\partial_t d_{g(t)}(x,y) \ge -D d_{g(t)}(x,y)$. Laplacian comparison along the minimising geodesic, using $\mathrm{Ric} \ge -C_n K g$, gives $\Delta d_{g(t)}(x,y) \le C_n(d_{g(t)}(x,y)^{-1} + \sqrt{K})$. Derivatives of χ occur only for $R/2 \le d_{g(t)}(x,y) \le R$. The chain rule and $\chi' \le 0$ then give $L\eta \le C_n(D + R^{-2} + \sqrt{K}/R)$ and the gradient bound.

At a cut point the same upper bounds are valid for a smooth lower support of η, which is sufficient at a positive maximum of ηF for $F \ge 0$. For clarity, fix a minimising geodesic from x to y_0 at t_0, choose q on it a small positive distance ε from x, and replace its distance by

$$\bar d(y,t) = d_{g(t)}(x,q) + d_{g(t)}(q,y).$$

The triangle inequality gives $\bar d \ge d_{g(t)}(x,y)$, with equality at (y_0,t_0), so $\chi(\bar d/R) \le \eta$, with equality there. The standard Calabi shift chooses q so the second distance is smooth near y_0; the first is smooth for sufficiently small ε. The $4R$ first-exit margin also keeps the comparison geodesics from q to y inside $B_0(s)$ for q sufficiently close to x and (y,t) near the contact point. At the contact point, its time derivative is bounded below by $-D\bar d$, and its Laplacian above by $C_n((d_{g(t_0)}(x,y_0) - \varepsilon)^{-1} + \sqrt{K})$. For $\varepsilon < R/4$ the bounds in (4.15) are therefore uniform. Replacing η by this lower support at a positive maximum, and then letting $\varepsilon \downarrow 0$, justifies the calculation below without an injectivity-radius hypothesis.

The local product inequality. Fix $k \ge 1$. Suppose lower levels have already been estimated on $B_0(s) \times (0,\tau]$, in the form $t^{(j-\ell)_+}E_j \le M_j$ for $0 \le j < k$. Set

$$\alpha_k = (k-\ell)_+, \quad \beta_k = (k-1-\ell)_+, \quad u = t^{\alpha_k}E_k, \quad v = t^{\beta_k}E_{k-1}, \quad a(t) = \begin{cases} 1, & k \le \ell, \\ t, & k > \ell. \end{cases}$$

Thus $0 \le v \le M := M_{k-1}$ and $t^{\alpha_k - \beta_k} = a(t)$. Apply (4.12) for $j = k$ and $j = k-1$. The latter requires only levels strictly below $k-1$, already available; for $k = 1$ its $j = 0$ case follows directly from $\Lambda_{\mathrm H} \le K$. Multiplying by the indicated powers of t, and using $t \le \tau_0$, gives

$$Lu \le -G + \frac{C}{a(t)}(u+1), \qquad Lv \le -\frac{u}{a(t)} + \frac{C}{a(t)}, \qquad G = t^{\alpha_k}E_{k+1}. \tag{4.16}$$

For example, when $k > \ell$ and $j = k-\ell$, $Lv \le -u/t + ((j-1)/t + C_1)v + C_2 \le -u/t + C/t$; when $k \le \ell$, no power of t is present. The gradient inequality (4.13) gives

$$|\nabla u| \le 2(uG)^{1/2}, \qquad |\nabla v| \le 2a(t)^{-1/2}(vu)^{1/2}.$$

Choose $A \ge \max\{1, 64M\}$ and put $F = (A+v)u$. The product rule, (4.16), and Young's inequality yield

$$\begin{aligned} LF &\le -(A+v)G - \frac{u^2}{a} + \frac{C}{a}(F+1) + 8a^{-1/2}v^{1/2}uG^{1/2} \\ &\le -(A/2+v)G - \frac{u^2}{2a} + \frac{C}{a}(F+1). \end{aligned}$$

In the last step, $8a^{-1/2}v^{1/2}uG^{1/2} \le (A/2)G + (32v/(Aa))u^2$, and $32v/A \le 1/2$. Since $A \le A+v \le A+M$, this implies

$$LF \le \frac{1}{a(t)}(-cF^2 + CF + C), \qquad c = \frac{1}{2(A+M)^2} > 0. \tag{4.17}$$

All constants depend only on the inductive bounds and the data in the theorem.

Localisation and induction. Fix the target order $m \ge 1$ and choose $r_j = r(1 - j/(2m))$, $0 \le j \le m$. We prove successively

$$t^{(j-\ell)_+}E_j \le M_j \quad \text{on } \overline{B_0(r_j)} \times (0,\tau], \qquad 0 \le j \le m. \tag{4.18}$$

The base case is $E_0 \le K^2$ on $B_0(r)$. For the step k, all preceding estimates are available on $B_0(r_{k-1})$, because their balls are larger. Thus the product inequality holds there. Fix any $x \in \overline{B_0(r_k)}$ and take the moving cutoff above with $s = r_{k-1}$ and $s' = r_k$. Its constants are uniform in x. Set $Z = \eta F$, extended by zero outside the cutoff support.

At a positive space-time maximum of Z with $t > 0$, using the lower support when necessary, $\nabla F = -F\nabla\eta/\eta$ and $LZ \ge 0$. By (4.17),

$$0 \le LZ \le \frac{-cZ^2 + C\eta Z + C\eta^2}{a\eta} + \frac{C_\eta Z}{\eta} + \frac{2Z|\nabla\eta|^2}{\eta^2}.$$

Multiply by $a\eta$ and use $a(t) \le \max\{1,\tau_0\}$ and (4.15):

$$0 \le -cZ^2 + \big(C + 3\max\{1,\tau_0\}C_\eta\big)Z + C.$$

The positive root bounds every positive maximum at $t > 0$. For $k \le \ell$, the initial maximum is bounded by $(A+M)K_\ell^2$; for $k > \ell$, it is zero because $u = t^{k-\ell}E_k$. Smoothness up to $t = 0$ ensures the required continuity, but no bound on $E_k(0)$ enters in this latter case. Consequently Z is uniformly bounded throughout the cylinder. Since $\eta(x,t) = 1$, $u(x,t) \le F(x,t)/A = Z(x,t)/A$. This proves the kth step of (4.18).

At $j = m$ the ball is $\overline{B_0(r/2)}$. Taking square roots and using $|\nabla^m \mathrm{Rm}| + |\nabla^{m+1}T| \le \sqrt{2E_m}$ gives the conclusion. When $m \le \ell$, continuity includes $t = 0$. The case $m = 0$ follows directly from $\Lambda_{\mathrm{H}} \le K$. □

Corollary 4.5 (Scale-invariant local retained estimate)**.** *Under the hypotheses of Theorem 4.4, assume instead that $\tau \le \alpha/K$ and the initial bounds on $B_0(r)$ are $|\nabla^j \mathrm{Rm}| + |\nabla^{j+1}T| \le \beta K^{1+j/2}$ for $j \le \ell$. Then on $\overline{B_0(r/2)} \times (0,\tau]$,*

$$|\nabla^m \mathrm{Rm}| + |\nabla^{m+1}T| \le CK^{1+m/2}(Kt)^{-(m-\ell)_+/2},$$

where C depends only on the fixed model, the coefficient bound, $n, \ell, m, \alpha, \beta$ and $\sqrt{K}r$.

Proof. Apply Theorem 4.4 to the parabolic rescaling $\widetilde{g}(s) = Kg(s/K)$. The control bound becomes one, the initial bounds become β, the time interval is at most α, and the initial ball has radius $\sqrt{K}r$. Scale back as in Corollary 4.3. □

The local estimate requires neither injectivity nor noncollapse. Its initial derivative bounds concern an already smooth solution. The different question of compactness for smooth sequences with finite initial bounds is treated in Theorem 4.9; existence from nonsmooth data remains open.

The retained estimates also complete the doubling and continuation argument for the whole fixed quadratic class. The mechanism is the one in [FFL+26, Proposition 5.1 and proof of Theorem 5.4]; we give the details, including the admissibility of the limiting structure.

Proposition 4.6 (Uniform doubling and continuation)**.** *Fix the tensorial model and a bounded family $\mathcal{K}$ of the natural quadratic maps in Definition 2.1. There is $C = C(n, \mathrm{H}, \xi_\circ, \sup_{\mathcal{K}}|Q|) > 0$ with the following property. For every smooth solution of one fixed equation (2.3) with $Q \in \mathcal{K}$ on a closed manifold, put $L(t) = \sup_M \Lambda_{\mathrm{H}}(t)$. If $L(s) > 0$, then*

$$L(t) \le \frac{L(s)}{1 - CL(s)(t-s)} \quad \text{whenever } s \le t < s + [CL(s)]^{-1} \tag{4.19}$$

within its interval of existence. In particular $L(t) \le 2L(s)$ when $0 \le t - s \le [2CL(s)]^{-1}$. If $L(s) = 0$, the solution is stationary from time s and extends for all later times.

A finite maximal time T satisfies

$$\lim_{t\uparrow T} L(t) = \infty, \qquad L(t) \geq \frac{1}{C(T-t)} \quad (0 \leq t < T). \tag{4.20}$$

Equivalently, a solution with $\sup_{M\times[0,T)} \Lambda_{\mathrm{H}} < \infty$ extends smoothly through every finite T.

Proof. Write $\mathscr{L} = \partial_t - \Delta$, $S = \Lambda_{\mathrm{H}}^2 = E_0 + |T|^4$ and $E_1 = |\nabla \mathrm{Rm}|^2 + |\nabla^2 T|^2$. The level-zero calculation in the proof of Theorem 4.2 gives

$$\mathscr{L} S \leq -E_1 + C_0 S^{3/2}, \tag{4.21}$$

with a constant depending only on the stated data. We spell out its homogeneous dependence. In the differentiated equations (4.9)–(4.10), the terms involving derivatives above level zero contribute at most

$$C_0|T|\big((|\mathrm{Rm}| + |\nabla T|)|\nabla^2 T| + |\nabla T||\nabla \mathrm{Rm}|\big) \leq E_1 + C_1 \Lambda_{\mathrm{H}}^3$$

to $\mathscr{L} E_0$. All other source and evolving-norm terms are bounded by $C_1\Lambda_{\mathrm{H}}^3$, since $|\mathrm{Rm}|, |\nabla T|, |T|^2 \leq \Lambda_{\mathrm{H}}$; the Laplacians supply $-2E_1$. For $f = |T|^2$, the undifferentiated torsion equation gives $\mathscr{L} f \leq -2|\nabla T|^2 + C_1 \Lambda_{\mathrm{H}} f$. Consequently $\mathscr{L}(f^2) = 2f\mathscr{L} f - 2|\nabla f|^2 \leq C_2 \Lambda_{\mathrm{H}}^3$. This proves (4.21). The coefficient models are bounded uniformly over $\mathcal{K}$ by (4.8).

Scalar maximum-principle comparison with $z' = C_0 z^{3/2}$ gives (4.19), with $C = C_0/2$, enlarged to be positive if needed. At zero initial maximum, comparison gives $S = 0$. Then $T = \mathrm{Rm} = 0$, and every term in the flow vanishes; short-time uniqueness makes the stationary continuation agree with the given solution. Short-time existence and uniqueness for this quadratic class are [FFL+26, Theorem 4.2].

Suppose next that $L \leq K$ up to a finite time T. On each $[\delta, T)$, $\delta > 0$, Theorem 4.2 with $\ell = 0$ bounds every covariant derivative of Rm and T. The metric equation gives $|\partial_t g|_g \leq CK$, so the metrics are uniformly equivalent to a fixed smooth metric. Moreover every $\nabla^r \partial_t g$ and every $\nabla^r \partial_t \Gamma$ is bounded there, using $\partial_t \Gamma = g^{-1} * \nabla \partial_t g$ and (4.8).

Here is the comparison with a fixed connection needed at the endpoint. Set $t_* = T/2$, $D = \nabla^{g(t_*)}$ and $A(t) = \nabla^{g(t)} - D$ on $[t_*, T)$. Since $\partial_t A = \partial_t \Gamma$ is bounded and the metrics are uniformly equivalent, integration first bounds A. Inductively, $D^r \partial_t \Gamma$ is a sum of $\nabla^r \partial_t \Gamma$ and products of lower covariant derivatives with $A, DA, \dots, D^{r-1}A$. Thus these derivatives are bounded and integration bounds $D^r A$. It follows from $\nabla g = 0$ and $\nabla \xi = T \diamond \xi$ that all fixed-connection derivatives of g and ξ are bounded. The flow equation then bounds $\partial_t g$ and $\partial_t \xi$ in every fixed C^r norm. Integrating in time proves that $g(t), \xi(t)$ converge in every such norm to smooth limits g_T, ξ_T, with g_T positive definite.

The limit retains its tensorial type. At a fixed point choose adapted $g(t)$-orthonormal frames along any sequence $t \uparrow T$. Uniform metric equivalence confines these frames to a compact subset of the invertible frame space. A subsequence converges to an invertible frame u, and $\xi_T = u \cdot \xi_\circ$. This argument does not require the entire $\mathrm{GL}(n, \mathbb{R})$-orbit to be closed. Restart [FFL+26, Theorem 4.2] at ξ_T. The old and new solutions paste smoothly: all spatial derivatives converge, and differentiating the smooth autonomous equation successively gives matching limits for every mixed time derivative. This proves continuation under a bounded L.

If a sequence $t_i \uparrow T$ at a finite maximal time had bounded $L(t_i)$, (4.19) would bound the entire remaining tail, contradicting continuation. Hence $L(t) \to \infty$. Finally, if $CL(t)(T-t) < 1$ at any time t, the same comparison would bound L on $[t, T)$ and again give continuation. This proves the lower bound in (4.20). □

4.2 Pointed limits

To pass from compactness of individual metrics to compactness of flows, the comparison maps must be kept independent of time. We choose them at one time and control the change of connection to estimate the evolving tensors in the same charts. This gives smooth convergence away from the initial boundary. Retained estimates then recover the initial trace, while rescaling and time translation give ancient and eternal limits.

Theorem 4.7 (Pointed smooth compactness for the quadratic class)**.** *Fix a tensorial model $\xi_\circ$ whose full stabiliser in $\mathrm{GL}(n, \mathbb{R})$ is a closed subgroup $\mathrm{H} \subset \mathrm{SO}(n)$. Let $Q^{(i)}$ be fixed natural quadratic terms as in Definition 2.1, with coefficient models in a bounded subset of their finite-dimensional space. Let $(M_i, \xi_i(t), p_i)$ be solutions of the corresponding quadratic Ricci–harmonic equations on closed M_i and a common interval (α, β), where $\alpha < 0 < \beta$. Suppose*

$$\sup_i \sup_{M_i \times (\alpha,\beta)} \Lambda_{\mathrm{H},i} \leq K, \qquad \inf_i \mathrm{inj}_{g_i(0)}(p_i) \geq i_0 > 0. \tag{4.22}$$

Then, after passing to a subsequence, the coefficient models converge $Q^{(i)} \to Q_\infty$ and the pointed flows converge smoothly on compact spacetime subsets, in time-independent comparison maps, to a complete solution $(M_\infty, \xi_\infty(t), p_\infty)$ of

$$\partial_t \xi_\infty = [-\mathrm{Ric} + \operatorname{div} T_\infty + Q_\infty(T_\infty)] \diamond \xi_\infty.$$

The limit is defined on $M_\infty \times (\alpha, \beta)$. In particular, sequences solving one fixed equation have limits solving that equation. Each $Q^{(i)}$ is independent of time.

Proof. Step 1. Spatial estimates and compactness at the base time. First extract a convergent subsequence of coefficient models in the finite-dimensional space; its limit Q_∞ is again a natural quadratic model. All constants below are uniform over the closure of this bounded family. Fix a compact interval $J \Subset (\alpha, \beta)$ containing 0, and put $K_0 = \max\{1, K, 2/\operatorname{dist}(J, \{\alpha, \beta\})\}$, with $2/\infty = 0$ when both endpoints are infinite. Since the equation is autonomous, we may apply Corollary 4.3, with $\ell = 0$ and unit rescaled time horizon, on each interval $[t - K_0^{-1}, t]$, for $t \in J$. For every m this gives a constant $C(J, m)$, independent of i, such that

$$\sup_{M_i \times J} (|\nabla^m \mathrm{Rm}_i| + |\nabla^{m+1} T_i|) \le C(J, m), \tag{4.23}$$

together with $|T_i| \le K^{1/2}$, since $|T|^4 \le \Lambda_{\mathrm{H}}^2$. At $t = 0$, Hamilton's static pointed compactness theorem [Ham95, Theorem 2.3], using (4.22) and these bounds, produces pointed exhaustion diffeomorphisms and a complete limit $g_\infty(0)$.

The defining tensors satisfy $\nabla \xi_i = T_i \diamond \xi_i$, and their pointwise norms are determined by the model orbit. Induction and (4.23) therefore bound all their spatial covariant derivatives.

Step 2. Estimates in time-independent charts. The pointed comparison maps are chosen at $t = 0$ and kept independent of time. The metric equation is $\partial_t g_i = -2\mathrm{Ric}_i + 2Q_\Sigma^{(i)}(T_i)$, so $|\partial_t g_i| \le CK$ and $g_i(t)$ is uniformly equivalent to $g_i(0)$ on J. The coefficient identity (4.8) and (4.23) bound every $g_i(t)$-covariant derivative of $\partial_t g_i$, uniformly in i. Put $D_i(t) = \nabla^{g_i(t)} - \nabla^{g_i(0)}$. The connection variation formula is

$$(\partial_t \Gamma_i)^k_{ab} = \tfrac{1}{2} g_i^{k\ell} \big(\nabla_a \partial_t g_{i,b\ell} + \nabla_b \partial_t g_{i,a\ell} - \nabla_\ell \partial_t g_{i,ab} \big).$$

Since $D_i(0) = 0$, integration bounds D_i. Differentiating with the fixed connection $\nabla^{g_i(0)}$, and using $\nabla^{g_i(0)} = \nabla^{g_i(t)} + D_i *$, gives by induction bounds for every fixed-connection derivative of D_i: at order k, the conversion terms involve only derivatives of D_i through order $k-1$, while the remaining term is controlled by $\nabla^{k+1} \partial_t g_i$. The identity $\nabla^{g_i(t)} g_i = 0$ then bounds every fixed-connection derivative of $g_i(t)$. The same connection comparison converts the covariant bounds for the defining tensors into fixed-connection bounds. Smooth convergence at $t = 0$ bounds the Christoffel symbols of the pulled-back $g_i(0)$ and all their coordinate derivatives on compact subsets. We therefore obtain coordinate bounds in the pointed charts. Finally, the operators are smooth quasilinear operators of order two on the admissible tensor bundle, with smooth dependence on their fixed coefficient models. In each compact pointed chart, the uniformly controlled metrics and fixed tensorial type confine the zero jets to a compact subset of that bundle. All spatial jets are bounded, as are the coefficient models. Thus the equation first bounds $\partial_t \xi_i$ and all its spatial derivatives. Repeated time differentiation then bounds every mixed derivative: at each step the new terms are products of already bounded spatial jets and lower time derivatives. There is no time derivative of the fixed model $Q^{(i)}$. The same argument applies to the metric equation. This proves uniform mixed space–time derivative bounds. Arzelà–Ascoli and a diagonal extraction now give smooth convergence of the metrics and defining tensors on compact spacetime sets in these same comparison maps.

Step 3. The limiting structure, equation, and completeness. In convergent $g_i(t)$-orthonormal frames, every ξ_i lies in the compact fixed-metric orbit $\xi_\circ \cdot \mathrm{SO}(n)$, so the limit is again an H-structure and induces $g_\infty(t)$. Since $\diamond \xi$ is injective on $\mathfrak{m}$, the torsion is a smooth algebraic function of $(\xi, \nabla\xi)$, so $T_i \to T_\infty$ with the structures. Passing to the limit in $\nabla \xi = T \diamond \xi$ identifies its intrinsic torsion. In adapted frames, $Q^{(i)}(T_i)$ is the fixed quadratic model evaluated on two copies of T_i. Smooth convergence of the structures and torsions, together with convergence of the models, therefore gives $Q^{(i)}(T_i) \to Q_\infty(T_\infty)$ in every local spacetime C^r norm. Passing to the limit in (2.3) proves the asserted equation. The bound $|\partial_t g_i| \le CK$ makes $g_\infty(t)$ uniformly equivalent to the complete $g_\infty(0)$ on each compact time interval, hence complete. Finally, exhausting (α, β) by compact intervals $J_k \ni 0$ and extracting diagonally in k gives the limit on $M_\infty \times (\alpha, \beta)$. □

Remark 4.8 (Analytic hypotheses and scope)**.** The compactness proof uses uniform positive-time spatial estimates, covariant derivative bounds for the metric velocity, a fixed tensorial orbit, and smooth convergence of the actual second-order operators in the comparison charts. The initial-boundary argument below additionally uses the retained finite-order bounds and the integrable velocity and connection estimates (4.28)–(4.29), together with their defining-tensor counterparts. For the fixed quadratic class these requirements follow from Theorem 4.2, the metric equation and (4.8); the coefficient bounds are uniform over bounded model families. The proofs also apply to other fixed operators when these same estimates and operator convergence have been verified for them. A schematic designation as reasonable, or a parabolic principal symbol alone, does not supply these bounds.

The noncollapse proposition below needs only the actual metric equation and its weighted error integral; it is independent of the tensorial group. Its ancient-limit consequence additionally uses exact parabolic scaling, the singularity lower rate and smooth compactness, all proved here for the fixed quadratic class. Neither a Type I rate alone nor an unverified monotonicity formula enters that conclusion. Exact MHR coefficients remain necessary for the soliton and dissipation identities in Section 5.

The same comparison maps can reach an initial-time boundary when only finitely many initial derivatives are controlled. Dwivedi's related finite-regularity compactness statement is [Dwi26, Theorem 5.6, (5.23)–(5.24)] for his particular G_2 equation. Here we specify the initial topology for the fixed quadratic class; the loss below is not asserted to be optimal.

The initial trace below is a limit of smooth solutions, rather than a rough-data existence theorem. Keeping the positive-time comparison maps fixed is what allows the trace and the limiting equation to be compared.

Theorem 4.9 (Compactness up to the initial boundary)**.** *Fix a tensorial model whose full* $\mathrm{GL}(n,\mathbb{R})$ *stabiliser is a closed subgroup* $\mathrm{H}\subset\mathrm{SO}(n)$*, an integer* $\ell\geq 0$*, and* $\tau, K, K_\ell, i_0>0$*. Let* $(M_i,\xi_i(t),p_i)$ *be smooth solutions on closed manifolds for* $0\leq t\leq\tau$*, with fixed quadratic models* $Q^{(i)}$ *in a bounded subset of the coefficient space in Definition 2.1. Suppose*

$$\Lambda_{\mathrm{H},i}\leq K \qquad \text{on } M_i\times[0,\tau], \tag{4.24}$$

$$|\nabla^j\mathrm{Rm}_i(0)|+|\nabla^{j+1}T_i(0)|\leq K_\ell \qquad (0\leq j\leq\ell), \tag{4.25}$$

$$\mathrm{inj}_{g_i(0)}(p_i)\geq i_0. \tag{4.26}$$

After extraction, $Q^{(i)}\to Q_\infty$*, and time-independent pointed comparison maps give a limit on a smooth manifold* M_∞ *such that the metrics and defining tensors converge in*

$$C\big([0,\tau];C^{\ell+1}(K')\big)\quad \text{for every compact } K'\subset M_\infty,$$

and smoothly on compact subsets of $M_\infty\times(0,\tau]$*. The limit is an admissible* H*-structure at every time, solves the* Q_∞ *equation for* $t>0$*, and induces complete length metrics, including at* $t=0$*. On each compact set, its metric and defining tensor are Lipschitz in time with values in* C^ℓ*, and* $1/2$*-Hölder with values in* $C^{\ell+1}$*. If* $\ell\geq 1$*, the equation also holds at zero, with a* $C^1([0,\tau];C^{\ell-1}_{\mathrm{loc}})$ *time trace.*

Proof. Step 1. Smooth comparison charts at a positive time. The metric velocity and (4.24) give a number $B\geq 1$, independent of i, such that

$$B^{-1}g_i(0)\leq g_i(t)\leq Bg_i(0),\qquad 0\leq t\leq\tau. \tag{4.27}$$

Set $t_*=\tau/2$. Choose $r_*<\min\{i_0/2,1/(4\sqrt{K+1})\}$. The curvature bound and initial injectivity bound give $\mathrm{Vol}_{g_i(0)}B_{g_i(0)}(p_i,r_*/\sqrt{B})\geq c_n(r_*/\sqrt{B})^n$ by normal-coordinate volume comparison. The inclusion of this ball in $B_{g_i(t_*)}(p_i,r_*)$, and the volume-form bound from (4.27), therefore give

$$\mathrm{Vol}_{g_i(t_*)}B_{g_i(t_*)}(p_i,r_*)\geq c_nB^{-n}r_*^n.$$

Together with $|\mathrm{Rm}_i(t_*)|\leq K$, this gives uniform injectivity at t_* by [CGT82, Theorem 4.7, pp. 47–48]. The retained estimates give all spatial bounds on $[\delta,\tau]$ for every $\delta>0$. The proof of Theorem 4.7, with base time t_*, then gives time-independent comparison maps and smooth convergence on all compact positive-time subintervals. Its derivative bounds are uniform up to τ, so this includes the upper endpoint. Extract the coefficient models at the same time. We keep these smooth charts, rather than constructing charts of finite regularity at zero.

Step 2. Integrable connection estimates. Pull back by the comparison maps and work on a fixed relatively compact coordinate set. For clarity suppress the pullbacks. Let $D_i=\nabla^{g_i(t_*)}$ and $A_i(t)=\nabla^{g_i(t)}-D_i$. The

coefficients of D_i and all their coordinate derivatives are uniformly bounded in these charts by the positive-time smooth convergence. By Theorem 4.2, the covariant derivatives of Rm_i and ∇T_i through order ℓ are bounded uniformly on $[0,\tau]$, and the next order is bounded by $Ct^{-1/2}$ for $t>0$. Differentiating the metric equation and the quadratic coefficient maps therefore gives

$$|\nabla^r \partial_t g_i| \le C \qquad (0 \le r \le \ell), \tag{4.28}$$

$$|\nabla^{\ell+1}\partial_t g_i| + |\nabla^k \partial_t \Gamma_i| \le C(1+t^{-1/2}) \qquad (0 \le k \le \ell). \tag{4.29}$$

Here $\partial_t\Gamma_i$ is the tensor given by the connection variation formula in Theorem 4.7. Its highest curvature derivative in the displayed range is $\nabla^{\ell+1}\mathrm{Rm}_i$; derivatives of $Q_\Sigma(T_i)$ use the algebraic identity (4.8) and the retained torsion bounds.

Since $A_i(t_*)=0$, integration first bounds A_i. Suppose $D_i^j A_i$ is bounded for $j<k\le\ell$. Converting $\nabla^k\partial_t\Gamma_i$ to $D_i^k\partial_t\Gamma_i$ introduces derivatives of A_i only through order $k-1$. Thus

$$|D_i^k \partial_t A_i| \le C(1+t^{-1/2}).$$

Integrating from t_* towards either endpoint proves, successively, uniform bounds for $D_i^k A_i$, $0\le k\le\ell$, all the way to zero. Only the integrable singularity $t^{-1/2}$ is used.

Step 3. Uniform control of the initial time trace. The identity $\nabla^{g_i(t)} g_i = 0$ and Step 2 give bounds for the D_i-derivatives of g_i through order $\ell+1$. Likewise, $\nabla\xi_i = T_i \diamond \xi_i$ and the retained torsion bounds control the covariant derivatives of ξ_i through order $\ell+1$; converting to D_i derivatives again uses A_i only through order ℓ. For the velocity of ξ_i, differentiate $\partial_t\xi_i = (-\mathrm{Ric}_i + \operatorname{div} T_i + Q^{(i)}(T_i))\diamond\xi_i$. Its covariant derivatives through order ℓ are uniformly bounded; at order $\ell+1$ the highest terms are $\nabla^{\ell+1}\mathrm{Rm}_i$ and $\nabla^{\ell+2}T_i$, bounded by $C(1+t^{-1/2})$. Step 2 converts these estimates, and (4.28)–(4.29), into fixed-coordinate bounds for both velocities. Hence, for either pulled-back field $u_i = g_i$ or $u_i=\xi_i$ and $0\le s\le t\le\tau$,

$$\|u_i(t)-u_i(s)\|_{C^\ell(K')} \le C_{K'}|t-s|, \tag{4.30}$$

$$\|u_i(t)-u_i(s)\|_{C^{\ell+1}(K')} \le C_{K'}|t-s|^{1/2}. \tag{4.31}$$

Indeed, $\int_s^t (1+u^{-1/2})\,du \le C_\tau\sqrt{t-s}$. Smoothness of each original solution justifies integration to zero; no uniform higher initial derivative bound has been used.

Step 4. Convergence and the limiting equation. For every fixed $\delta>0$, the already chosen subsequence converges smoothly on $[\delta,\tau]$. On $[0,\delta]$, (4.31) bounds the distance in $C^{\ell+1}(K')$ from the value at δ by $C_{K'}\sqrt{\delta}$. First taking $i,j\to\infty$ at δ and then $\delta\downarrow 0$ shows that the sequence is Cauchy in $C([0,\tau];C^{\ell+1}(K'))$. A diagonal exhaustion handles all compact sets. The time moduli pass to the limit. Compactness of the fixed-metric tensor orbit, as in Theorem 4.7, preserves admissibility even at zero. The limiting metric is uniformly equivalent to the complete metric at t_* by (4.27), hence is complete as a length metric at every time.

The positive-time smooth convergence identifies the torsion and proves the Q_∞ equation there. If $\ell\ge 1$, convergence in $C^{\ell+1}$ also gives uniform convergence of the order-two right-hand sides in $C^{\ell-1}$ down to zero. Integrating the equations in that space proves the claimed C^1 trace and the equation at zero. For $\ell=0$ we claim only the C^1 spatial initial trace and the positive-time equation, not a classical curvature equation at zero. □

The injectivity hypothesis is separate from curvature–torsion control. For metric evolutions $\partial_t g = -2\mathrm{Ric}+E$, Chen proves relative noncollapse on scales $r<\rho$, under control of a weighted time integral of $\sup_M|E|^2$ [Che18, Definition 4.1, Theorem 4.2 and (62)]. Here $E = 2Q_\Sigma(T)$ and $|E|\le C|T|^2$; for MHR this is $E = 2a(T\star T)+2b|T|^2 g$. Uniform application along a blow-up sequence requires uniform control of that additional quantity. The collapsing example in Corollary 5.23 shows why an injectivity conclusion cannot be inferred from the Type I bound alone for the entire MHR family.

At record maxima of Λ_{H}, parabolic rescaling gives a uniform bound on an expanding backward time interval. The next result turns these rescalings into an ancient solution when the base points do not collapse.

Theorem 4.10 (Ancient quadratic-flow blow-up under injectivity)**.** *Let a solution of one fixed equation (2.3) on a closed manifold have finite maximal time T_0. There are record times $t_i\uparrow T_0$ and $p_i\in M$ such that*

$$Q_i := \Lambda_{\mathrm{H}}(p_i,t_i) = \max_{0\le t\le t_i}\Lambda_{\mathrm{H}}(t) \longrightarrow \infty. \tag{4.32}$$

For each defining tensor component of type (p,q), with p contravariant and q covariant indices, set

$$g_i(s) = Q_i g(t_i + s/Q_i), \qquad \xi_i^{(p,q)}(s) = Q_i^{(q-p)/2}\xi^{(p,q)}(t_i + s/Q_i). \tag{4.33}$$

If $\mathrm{inj}_{g_i(0)}(p_i) \geq i_0 > 0$, then a subsequence converges pointed smoothly to a complete nontrivial ancient solution of the same fixed quadratic equation, defined on $(-\infty, \varepsilon)$ for some $\varepsilon > 0$, normalised by $\Lambda_{\mathrm{H},\infty}(p_\infty, 0) = 1$.

Proof. By Proposition 4.6, the function $\sup_M \Lambda_{\mathrm{H}}(t)$ is continuous and tends to infinity. Choose increasing levels $Q_i \to \infty$ above $\sup_M \Lambda_{\mathrm{H}}(0)$, let t_i be their first hitting times, and choose p_i attaining the spatial maximum. Then $t_i \uparrow T_0$ and (4.32) holds. The rescaling (4.33) induces g_i and preserves the fixed quadratic equation by the scaling calculation in Corollary 4.3. Since Λ_{H} is scale-covariant,

$$\Lambda_{\mathrm{H},i}(x,s) = Q_i^{-1}\Lambda_{\mathrm{H}}(x, t_i + s/Q_i), \qquad \Lambda_{\mathrm{H},i} \leq 1 \ (s \leq 0), \qquad \Lambda_{\mathrm{H},i}(p_i, 0) = 1. \tag{4.34}$$

The backward domains exhaust $(-\infty, 0]$ because $Q_i \to \infty$ and $t_i \to T_0 > 0$, so $Q_i t_i \to \infty$. Let C be the constant of Proposition 4.6. Its singularity lower bound gives $Q_i(T_0 - t_i) \geq C^{-1}$, so every rescaled solution exists on $[0, C^{-1})$. Its doubling estimate, starting from $\sup_M \Lambda_{\mathrm{H},i}(0) = 1$, gives $\Lambda_{\mathrm{H},i} \leq 2$ on $[0, (2C)^{-1}]$. Fix $0 < \varepsilon < (2C)^{-1}$. Apply Theorem 4.7 on $(-A, \varepsilon)$ and diagonalise as $A \to \infty$. Completeness follows from that compactness theorem, while the last equality in (4.34) passes to the limit and rules out the flat torsion-free solution. □

One sufficient condition for the injectivity hypothesis is a weighted bound for the metric error. The scale restriction in Chen's theorem is strict; the following choice leaves room between the test radius and the comparison scale. The weighted torsion hypothesis below also occurs in [Che18, Theorems 5.3–5.4, (75) and (89)] and [Dut26, Theorems 7.6–7.7, (7.29) and (7.67)]. Those results use an additional growth condition to obtain torsion-free time-slice limits; here the fixed-quadratic compactness theorem yields an ancient solution of the same equation.

Proposition 4.11 (Relative noncollapse at parabolic scales)**.** *Let $g(t)$, $0 \leq t < T_0 < \infty$, be a smooth metric evolution on a closed n-manifold satisfying $\partial_t g = -2\mathrm{Ric} + E$. If*

$$J_E := \int_0^{T_0} (T_0 - t) \sup_M |E(t)|^2_{g(t)}\, dt < \infty, \tag{4.35}$$

then for every $L > 0$ there is $\kappa_L > 0$, depending only on $n, g(0), T_0, L, J_E$, with the following property. For $T_0/2 \leq t < T_0$ and $0 < r \leq L\sqrt{T_0 - t}$,

$$\sup_{B_{g(t)}(p,r)} R(t) \leq r^{-2} \quad \Longrightarrow \quad \mathrm{Vol}_{g(t)}\, B_{g(t)}(p,r) \geq \kappa_L r^n.$$

Proof. Put $\rho = (L+1)\sqrt{T_0 - t}$ and $\rho_0 = (L+1)\sqrt{T_0}$. Then $r < \rho \leq \rho_0$, and for $0 \leq u \leq t$,

$$t + \rho^2 - u \leq (T_0 - u) + (L+1)^2(T_0 - t) \leq \big(1 + (L+1)^2\big)(T_0 - u).$$

Consequently Chen's error integral satisfies

$$\int_0^t (t + \rho^2 - u) \sup_M |E(u)|^2\, du \leq \big(1 + (L+1)^2\big) J_E. \tag{4.36}$$

Apply [Che18, Definition 4.1 and Theorem 4.2, (62)] on the closed time interval $[0, t]$. This avoids assuming smooth extension to T_0. The constant is uniform as $t \uparrow T_0$: in the proof's initial entropy bound [Che18, (67)], the entropy parameter lies in the fixed compact positive interval $[T_0/4, T_0 + \rho_0^2]$. Its lower bound depends only on $g(0), T_0, \rho_0$; (4.36) controls the entire error subtraction. The remaining cut-off and volume comparison in that proof therefore give one positive κ_L for all the indicated t and r. □

Corollary 4.12 (Ancient limits under weighted metric-error control)**.** *Fix a tensorial model whose full $\mathrm{GL}(n,\mathbb{R})$ stabiliser is a closed subgroup $\mathrm{H} \subset \mathrm{SO}(n)$. Let a solution of one fixed equation (2.3) on a closed manifold have finite maximal time T_0. If (4.35) holds with $E = 2Q_\Sigma(T)$, then the record-point rescalings of Theorem 4.10 have uniformly positive base-point injectivity radii and admit a complete nontrivial ancient limit of the same equation. In particular, it suffices that*

$$\int_0^{T_0} (T_0 - t) \sup_M |T(t)|^4\, dt < \infty. \tag{4.37}$$

Proof. At the record times, $|\mathrm{Rm}(g(t_i))| \le Q_i$. Choose a dimensional constant c_n with $|R| \le c_n|\mathrm{Rm}|$, and put

$$a_0 = \min\{1/4, (1+c_n)^{-1/2}\}, \qquad r_i = a_0 Q_i^{-1/2}.$$

Then $R(g(t_i)) \le r_i^{-2}$ on M. The singularity lower bound in Proposition 4.6 gives $Q_i(T_0 - t_i) \ge C^{-1}$, hence $r_i \le a_0\sqrt{C}\sqrt{T_0 - t_i}$. Apply Proposition 4.11 with the fixed number $L = a_0\sqrt{C}$; after discarding finitely many times below $T_0/2$,

$$\mathrm{Vol}_{g(t_i)} B_{g(t_i)}(p_i, r_i) \ge \kappa_L r_i^n.$$

For $\widehat{g}_i = Q_i g(t_i)$ this becomes

$$|\mathrm{Rm}(\widehat{g}_i)| \le 1, \qquad \mathrm{Vol}_{\widehat{g}_i} B_{\widehat{g}_i}(p_i, a_0) \ge \kappa_L a_0^n.$$

The metrics are complete. The curvature and ball-volume bounds give $\mathrm{inj}_{\widehat{g}_i}(p_i) \ge i_* > 0$ by [CGT82, Theorem 4.7, pp. 47–48]. Now apply Theorem 4.10. Finally, $|2Q_\Sigma(T)| \le C_Q|T|^2$ for the fixed quadratic model, so (4.37) implies (4.35). □

For selected time slices, Duthie proves that under (4.37) and $\sup_M(|R| + |T|^2) = o((T_0 - t)^{-1})$ suitable rescaled time slices of a reasonable $\mathrm{Spin}(7)$-flow converge to a complete torsion-free $\mathrm{Spin}(7)$-structure with maximal volume growth [Dut26, Theorems 7.6–7.7]; by Proposition 2.4 this applies to every fixed $\mathrm{Spin}(7)$ MHR equation. Chen proves the corresponding statement for reasonable G_2-flows [Che18, Theorems 5.3–5.4]; whether G_2 MHR satisfies Chen's structural conditions is not verified here. For general tensorial H the analogous conclusion is not asserted here; transferring these arguments would require checking that the rescaled lower-order terms vanish and that the limiting reduction remains an H-structure, and justifying the Ricci-flat and volume-ratio steps that the sources derive from torsion-freeness.

For an immortal flow with uniformly bounded Λ_{H}, time translation gives eternal limits by the same argument, without rescaling.

Corollary 4.13 (Eternal limits of immortal quadratic flows)**.** *Let a solution of one fixed equation* (2.3) *on a closed manifold be immortal and suppose*

$$\sup_{M\times[0,\infty)} \Lambda_{\mathrm{H}} < \infty, \qquad \mathrm{inj}_{g(t_i)}(p_i) \ge i_0 > 0 \quad \textit{along a sequence } t_i \longrightarrow \infty.$$

A subsequence of $(M, \xi(t_i + s), p_i)$ converges smoothly on compact spacetime subsets to a complete eternal solution of the same fixed quadratic equation.

Proof. The translated flows are defined on $(-t_i, \infty)$, with the same uniform curvature–torsion bound. Apply Theorem 4.7 on each $(-A, B)$ and extract diagonally as $A, B \to \infty$. The comparison maps are chosen at $s = 0$ and kept independent of time, so the nested limits define one eternal solution. □

4.3 Continuous endpoints and an integral criterion

The blow-up argument follows the Euclidean-volume-rigidity method of Šešum [Šeš05, §3], whose Ricci-flow theorem assumes bounded Ricci curvature. For Ricci flow, the corresponding time-integral obstruction is due to Wang [Wan12, Corollary 4.1, (58)]. The bounded-torsion versions for Laplacian flow and reasonable $\mathrm{Spin}(7)$-flows are [LW17, Theorem 8.1] and [Dut26, Theorem 7.1]; the argument below uses $\int_0^{T_0} \sup_M |T|^2\, dt < \infty$ in their place.

A positive continuous metric looks Euclidean at sufficiently small scales. The following lemma makes this observation uniform for a convergent sequence of metrics. It supplies both the injectivity bound needed for smooth blow-up and the rigidity of the limiting metric.

Lemma 4.14 (Blowing up a positive continuous metric endpoint)**.** *Let M be closed and let g_i be smooth metrics converging uniformly, with respect to a fixed smooth background metric, to a continuous positive definite metric g_*. If $Q_i \to \infty$ and $p_i \in M$, then $(M, Q_i g_i, p_i)$ converges in pointed Gromov–Hausdorff sense to $(\mathbb{R}^n, g_{\mathrm{Euc}}, 0)$. For every fixed $r > 0$,*

$$\mathrm{Vol}_{Q_i g_i} B_{Q_i g_i}(p_i, r) \longrightarrow \omega_n r^n.$$

If in addition $|\mathrm{Rm}(Q_i g_i)| \le C$, then $\inf_i \mathrm{inj}_{Q_i g_i}(p_i) > 0$, after discarding finitely many terms.

Proof. From any subsequence, extract one with $p_i \to p$ and choose a coordinate chart $x : U \to \mathbb{R}^n$ about p, with p_i in a relatively compact subchart. Let G_i be the matrix of g_* at p_i and set $A_i = G_i^{-1/2}$. Both A_i and A_i^{-1} are uniformly bounded. The maps

$$F_i(y) = x^{-1}\big(x(p_i) + Q_i^{-1/2} A_i y\big)$$

are defined on Euclidean balls of radii tending to infinity. For each fixed R, their pulled-back metrics are

$$F_i^*(Q_i g_i)(y) = A_i^{\mathsf{t}} g_i(F_i(y)) A_i \longrightarrow I \quad \text{uniformly on } B(0,R).$$

This follows from uniform convergence of g_i and uniform continuity of g_* in the chart.

The metrics g_i are uniformly equivalent to the fixed background metric. Consequently every curve starting at p_i and leaving a fixed smaller coordinate chart has $Q_i g_i$-length tending to infinity. On bounded scales no distance-minimising curve can leave that chart. More explicitly, to compare distances on $B(0,R)$ choose $L > 4R$: uniform metric comparison on $B(0,L)$ makes every curve exiting that ball longer than the straight-line competitors between points of $B(0,R)$. Thus outside shortcuts cannot change these distances. The preceding uniform comparison with the Euclidean metric therefore gives the pointed distance convergence. The same comparison squeezes each metric ball between Euclidean balls of radii tending to r; its volume density tends uniformly to one, proving the volume assertion. The argument applies to a further subsequence of every subsequence and gives the same pointed Euclidean limit and ball volumes. Therefore the asserted convergence holds for the full sequence.

The curvature bound and the resulting positive lower bound for the volume of one fixed sufficiently small ball give the injectivity conclusion by [CGT82, Theorem 4.7, pp. 47–48]. □

For a fixed quadratic flow, the remaining obstruction is torsion. If its squared norm is integrable in time, torsion vanishes in every smooth singularity blow-up. Combined with the Euclidean metric limit, this contradicts the normalisation $\Lambda_{\mathrm{H}} = 1$ at the base point.

Theorem 4.15 (Continuation from a continuous metric endpoint)**.** *Let $\xi(t)$ solve one fixed equation (2.3) on a closed manifold, on $[0,T_0)$ with $T_0 < \infty$. Suppose $g(t)$ converges uniformly to a continuous positive definite metric g_* and*

$$\int_0^{T_0} \sup_M |T(t)|^2 \, dt < \infty. \tag{4.38}$$

Then $\xi(t)$ extends smoothly through T_0. In particular, uniform boundedness of T suffices in place of (4.38).

Proof. Suppose that T_0 is a finite maximal time. Take record points and rescale as in (4.32)–(4.33). At $s = 0$ the curvature satisfies $|\mathrm{Rm}(g_i(0))| \le 1$. By Lemma 4.14, the rescaled base-point injectivity radii have a uniform positive lower bound and the metric spaces at $s = 0$ converge pointedly to Euclidean space. Theorem 4.10 gives smooth convergence on an interval $(-\infty, \varepsilon)$, for some $\varepsilon > 0$, to a complete solution of the same fixed quadratic equation, with

$$\Lambda_{\mathrm{H},\infty}(p_\infty, 0) = 1.$$

Smooth pointed convergence implies pointed Gromov–Hausdorff convergence at $s = 0$, so uniqueness of the pointed limit shows that $(M_\infty, g_\infty(0))$ is isometric to Euclidean space as a metric space, hence, by Myers–Steenrod, as a Riemannian manifold. In particular, $\mathrm{Rm}_\infty(0) = 0$. This flatness follows from the metric endpoint; no Ricci-flatness assumption on arbitrary torsion-free H-structures is used.

Write T_i for the rescaled torsion. For each $A > 0$, its scaling law gives

$$\int_{-A}^{0} \sup_M |T_i(s)|^2 \, ds = \int_{t_i - A/Q_i}^{t_i} \sup_M |T(t)|^2 \, dt \longrightarrow 0.$$

The final limit follows from (4.38), because both endpoints tend to T_0. On each compact subset of the limit manifold, smooth spacetime convergence makes the integral of the limiting nonnegative torsion norm vanish. Thus $T_\infty = 0$ for $s < 0$, and by continuity also at $s = 0$. Its spatial covariant derivative at $s = 0$ is zero as well. Hence $\Lambda_{\mathrm{H},\infty}(p_\infty, 0) = 0$, a contradiction. Therefore T_0 cannot be a finite maximal time. □

The metric equation gives a useful sufficient condition for a positive continuous endpoint: integrability of $|\mathrm{Ric}| + |T|^2$ controls the total change of the metric and prevents degeneration.

Corollary 4.16 (Integral continuation criterion for general H)**.** *For a solution of one fixed equation (2.3) on a closed manifold, every finite maximal time T_0 satisfies*

$$\int_0^{T_0} \sup_M \big(|\mathrm{Ric}| + |T|^2\big)\, dt = \infty.$$

Proof. Suppose the integral is finite and put $a_0(t) = \sup_M(|\mathrm{Ric}| + |T|^2)$. By the metric equation,

$$|\partial_t g|_g \le C(n, \mathrm{H}, Q) a_0(t),$$

since $|Q_\Sigma(T)| \le C|T|^2$ for the fixed natural quadratic term. Integrating the corresponding inequalities for $g(t)(v, v)$ gives uniform upper and positive lower bounds for $g(t)$ relative to $g(0)$. Consequently

$$\|\partial_t g(t)\|_{C^0(g(0))} \le C' a_0(t).$$

The integrability of a_0 makes $g(t)$ Cauchy in C^0 as $t \uparrow T_0$. Its limit is continuous and positive definite. The torsion integral in (4.38) is also finite, so Theorem 4.15 contradicts maximality. □

There is a second way to remove torsion from a Euclidean blow-up: retain the continuous endpoint of the defining tensor itself. The issue is that the coordinates in which this endpoint is continuous need not be smooth convergence coordinates. The following comparison uses only first derivatives of the transition maps.

The lemma separates these two kinds of charts. Uniform convergence is used in the endpoint charts; smooth convergence is used in the pointed charts. Their transition maps transfer the constant limiting tensor without requiring bounds for second derivatives of those maps.

Lemma 4.17 (Continuous tensors in a smooth Euclidean tangent limit)**.** *In the setting of Lemma 4.14, suppose also that the defining tensors ξ_i converge uniformly to a continuous tensor collection ξ_*. If a subsequence of $(M, Q_i g_i, \delta_{Q_i}\xi_i, p_i)$ converges smoothly and pointedly to a complete limit, then its metric is Euclidean and its defining tensors are parallel. Here each tensor component of net covariant degree w is multiplied by $Q_i^{w/2}$ in δ_{Q_i}.*

Proof. Pass to a subsequence with $p_i \to p$ and use the chart and maps F_i from the proof of Lemma 4.14. On each fixed Euclidean ball,

$$h_i = F_i^*(Q_i g_i) \longrightarrow I, \qquad \eta_i = F_i^*(\delta_{Q_i}\xi_i) \longrightarrow \eta \quad \text{uniformly},$$

where η is constant. Indeed, the derivative of F_i is $Q_i^{-1/2}A_i$ in the fixed chart, and the powers of Q_i cancel for every tensor component. The remaining factors are A_i and A_i^{-1}, which converge, and the components of ξ_i converge uniformly to those of ξ_* at p.

The metric limit is Euclidean by Lemma 4.14 and uniqueness of the pointed Gromov–Hausdorff limit. Identify it smoothly with $\mathbb{R}^n$ by Myers–Steenrod. Let Ψ_i be its smooth pointed comparison maps, so $\Psi_i(0) = p_i$ and, on each compact set,

$$k_i = \Psi_i^*(Q_i g_i) \longrightarrow I, \qquad \Psi_i^*(\delta_{Q_i}\xi_i) \longrightarrow \xi_\infty \quad \text{smoothly}.$$

Fix $R > 0$. For large i, F_i and Ψ_i are embeddings on neighbourhoods of $\overline{B_{8R}}$ and $\overline{B_{4R}}$, respectively. Choose $\varepsilon_i \downarrow 0$ such that both h_i on B_{8R} and k_i on B_{4R} lie between $(1-\varepsilon_i)^2 I$ and $(1+\varepsilon_i)^2 I$.

Every Ψ_i-image of a radial segment in B_{4R} has length at most $4R(1+\varepsilon_i)$, whereas a curve from p_i exiting $F_i(B_{8R})$ has first-exit length at least $8R(1-\varepsilon_i)$. Hence $\Psi_i(B_{4R}) \subset F_i(B_{8R})$, and $X_i = F_i^{-1} \circ \Psi_i$ is defined on B_{4R}. Put $c_i = (1+\varepsilon_i)/(1-\varepsilon_i)$. Radial length comparison gives $|X_i(x)| \le c_i|x|$. For $x, y \in B_R$, the Ψ_i-image of the segment $[x, y]$ then gives

$$|X_i(x) - X_i(y)| \le c_i|x - y|.$$

For the reverse inequality, the segment joining $X_i(x)$ to $X_i(y)$ lies in $B_{c_i R}$, and its F_i-image has length at most $2Rc_i(1+\varepsilon_i)$. Any curve from $\Psi_i(x)$ exiting $\Psi_i(B_{4R})$ has first-exit length at least $(1-\varepsilon_i)(4R-|x|) \ge 3R(1-\varepsilon_i)$: lift it up to the first exit and use the lower bound for k_i. Since $2c_i(1+\varepsilon_i) < 3(1-\varepsilon_i)$ for large i, the F_i-image segment stays in $\Psi_i(B_{4R})$. Its lift joins x to y, so

$$c_i^{-1}|x - y| \le |X_i(x) - X_i(y)| \le c_i|x - y|, \qquad x, y \in B_R. \tag{4.39}$$

The first-exit assertions apply because both maps are embeddings on neighbourhoods of the closed balls; the boundary of each image is the image of the boundary.

By Arzelà–Ascoli and (4.39), after extraction X_i converges uniformly on $\overline{B_R}$ to a distance-preserving map fixing zero. Polarisation of the distance identity shows that this map is $x \mapsto Ox$ for some $O \in \mathrm{O}(n)$. Moreover,

$$(DX_i)^{\mathsf{t}} h_i(X_i) DX_i = k_i.$$

Thus DX_i and $(DX_i)^{-1}$ are uniformly bounded, and $|DX_i|^2 \to n$ uniformly on B_R. Uniform convergence of X_i gives $DX_i \rightharpoonup O$ in $L^2(B_R)$; convergence of the squared norms then gives strong L^2 convergence. The inverses converge strongly as well, since inversion is Lipschitz on this uniformly bounded set of matrices with uniformly bounded inverses.

Tensor pullback depends on finitely many products of DX_i and $(DX_i)^{-1}$. Their uniform bounds, strong L^2 convergence and $\eta_i \to \eta$ uniformly therefore imply

$$\Psi_i^*(\delta_{Q_i}\xi_i) = X_i^*\eta_i \longrightarrow O^*\eta \quad \text{in } L^2(B_R).$$

The left-hand side also converges smoothly to ξ_∞. Hence ξ_∞ is constant on B_R. As R was arbitrary, it is parallel on the whole Euclidean limit. □

Theorem 4.18 (Continuation from a continuous structure endpoint)**.** *Fix a tensorial model whose full* $\mathrm{GL}(n,\mathbb{R})$ *stabiliser is a closed subgroup* $\mathrm{H} \subset \mathrm{SO}(n)$*. Let* $\xi(t)$ *solve one fixed equation* (2.3) *on a closed manifold for* $0 \le t < T_0 < \infty$*. Suppose that* $g(t)$ *and* $\xi(t)$ *converge uniformly to continuous limits* g_* *and* ξ_**, respectively, and that* g_* *is positive definite. Then the solution extends smoothly through* T_0*.*

Proof. If T_0 were a finite maximal time, take the record-point rescalings (4.32)–(4.33). The curvature bound at time zero and Lemma 4.14 give a uniform rescaled base-point injectivity bound. Thus Theorem 4.10 provides a complete smooth limit with $\Lambda_{\mathrm{H},\infty}(p_\infty, 0) = 1$. By Lemma 4.17, its metric at zero is flat and its defining tensor is parallel. Hence $T_\infty(0) = 0$ and $\nabla T_\infty(0) = 0$, contradicting that normalisation. □

For Dwivedi's particular G_2 equation, bounded full velocity gives continuation by [Dwi26, Theorem 5.9, (5.30)]. The preceding tensor comparison gives the following integral version for every fixed quadratic Ricci–harmonic equation. Both the metric and the vertical part of the velocity are controlled.

Corollary 4.19 (Integral full-velocity continuation)**.** *Fix a tensorial model whose full* $\mathrm{GL}(n,\mathbb{R})$ *stabiliser is a closed subgroup* $\mathrm{H} \subset \mathrm{SO}(n)$*. Let* $\xi(t)$ *solve one fixed equation* (2.3) *on a closed manifold for* $0 \le t < T_0 < \infty$*. Write uniquely* $\partial_t \xi = (h + \mathcal{A}) \diamond \xi$*, where* h *is symmetric and* $\mathcal{A} \in \mathfrak{m}$*. If*

$$\int_0^{T_0} \sup_M \big(|h| + |\mathcal{A}|\big)\, dt < \infty, \tag{4.40}$$

then the solution extends smoothly through T_0*. Equivalently, one may assume* $\int_0^{T_0} \sup_M |\partial_t \xi|_{g(t)}\, dt < \infty$*. In particular, bounded full velocity suffices.*

Proof. At the fixed tensor model, the diamond action on $\Sigma^2 \oplus \mathfrak{m}$ is injective, since the full $\mathrm{GL}(n)$ stabiliser is H. Finite-dimensional norm equivalence and adapted orthonormal frames therefore give constants depending only on the model such that

$$c\big(|h| + |\mathcal{A}|\big) \le |(h + \mathcal{A}) \diamond \xi|_g \le C\big(|h| + |\mathcal{A}|\big).$$

Since $\partial_t g = 2h$, (4.40) first gives uniform metric equivalence to $g(0)$ and a continuous positive definite limit g_*, by the same integration argument as in Corollary 4.16. Each component of ξ has constant pointwise norm in its induced metric. Metric equivalence and the upper diamond-action bound give

$$\|\partial_t \xi(t)\|_{C^0(g(0))} \le C' \sup_M \big(|h| + |\mathcal{A}|\big).$$

Thus $\xi(t)$ is Cauchy in C^0 and has a continuous limit ξ_*. Apply Theorem 4.18. □

5 MHR solitons, dissipation and explicit models

Self-similar solutions evolve only by diffeomorphisms and a uniform change of scale. Their equations separate into a metric equation and a torsion equation. The trace of the metric equation gives sign restrictions on closed solitons and, along the flow, controls volume and the total torsion energy. We first use these identities to identify stationary limits of immortal solutions, then construct explicit models. Write $\mu = a + nb$ for the parameter governing the trace and volume identities; the full soliton equation still depends separately on a and b.

5.1 The soliton system and its symmetries

We first separate the two soliton equations and explain how they reconstruct a self-similar flow. We then fix the diffeomorphism freedom to obtain an elliptic system. The final propositions describe the remaining freedoms and the seed geometries used in the examples.

Soliton equations. A diffeomorphism changes both the metric and the position of the defining tensor in its fixed-metric orbit. For a vector field Y, denote the second part by $\mathcal{K}_\xi(Y) \in \Lambda^2_{\mathfrak{m}}$, so that

$$\mathcal{L}_Y\xi = \big(\tfrac{1}{2}\mathcal{L}_Y g + \mathcal{K}_\xi(Y)\big) \diamond \xi. \tag{5.1}$$

Definition 5.1 (MHR soliton)**.** An MHR soliton with infinitesimal metric dilation rate ρ is a triple (ξ, Y, ρ) satisfying

$$-\mathrm{Ric} + a(T \star T) + b|T|^2 g = \rho g + \tfrac{1}{2}\mathcal{L}_Y g, \tag{5.2}$$

$$\operatorname{div} T = \mathcal{K}_\xi(Y). \tag{5.3}$$

With this convention, $\rho > 0$, $\rho = 0$, and $\rho < 0$ mean expanding, steady, and shrinking, respectively.

The self-similar reconstruction has metric scale $s(t) = 1 + 2\rho t$. If Y is complete, let F_t be generated by $Y/s(t)$, with $F_0 = \mathrm{Id}$. For a defining tensor of net degree $w = q - p$, put

$$\xi(t) = s(t)^{w/2} F_t^* \xi, \qquad g(t) = s(t) F_t^* g, \qquad s(t) > 0. \tag{5.4}$$

For a tensor tuple this formula is applied componentwise; a covariant k-tensor has $w = k$. Naturality and parabolic scaling give $\mathcal{F}_{a,b}(s^{w/2}\xi) = s^{w/2-1}\mathcal{F}_{a,b}(\xi)$. Differentiating (5.4), using $s' = 2\rho$, therefore gives the MHR equation exactly when (5.2)–(5.3) hold. Completeness of Y is automatic on a closed manifold; without it, the reconstruction is valid on the domain of its local flow.

The following identity makes the vertical part of a Lie derivative explicit. It is needed to check the full soliton equation: solving the metric equation alone does not determine the motion of the structure.

Lemma 5.2 (Lie derivatives of H-structures)**.** *Assume that the stabiliser of $\xi_\circ$ in $\mathfrak{gl}(n)$ is $\mathfrak{h}$. For every vector field Y,*

$$\mathcal{L}_Y\xi = \big(\tfrac{1}{2}\mathcal{L}_Y g + \tfrac{1}{2}dY^\flat + T_Y\big) \diamond \xi, \qquad \textit{so} \qquad \mathcal{K}_\xi(Y) = \pi_{\mathfrak{m}}\big(\tfrac{1}{2}dY^\flat\big) + T_Y. \tag{5.5}$$

Consequently (ξ, Y, ρ) is an MHR$_{a,b}$ soliton if and only if

$$\mathcal{S}_{a,b}(\xi, Y, \rho) := \mathcal{F}_{a,b}(\xi) - \rho\, g \diamond \xi - \mathcal{L}_Y\xi = 0. \tag{5.6}$$

Here $g \diamond \xi = (q-p)\xi$ on a tensor of type (p,q); the action is componentwise on a tuple.

Proof. $(\mathcal{L}_Y\xi)(v_1,\dots,v_k) = (\nabla_Y\xi)(v_1,\dots,v_k) + \sum_s \xi(\dots,\nabla_{v_s}Y,\dots)$. The first term is $T_Y \diamond \xi$. The second is the diamond action of the 2-tensor $(v,v') \mapsto g(\nabla_v Y, v')$, whose symmetric part is $\frac{1}{2}\mathcal{L}_Y g$ and whose skew part is $\frac{1}{2}dY^\flat$; its $\mathfrak{h}$-component acts trivially. Contravariant slots contribute the dual action, giving the same formula for mixed tensors. The kernel of $(S,C) \mapsto (S+C)\diamond\xi$ on $\Sigma^2 \oplus \Lambda^2_{\mathfrak{m}}$ lies in the $\mathfrak{gl}$-stabiliser $\mathfrak{h}$, which meets $\Sigma^2 \oplus \Lambda^2_{\mathfrak{m}}$ only in 0; comparison with (5.1) gives $\mathcal{K}_\xi$. If (5.2)–(5.3) hold, then $\mathcal{F}_{a,b}(\xi) = (\rho g + \frac{1}{2}\mathcal{L}_Y g + \mathcal{K}_\xi(Y))\diamond\xi = \rho g\diamond\xi + \mathcal{L}_Y\xi$; conversely the Σ^2 and $\Lambda^2_{\mathfrak{m}}$ components of (5.6) are the two soliton equations, by the same injectivity. For $Y = \nabla f$ this recovers $\mathcal{K}_\xi(\nabla f) = T_{\nabla f}$ (Proposition 5.9). □

As for the evolution equation, diffeomorphisms account for the degeneracy of the soliton operator. The next theorem identifies that degeneracy at the symbol level and removes it by the standard DeTurck gauge; the background vector field is fixed when taking the symbol.

Theorem 5.3 (The DeTurck-gauged soliton system)**.** *Assume the full* $\mathrm{GL}(n)$*-stabiliser is* $\mathrm{H} \subset \mathrm{SO}(n)$*. Let* M *be a manifold, fix a background metric* $\bar{g}$*, a vector field* $\bar{Y}$ *and* $\rho \in \mathbb{R}$*, and let* $W(g,\bar{g})^k = g^{ij}\big(\Gamma(g)^k_{ij} - \Gamma(\bar{g})^k_{ij}\big)$ *be the standard DeTurck field (the field of* (3.2)*; it equals* $2X_{\mathrm{DT}}$ *in the notation of* [FFL+26, Proposition 4.1(1)]*, cf. Section 2). Put*

$$\mathcal{N}(\xi) := \mathcal{S}_{a,b}\big(\xi, \bar{Y} - W(g_\xi,\bar{g}), \rho\big) = \mathcal{P}_{\bar{g}}(\xi) - \rho\, g_\xi \diamond \xi - \mathcal{L}_{\bar{Y}}\xi, \tag{5.7}$$

where $\mathcal{P}_{\bar{g}}(\xi) = \mathcal{F}_{a,b}(\xi) + \mathcal{L}_W\xi$*, with* $W = 2X_{\mathrm{DT}}$ *and sign* $+$*, is the gauged operator of* [FFL+26, Proposition 4.1(1) and the proof of Theorem 4.2] *(by* (5.5)*,* $\mathcal{L}_{2X_{\mathrm{DT}}}\xi = (\mathcal{L}_{X_{\mathrm{DT}}}g + dX^\flat_{\mathrm{DT}} + T_{2X_{\mathrm{DT}}}) \diamond \xi$*).*

(i) $\mathcal{N}(\xi) = 0$ *if and only if* $(\xi, \bar{Y} - W(g_\xi,\bar{g}), \rho)$ *is an MHR*$_{a,b}$ *soliton. If* $(\bar{\xi}, \bar{Y}, \rho)$ *is a soliton and* $\bar{g} = g_{\bar{\xi}}$*, then* $\mathcal{N}(\bar{\xi}) = 0$*.*

(ii) In the orbit chart $(S,C) \mapsto \xi \cdot \exp(S+C)$ *the linearisation of* $\mathcal{N}$ *has principal symbol* $\sigma(D\mathcal{N})(x,\chi) = |\chi|^2\,\mathrm{Id}$ *on* $\Sigma^2 \oplus \Lambda^2_{\mathfrak{m}}$*, for every* (a,b) *and every closed tensorial* H*; thus* $\mathcal{N}$ *is a strongly elliptic second-order system.*

(iii) Without the gauge, with the convention $\partial_i\partial_j \mapsto \chi_i\chi_j$*,* $\chi \odot v = \frac{1}{2}(\chi\otimes v + v\otimes\chi)$ *and* $(\chi\wedge v)_{ab} = \chi_a v_b - \chi_b v_a$*,*

$$\begin{aligned} \sigma(D_\xi \mathcal{S}_{a,b})(\chi)(S,C) &= \big(A_\chi(S), B_\chi(S,C)\big) = |\chi|^2(\mathrm{Id} - \Pi_\chi)(S,C), \\ A_\chi(S) &= |\chi|^2 S - \chi\otimes S\chi - S\chi\otimes\chi + (\operatorname{tr} S)\chi\otimes\chi, \\ B_\chi(S,C) &= |\chi|^2 C - \pi_{\mathfrak{m}}(\chi\wedge S\chi). \end{aligned} \tag{5.8}$$

where $\Pi_\chi(S,C) = \big(\chi\odot v, \frac{1}{2}\pi_{\mathfrak{m}}(\chi\wedge v)\big)$ *with* $v = |\chi|^{-2}\big(2S\chi - (\operatorname{tr} S)\chi\big)$*.* Π_χ *is idempotent; its image, of dimension* n*, is the set of symbols of infinitesimal diffeomorphisms* $\xi \mapsto \mathcal{L}_v\xi$*, and its kernel is the linearised gauge condition* $S\chi = \frac{1}{2}(\operatorname{tr} S)\chi$*. Hence the ungauged soliton operator is elliptic exactly modulo diffeomorphisms: its symbol vanishes on the diffeomorphism directions and is* $|\chi|^2$ *on the gauge complement. The symbol is independent of* (a,b)*,* ρ *and* $\bar{Y}$*.*

Proof. (i) $\mathcal{L}_Y\xi$ is linear in Y, so $\mathcal{S}(\xi, \bar{Y} - W, \rho) = \mathcal{S}(\xi,\bar{Y},\rho) + \mathcal{L}_W\xi$; apply Lemma 5.2. $W(\bar{g},\bar{g}) = 0$. (ii) $\mathcal{L}_{\bar{Y}}$ is of first order and $g \diamond \xi$ of order zero in ξ, so $\sigma(D\mathcal{N}) = \sigma(D\mathcal{P}_{\bar{g}})$, which is the scalar principal symbol [FFL+26, Theorem 4.2]. (iii) With $\delta g = 2S$, $DW(2S)_k = 2(\nabla^i S_{ik} - \frac{1}{2}\nabla_k \operatorname{tr} S)$ modulo order zero, and by (5.5) the second-order part of $\mathcal{L}_{DW}\xi$ is $(\nabla DW)^{\mathrm{sym}} + \pi_{\mathfrak{m}}(\nabla DW)^{\mathrm{skew}}$; its symbol is $|\chi|^2\Pi_\chi$ (here $|\chi|^2 v = 2S\chi - (\operatorname{tr} S)\chi$, with v as in (iii)). The term $\mathcal{L}_W(\delta\xi)$ is of first order. Hence $\sigma(D\mathcal{F}) = \sigma(D\mathcal{P}) - |\chi|^2\Pi_\chi$, and the torsion-quadratic terms and $\rho g\diamond\xi + \mathcal{L}_{\bar{Y}}\xi$ are of order at most one. If $S = \chi\odot v$ then $2S\chi - (\operatorname{tr} S)\chi = |\chi|^2 v$, so $\Pi_\chi^2 = \Pi_\chi$ and the image is $\{(\chi\odot v, \frac{1}{2}\pi_{\mathfrak{m}}(\chi\wedge v))\}$, the symbol of $v \mapsto \mathcal{L}_v\xi$ by (5.5); it has dimension n since $\chi \odot v \neq 0$ for $v \neq 0$. □

Proposition 5.4 (Freedoms of the soliton equations)**.** *Let* (ξ, Y, ρ) *be an MHR*$_{a,b}$ *soliton, under the hypothesis of Lemma 5.2.*

(a) (Diffeomorphisms) $(\phi^*\xi, \phi^*Y, \rho)$ *is a soliton for every diffeomorphism* ϕ*.*

(b) (Automorphisms) $(\xi, Y + Z, \rho)$ *is a soliton for every* Z *with* $\mathcal{L}_Z\xi = 0$*. Conversely, if* (ξ, Y', ρ') *is also a soliton, then* $\mathcal{L}_{Y'-Y}\xi = (\rho - \rho')\, g\diamond\xi$*. If* M *is closed,* $\rho' = \rho$*, so* $Y' - Y$ *is an infinitesimal automorphism, hence Killing. On flat* $\mathbb{R}^n$ *the torsion-free model gives the gradient solitons* $(\xi_{\mathrm{o}}, -\rho\, x^i\partial_i, \rho)$ *for every* ρ*, so the rate is not unique in general.*

(c) (Scaling) $(\lambda^{w/2}\xi, \lambda^{-1}Y, \lambda^{-1}\rho)$ *is a soliton for every* $\lambda > 0$ *(*$w = q - p$ *for a component of type* (p,q)*); the type is unchanged.*

(d) (Infinitesimal version) $D\mathcal{S}_{a,b}$ *at* (ξ, Y, ρ) *annihilates* $(\mathcal{L}_V\xi, [V,Y], 0)$ *for every vector field* V*,* $(0, Z, 0)$ *for* Z *as in (b), and* $(\frac{1}{2}g\diamond\xi, -Y, -\rho)$*.*

The symbol calculation does not by itself provide a local slice for solitons; the gauge in Theorem 5.3 is part of that elliptic system.

Proof. (a) Naturality: $\mathcal{F}(\phi^*\xi) = \phi^*\mathcal{F}(\xi)$, $g_{\phi^*\xi} = \phi^* g$ and $\mathcal{L}_{\phi^*Y}\phi^*\xi = \phi^*\mathcal{L}_Y\xi$. (b) $\mathcal{S}$ depends on Y only through $\mathcal{L}_Y\xi$; subtract the two equations (5.6). On a closed manifold the trace identity (5.9) integrates to $n\rho\,\mathrm{Vol} = \int(-R + \mu|T|^2)$, which does not involve Y. An automorphism preserves g_ξ. On $\mathbb{R}^n$, $\mathcal{F}(\xi_\circ) = 0$ and $\mathcal{L}_{x^i\partial_i}\xi_\circ = g \diamond \xi_\circ$. (c) $\mathcal{F}(\lambda^{w/2}\xi) = \lambda^{w/2-1}\mathcal{F}(\xi)$ (as in (5.4)), $g_\lambda \diamond \lambda^{w/2}\xi = \lambda^{w/2} g \diamond \xi$ and $\mathcal{L}_{\lambda^{-1}Y}\lambda^{w/2}\xi = \lambda^{w/2-1}\mathcal{L}_Y\xi$. (d) Differentiate the identities $\mathcal{S}(\phi_s^*\xi, \phi_s^*Y, \rho) = \phi_s^*\mathcal{S}(\xi, Y, \rho) = 0$, $\mathcal{S}(\xi, Y + sZ, \rho) = 0$ and $\mathcal{S}(\lambda^{w/2}\xi, \lambda^{-1}Y, \lambda^{-1}\rho) = 0$; note $\frac{d}{d\lambda}\lambda^{w/2}\xi|_1 = \frac{w}{2}\xi = \frac{1}{2}g \diamond \xi$. □

The seed cases help distinguish the metric and structure conditions. In particular, torsion-freeness does not imply Ricci-flatness for an arbitrary structure group, so the groups are specified in each case.

Proposition 5.5 (Torsion-free, Einstein and Ricci-soliton seeds)**.** *(i) Let ξ be torsion-free. For every (a, b), (ξ, Y, ρ) is an MHR$_{a,b}$ soliton if and only if $-\mathrm{Ric} = \rho g + \frac{1}{2}\mathcal{L}_Y g$ and $\pi_{\mathfrak{m}}(dY^\flat) = 0$. In particular every gradient Ricci soliton metric carrying a torsion-free* H*-structure gives an MHR soliton for every (a, b), and ξ is stationary if and only if* $\mathrm{Ric} = 0$.

(ii) If $\mathrm{H} \subseteq \mathrm{SU}(m)$, $\mathrm{Sp}(k)$, G_2 *or* $\mathrm{Spin}(7)$ *in its standard representation, torsion-free implies* $\mathrm{Ric} = 0$ *(*[Bon66, Propositions 7 and 8]*for* G_2 *and* $\mathrm{Spin}(7)$*; for* $\mathrm{SU}(m) \supseteq \mathrm{Sp}(k)$ *the Ricci form is the contraction of the curvature with J, which vanishes when the curvature takes values in* $\mathfrak{su}(m)$*). For other groups, e.g.* $\mathrm{U}(m)$ *or* $\mathrm{Sp}(k)\mathrm{Sp}(1)$*, it does not: a torsion-free* $\mathrm{U}(m)$*-structure whose metric is Kähler–Einstein with* $\mathrm{Ric} = \lambda g$*, $\lambda \neq 0$, is a Ricci-soliton seed $(\xi, 0, -\lambda)$ that is not stationary.*

(iii) If M is closed and ξ is torsion-free with $\mathrm{Ric} = 0$*, every soliton has $\rho = 0$ and Y parallel; in particular $\mathcal{L}_Y\xi = 0$. On flat $\mathbb{R}^n$ the Gaussian solitons of Proposition 5.4(b) are complete non-stationary torsion-free seeds.*

(iv) (Inherited at $a = 0$.) If $|T|^2$ is constant, $-\mathrm{Ric} = \lambda g + \frac{1}{2}\mathcal{L}_Y g$ and $\operatorname{div} T = \mathcal{K}_\xi(Y)$*, then $(\xi, Y, \lambda + b|T|^2)$ is an MHR$_{0,b}$ soliton for every b. This contains the inherited co-closed nilsoliton rows of Remark 5.34.*

(v) (Einstein seeds with torsion.) If $\mathrm{Ric} = \lambda g$*, then $(\xi, 0, \rho)$ is a soliton if and only if* $\operatorname{div} T = 0$ *and $aT \star T + b|T|^2 g = (\lambda + \rho)g$. Nearly parallel* G_2*-structures (Proposition 5.15, $\rho = \frac{2}{3}(\mu - 9)c^2$) and nearly Kähler* $\mathrm{SU}(3)$*-structures (Corollary 5.40, velocity $\frac{\mu-5}{6}|T|^2 g$) satisfy both equations for every (a, b); they are stationary exactly at $\mu = 9$, respectively $\mu = 5$. No* $\mathrm{Spin}(7)$ *Einstein seed with torsion is claimed.*

Proof. (i) $T = 0$ kills $T \star T$, $|T|^2$ and $\operatorname{div} T$, and $\mathcal{K}_\xi(Y) = \pi_{\mathfrak{m}}(\frac{1}{2}dY^\flat)$ by (5.5). The velocity is $-\mathrm{Ric} \diamond \xi$, injective in Ric. (ii) Sources as stated; the $\mathrm{SU}(m)$ and $\mathrm{Sp}(k)$ case is the one-line Kähler argument given. (iii) Integrating the trace $\operatorname{div} Y = -n\rho$ gives $\rho = 0$; then $\mathcal{L}_Y g = 0$, and for a Killing field on a closed Ricci-flat manifold $\int |\nabla Y|^2 = \int \mathrm{Ric}(Y, Y) = 0$. So $\nabla Y = 0$, $dY^\flat = 0$ and $\mathcal{L}_Y\xi = \nabla_Y\xi + (\nabla Y) \diamond \xi = 0$. (iv), (v) Substitute in (5.2)–(5.3). □

5.2 Trace, volume and gradient identities

The dilation rate of a closed soliton is determined by its average scalar curvature and torsion energy. A definite sign for the resulting quadratic torsion expression therefore rules out one direction of self-similar scaling.

Proposition 5.6 (Trace identity and compact sign rigidity)**.** *Every MHR soliton satisfies*

$$-R + \mu|T|^2 = n\rho + \operatorname{div} Y, \qquad \mu = a + nb. \tag{5.9}$$

Suppose M is closed and a scalar-torsion identity gives

$$-R + \mu|T|^2 = \operatorname{div}\Theta + q_\mu(T). \tag{5.10}$$

If $q_\mu \leq 0$, there are no expanding solitons on M; if $q_\mu \geq 0$, there are no shrinking solitons. If either inequality is strictly coercive, in the sense that $q_\mu \leq -c|T|^2$ or $q_\mu \geq c|T|^2$ for a constant $c > 0$, then every steady soliton is torsion-free.

Proof. Trace (5.2) and use $\mathrm{tr}(T \star T) = |T|^2$ to obtain (5.9). Integrating it together with (5.10) gives

$$n\rho\,\mathrm{Vol}(M) = \int_M q_\mu(T)\, dV.$$

The sign conclusions follow. □

For G_2, write $\psi = *\varphi$. The torsion forms are defined by

$$\begin{aligned} d\varphi &= \tau_0\psi + 3\tau_1 \wedge \varphi + *\tau_3, \\ d\psi &= 4\tau_1 \wedge \psi + \tau_2 \wedge \varphi, \end{aligned} \tag{5.11}$$

where τ_0 is a function, τ_1 is a one-form, $\tau_2 \in \Omega^2_{14}$ and $\tau_3 \in \Omega^3_{27}$. The subscripts denote the dimensions of the irreducible G_2 modules. This is the decomposition of [Bry06, Proposition 1, (3.7)], with the Hodge star taken in our declared orientation: relative to Bryant's convention, τ_0, τ_2, τ_3 change sign and τ_1 does not. The scalar and norm identities below are unaffected. For $\mathrm{Spin}(7)$, the two torsion components are determined by

$$d\Phi = T^1_8 \wedge \Phi + T^5_{48}, \qquad (*T^5_{48}) \wedge \Phi = 0,$$

where T^1_8 is a one-form and T^5_{48} is a five-form in the 48-dimensional irreducible summand. For $\mathrm{Spin}(7)$ these are the normalisations of [Fow20, (2.5)]. All torsion-form norms below are exterior-form norms, and δ denotes the codifferential.

With the intrinsic-torsion norm fixed in (2.2), the scalar curvature and torsion norm are [FFL+26, Section 3.3.2, pp. 28–29]

$$\begin{aligned} R_{\mathrm{G}_2} &= 12\delta\tau_1 + \tfrac{21}{8}\tau_0^2 + 30|\tau_1|^2 - \tfrac{1}{2}|\tau_2|^2 - \tfrac{1}{2}|\tau_3|^2, \\ |T|^2_{\mathrm{G}_2} &= \tfrac{7}{24}\tau_0^2 + 4|\tau_1|^2 + \tfrac{1}{3}(|\tau_2|^2 + |\tau_3|^2), \end{aligned} \tag{5.12}$$

and

$$\begin{aligned} R_{\mathrm{Spin}(7)} &= \tfrac{7}{2}\delta T^1_8 + \tfrac{21}{8}|T^1_8|^2 - \tfrac{1}{2}|T^5_{48}|^2, \\ |T|^2_{\mathrm{Spin}(7)} &= \tfrac{7}{32}|T^1_8|^2 + \tfrac{1}{4}|T^5_{48}|^2. \end{aligned} \tag{5.13}$$

Consequently the quadratic terms in (5.10) are

$$q^{\mathrm{G}_2}_\mu(T) = \tfrac{7(\mu-9)}{24}\tau_0^2 + (4\mu - 30)|\tau_1|^2 + \tfrac{2\mu+3}{6}(|\tau_2|^2 + |\tau_3|^2), \tag{5.14}$$

$$q^{\mathrm{Spin}(7)}_\mu(T) = \tfrac{7(\mu-12)}{32}|T^1_8|^2 + \tfrac{\mu+2}{4}|T^5_{48}|^2, \tag{5.15}$$

with divergence terms $-12\delta\tau_1$ and $-\frac{7}{2}\delta T^1_8$, respectively.

Corollary 5.7 (Compact MHR sign rigidity)**.** *On a closed manifold, the following exclusions hold:*

H	*no expanders*	*no shrinkers*
G_2	$\mu \le -\frac{3}{2}$	$\mu \ge 9$
$\mathrm{Spin}(7)$	$\mu \le -2$	$\mu \ge 12$.

Every steady soliton is torsion-free when either displayed inequality for its group is strict. In particular this applies globally to the G_2 *bridge parameters* (2.15)*, for which* $\mu = -6$*, without a normal-torsion assumption.*

Proof. For G_2, the coefficients in (5.14) are all nonpositive when $\mu \le -3/2$ and all nonnegative when $\mu \ge 9$. For $\mathrm{Spin}(7)$, (5.15) gives the corresponding thresholds -2 and 12. Away from the endpoints the quadratic form is definite. The conclusions follow from Proposition 5.6. □

At the critical parameters where the quadratic form has a kernel, the steady trace equation forces $\tau_0 = \tau_1 = 0$ for G_2 at $\mu = -3/2$, $T^1_8 = 0$ for $\mathrm{Spin}(7)$ at $\mu = -2$, and $T^5_{48} = 0$ for $\mathrm{Spin}(7)$ at $\mu = 12$. These restrictions alone do not solve the remaining soliton equations. The circle products in Corollary 5.18 realise the last case with nonzero torsion. At the G_2 endpoint $\mu = 9$, the exterior torsion identities give the classification in Proposition 5.16.

Volume dissipation. In the parameter ranges where q_μ is negative definite, volume is a nonincreasing quantity whose loss controls the total torsion energy. For an immortal solution satisfying the compactness hypotheses below, finite total torsion energy will force every translated limit to be torsion-free.

Proposition 5.8 (Volume dissipation for MHR)**.** *Along an MHR*$_{a,b}$ *solution on a closed manifold,*

$$\frac{d}{dt}\,\mathrm{Vol}(g(t)) = \int_M q_\mu(T)\, dV_{g(t)} \tag{5.16}$$

for $\mathrm{H} = \mathrm{G}_2$ *or* $\mathrm{Spin}(7)$*, with* q_μ *as in* (5.14)–(5.15)*. If*

$$c_* = \begin{cases} -\mu - \frac{3}{2}, & \mathrm{H} = \mathrm{G}_2, \quad \mu < -\frac{3}{2}, \\ -\mu - 2, & \mathrm{H} = \mathrm{Spin}(7), \quad \mu < -2, \end{cases}$$

then $c_* > 0$ *and*

$$c_* \int_0^\tau \int_M |T|^2 \, dV_{g(t)} dt \le \mathrm{Vol}(g(0)) - \mathrm{Vol}(g(\tau)) \le \mathrm{Vol}(g(0)) \tag{5.17}$$

for every time τ *in the interval of existence.*

Proof. The metric equation gives $\partial_t dV = (-R + \mu|T|^2) dV$. The divergence terms in the displayed scalar-torsion formulae integrate to zero, which proves (5.16). For G_2, the ratios of the three coefficients of q_μ to those of $|T|^2$ are $\mu - 9$, $\mu - 15/2$, and $\mu + 3/2$. For $\mathrm{Spin}(7)$ they are $\mu - 12$ and $\mu + 2$. Thus $q_\mu \le -c_*|T|^2$ in the stated ranges. Integration in time proves (5.17). □

Gradient identities. For gradient solitons, the skew equation becomes a weighted divergence identity. The metric equation gives a companion identity for scalar curvature and the potential, with explicit torsion terms. Both hold for an arbitrary structure group H.

Proposition 5.9 (Gradient MHR identities)**.** *Let* $(\xi, \nabla f, \rho)$ *be an MHR*$_{a,b}$ *soliton, put* $S = T \star T$*,* $u = |T|^2$*, and* $\mu = a + nb$*. Then*

$$\operatorname{div} T = T_{\nabla f}, \qquad \operatorname{div}(e^{-f} T) = 0, \tag{5.18}$$

and

$$d\big(R + |\nabla f|^2 + 2\rho f - 2(\mu - b)u\big) = 2a\big(S(\nabla f, \cdot) - \operatorname{div} S\big) + 2bu\, df. \tag{5.19}$$

Here $(\operatorname{div} S)_j = \nabla^i S_{ij}$.

Proof. For a gradient field the infinitesimal tensorial Lie derivative is

$$\mathcal{L}_{\nabla f} \xi = \big(\nabla^2 f + T_{\nabla f}\big) \diamond \xi.$$

Comparison with (5.1) proves $\mathcal{K}_\xi(\nabla f) = T_{\nabla f}$, and the skew soliton equation gives (5.18). The symmetric equation and its trace are

$$\mathrm{Ric} + \nabla^2 f + \rho g = aS + bug,$$
$$R + \Delta f + n\rho = \mu u.$$

Take the divergence of the first equation, use the contracted Bianchi identity and $\operatorname{div}(\nabla^2 f) = d\Delta f + \mathrm{Ric}(\nabla f, \cdot)$, and substitute the second equation. This yields

$$dR = 2(\mu - b) du + 2\mathrm{Ric}(\nabla f, \cdot) - 2a \operatorname{div} S.$$

Adding $d(|\nabla f|^2 + 2\rho f - 2(\mu - b)u)$ and using the symmetric equation once more gives (5.19). □

On the normal-torsion locus, the bridge parameters recover Dwivedi's Hamilton identity, with his tensor dilation rate $\lambda_s = 3\rho$ [Dwi26, Lemma 6.3, (6.10)].

To organise the remaining gradient identities, we first work with the error in the Ricci-soliton equation for an arbitrary metric and potential. Substituting the MHR torsion tensor into this error then gives the identities in the following corollary.

Lemma 5.10 (Soliton-error identities)**.** *For any metric* g*, function* f *and constant* ρ*, put* $E := \mathrm{Ric} + \nabla^2 f + \rho g$ *and* $\Delta_f = \Delta - \nabla_{\nabla f}$*. Then*

$$d\big(R + |\nabla f|^2 + 2\rho f\big) = 2E(\nabla f, \cdot) - 2 \operatorname{div} E + 2\, d \operatorname{tr} E, \tag{5.20}$$
$$\Delta_f R = -2|\mathrm{Ric}|^2 - 2\rho R + 2\langle \mathrm{Ric}, E\rangle + 2\big(\Delta \operatorname{tr} E - \operatorname{div}\operatorname{div} E\big). \tag{5.21}$$

Proof. By the contracted Bianchi identity and $\operatorname{div} \nabla^2 f = d\Delta f + \mathrm{Ric}(\nabla f)$: $\frac{1}{2} dR + d\Delta f + \mathrm{Ric}(\nabla f) = \operatorname{div} E$, with $\Delta f = \operatorname{tr} E - R - n\rho$. So $dR = 2d \operatorname{tr} E + 2\mathrm{Ric}(\nabla f) - 2 \operatorname{div} E$. Add $d|\nabla f|^2 = 2E(\nabla f) - 2\mathrm{Ric}(\nabla f) - 2\rho\, df$ to get (5.20). Taking div once more, $\frac{1}{2}\Delta R + \Delta\Delta f + \frac{1}{2}\langle \nabla R, \nabla f\rangle + \langle \mathrm{Ric}, \nabla^2 f\rangle = \operatorname{div}\operatorname{div} E$. Subtract this from $\Delta R + \Delta\Delta f = \Delta \operatorname{tr} E$ and substitute $\nabla^2 f = E - \rho g - \mathrm{Ric}$. □

Corollary 5.11 (Gradient MHR soliton package)**.** *Let $(\xi, \nabla f, \rho)$ be a gradient MHR$_{a,b}$ soliton, $S = T \star T$, $u = |T|^2$, $\mu = a + nb$. Then $E = aS + bug$, and*

(a) $\nabla^2 f - \frac{\Delta f}{n} g = -\big(\mathrm{Ric} - \frac{R}{n} g\big) + a\big(S - \frac{u}{n} g\big)$;

(b) $\Delta_f R = -2|\mathrm{Ric}|^2 - 2\rho R + 2a\langle \mathrm{Ric}, S\rangle + 2bRu + 2(\mu - b)\Delta u - 2a \operatorname{div}\operatorname{div} S$;

(c) (5.19)*; for $a = b = 0$, $R + |\nabla f|^2 + 2\rho f$ is constant on connected M;*

(d) $\Delta_f f = \mu u - R - n\rho - |\nabla f|^2$;

(e) if M is closed, then $\int_M (R - \mu u)\, dV = -n\rho \operatorname{Vol}$ and $\int_M (R + |\nabla f|^2 + n\rho - \mu u) e^{-f} dV = 0$, and $\int_M \langle T_p, \nabla_p C\rangle e^{-f} dV = 0$ for every section C of $\Lambda^2_{\mathfrak{m}}$.

The dependence on H *enters only through S and u (e.g. $S = \frac{2}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}}$, $u = \frac{2}{3}|\mathcal{T}|^2$ for G_2; $S = 6\theta \otimes \theta + 4\mathcal{T}\mathcal{T}^{\mathsf{t}}$, $u = 6|\theta|^2 + 4|\mathcal{T}|^2$ for $\mathrm{SU}(3)$).*

Proof. $E = aS + bug$ is (5.2) with $Y = \nabla f$; $\operatorname{tr} E = \mu u$, $\operatorname{div}\operatorname{div} E = a \operatorname{div}\operatorname{div} S + b\Delta u$ and $\langle \mathrm{Ric}, E\rangle = a\langle \mathrm{Ric}, S\rangle + bRu$. (a) is the trace-free part; (b), (c) follow from Lemma 5.10; (d) is the trace equation. (e) Integrate the trace and $\Delta_f f$ against dV, respectively $e^{-f} dV$ (as Δ_f is $e^{-f} dV$-symmetric). For the last identity, integrate $\langle \operatorname{div}(e^{-f} T), C\rangle = 0$ ((5.18)) by parts. The last identity also holds for arbitrary skew C, since the weighted divergence vanishes as a two-form. □

5.3 Stationarity of immortal limits

Finite total torsion energy leaves no torsion in a smooth limit obtained by translating time to infinity. Pointed compactness supplies such limits when the curvature–torsion quantity stays bounded and the selected time slices have a uniform injectivity-radius bound.

Theorem 5.12 (Stationary immortal limits)**.** *Let an MHR$_{a,b}$ solution on a closed manifold be immortal, with* $\mathrm{H} = \mathrm{G}_2$*, $\mu < -3/2$, or* $\mathrm{H} = \mathrm{Spin}(7)$*, $\mu < -2$. Assume*

$$\sup_{M \times [0,\infty)} \Lambda_{\mathrm{H}} < \infty, \qquad \mathrm{inj}_{g(t_i)}(p_i) \geq i_0 > 0 \quad \text{for a sequence } t_i \longrightarrow \infty.$$

Every subsequential limit of the time-translated flows $(M, \xi(t_i + s), p_i)$ in the pointed smooth sense of Theorem 4.7 is torsion-free, Ricci-flat, and stationary. Such a complete eternal subsequential limit exists by Corollary 4.13.

Proof. By Proposition 5.8,

$$\int_0^\infty \int_M |T|^2 \, dV_{g(t)} dt < \infty.$$

Fix $A > 0$ and a relatively compact domain U in any smooth pointed limit. On the exhaustion maps defining convergence, nonnegativity gives

$$\int_{-A}^{A} \int_U |T_\infty(s)|^2 \, dV_{g_\infty(s)} ds \leq \liminf_{i \to \infty} \int_{t_i - A}^{t_i + A} \int_M |T(t)|^2 \, dV_{g(t)} dt = 0.$$

The passage to the limit on $U \times [-A, A]$ is justified by smooth convergence of the torsion and volume forms. The last equality follows from integrability and $t_i - A \to \infty$. Exhaustion in U and arbitrariness of A imply $T_\infty \equiv 0$. Torsion-free G_2 and $\mathrm{Spin}(7)$ structures have Ricci-flat metrics, so their MHR velocity vanishes identically. Thus the eternal limit is stationary. □

The volume bound also upgrades pointed convergence to global convergence along a subsequence. If a limit has full holonomy, stability then gives convergence of the whole flow.

Corollary 5.13 (Compact limits and whole-flow convergence)**.** *Under the hypotheses of Theorem 5.12, with M connected, every pointed limit is compact and the comparison maps are eventually diffeomorphisms onto M. The subsequential convergence is therefore global and smooth on compact time intervals. If one limit has full holonomy* H*, there is a torsion-free structure ξ_* on M such that, for every $k \geq 0$,*

$$\|\xi(t) - \xi_*\|_{C^k} \leq C_k e^{-\delta_k t} \qquad (t \geq 1)$$

for constants $C_k, \delta_k > 0$. Here the norms use a fixed smooth background metric on M.

Proof. Let $(N, \xi_\infty, p_\infty)$ be a stationary pointed limit. Volume is nonincreasing, and smooth convergence on each relatively compact domain $U \Subset N$ gives

$$\mathrm{Vol}_{g_\infty}(U) \le \mathrm{Vol}_{g(0)}(M).$$

Exhaustion yields $\mathrm{Vol}_{g_\infty}(N) < \infty$. The Calabi–Yau infinite-volume property for complete noncompact manifolds with nonnegative Ricci curvature [Yau76, p. 659] therefore forces N to be compact. We recall the short comparison argument. If N were noncompact, take a unit-speed minimising ray γ from $p = p_\infty$, and set $r_j = 3^j$, $q_j = \gamma(3r_j)$ for $j \ge 0$. The balls $B(q_j, r_j)$ lie in the pairwise disjoint annuli $2r_j < d(p, \cdot) < 4r_j$. Since $B(p, 1) \subset B(q_j, 4r_j)$, Bishop–Gromov comparison gives

$$\mathrm{Vol}\, B(q_j, r_j) \ge 4^{-n}\, \mathrm{Vol}\, B(q_j, 4r_j) \ge 4^{-n}\, \mathrm{Vol}\, B(p, 1) > 0.$$

Summing over these disjoint balls contradicts finite volume.

Since N is compact, the comparison embeddings are eventually defined on all of N. Each has open image by the inverse function theorem and closed image by compactness. Connectedness of M makes it surjective, hence a diffeomorphism. This proves global subsequential convergence.

Suppose now that $\mathrm{Hol}(g_\infty) = \mathrm{H}$. Choose one sufficiently late comparison diffeomorphism $F : N \to M$. Global smooth convergence places $F^*\xi(t_i)$ in the $W^{2-2/p,p}$ neighbourhood of ξ_∞ supplied by Theorem 3.1, for a fixed $p > n + 2$. Apply that theorem and reconstruct its ungauged solution. Naturality and uniqueness identify this solution with $F^*\xi(t_i + t)$ for all $t \ge 0$. The proof of stability gives exponential convergence of the gauged solution and its DeTurck vector field in every C^k norm. Integrating the vector-field flow from t to infinity, and using its uniform derivative and inverse-derivative bounds, gives exponential convergence of the reconstruction diffeomorphisms as well. Thus $F^*\xi(t_i + t)$ converges smoothly and exponentially to a torsion-free structure on N. Pushing forward by the fixed F proves the assertion on M, with constants enlarged to cover $1 \le t \le t_i$. □

Corollary 5.14 (Collapse when torsion-free limits are obstructed)**.** *Let M be connected and closed and admit no torsion-free* H*-structure, where* $\mathrm{H} = \mathrm{G}_2$ *or* $\mathrm{Spin}(7)$*. An immortal MHR solution with uniformly bounded Λ_{H} and with $\mu < -3/2$ or $\mu < -2$, respectively, satisfies*

$$\sup_{p \in M} \mathrm{inj}_{g(t)}(p) \longrightarrow 0 \qquad \text{as } t \longrightarrow \infty.$$

Proof. Otherwise some sequence $t_i \to \infty$, $p_i \in M$ has $\mathrm{inj}_{g(t_i)}(p_i) \ge i_0 > 0$. Theorem 5.12 and Corollary 5.13 would give a compact torsion-free limit diffeomorphic to M. Its defining tensor pulls back to a torsion-free structure on M, contrary to the hypothesis. □

We now turn to explicit solutions. Nearly parallel G_2 seeds give homothetic flows and determine the critical steady case; their products with a line or circle give complementary $\mathrm{Spin}(7)$ models. Contact Calabi–Yau geometry allows independent Reeb and transverse scales. On the Heisenberg group this family gives complete solitons, while compact quotients provide the collapsing Type I example.

5.4 Nearly parallel and cylindrical solutions

A nearly parallel structure has full torsion proportional to the metric, so every quadratic torsion term in MHR is also proportional to the metric. The resulting evolution is homothetic for every parameter pair, extending Dwivedi's Ricci–harmonic example [Dwi26, Proposition 3.13]. Its stationary case exhausts the closed steady solutions at $\mu = 9$.

Proposition 5.15 (Nearly parallel MHR solutions)**.** *Let φ_0 be a nearly parallel* G_2*-structure with $\mathcal{T}_0 = cg_0$, where c is constant. Put $\mu = a + 7b$ and*

$$s(t) = 1 + \tfrac{4}{3}(\mu - 9)c^2 t.$$

On the interval where $s(t) > 0$, the forms and metrics

$$\varphi(t) = s(t)^{3/2}\varphi_0, \qquad g(t) = s(t)g_0$$

solve MHR$_{a,b}$. Their soliton rate is $\rho = \frac{2}{3}(\mu - 9)c^2$. If $c \neq 0$, the solution is shrinking for $\mu < 9$, stationary for $\mu = 9$, and expanding for $\mu > 9$.

Proof. The nearly parallel identity gives $\mathrm{Ric}_{g_0} = 6c^2 g_0$ [Bry06, (4.30)][Kar09, Proposition 4.15]. By (2.12),

$$T_0 \star T_0 = \tfrac{2}{3}c^2 g_0, \qquad |T_0|^2 = \tfrac{14}{3}c^2, \qquad \operatorname{div} T_0 = 0.$$

The last equality follows from $\operatorname{div} \mathcal{T}_0 = 0$ and (2.13). Thus the symmetric velocity is ρg_0. For the scaled structure, the full torsion is $\mathcal{T}(t) = cs(t)^{-1/2} g(t)$, so its symmetric velocity is $\rho s(t)^{-1} g(t)$ and its skew velocity vanishes. Since $g(t) \diamond \varphi(t) = 3\varphi(t)$, the MHR equation becomes

$$\partial_t \varphi(t) = 3\rho s(t)^{-1} \varphi(t).$$

This is precisely the derivative of $s(t)^{3/2}\varphi_0$, because $s' = 2\rho$. □

For a closed seed with $c \neq 0$, the control quantity satisfies $\Lambda_{\mathrm{H}}(t) = s(t)^{-1}\Lambda_{\mathrm{H}}(0)$. Thus a shrinking example has a finite Type I singularity; an expanding example is immortal of Type III. At $\mu = 9$ the structure is stationary with nonzero torsion, showing why the strict upper inequality in the torsion-free rigidity statement is necessary.

Proposition 5.16 (The critical G_2 parameter)**.** *Let M be connected and closed and let $a + 7b = 9$. An MHR$_{a,b}$ steady soliton is a nearly parallel G_2-structure, allowing the torsion-free case, with a vector field Y satisfying $\mathcal{L}_Y \varphi = 0$. Conversely, every such pair is a steady soliton.*

Proof. Since the soliton is steady, $\rho = 0$. The integrated trace equation and (5.14) give

$$0 = \int_M \left(6|\tau_1|^2 + \tfrac{7}{2}|\tau_2|^2 + \tfrac{7}{2}|\tau_3|^2\right) dV.$$

Consequently $\tau_1 = \tau_2 = \tau_3 = 0$, so the defining torsion-form identities reduce to $d\varphi = \tau_0 \psi$ and $d\psi = 0$. Applying d gives $d\tau_0 \wedge \psi = 0$. The map $\alpha \mapsto \alpha \wedge \psi$ is injective: its Hodge dual, up to sign, is $\alpha^\sharp \lrcorner \varphi$, and the contraction identity $\varphi_{ipq}\varphi_{jpq} = 6g_{ij}$ gives injectivity. Hence τ_0 is constant and $\mathcal{T} = (\tau_0/4)g$ by the torsion decomposition [Kar09, Theorem 2.27], in the convention above. Proposition 5.15 shows that the MHR velocity vanishes when $\mu = 9$. The steady soliton equation therefore says exactly $\mathcal{L}_Y \varphi = 0$. The same calculation proves the converse. □

The product of a nearly parallel G_2 manifold with a line gives a complementary family for $\mathrm{Spin}(7)$. The product structure is locally conformally parallel and harmonic; the MHR metric equation then determines its soliton rate and gradient potential explicitly.

Proposition 5.17 (Cylindrical $\mathrm{Spin}(7)$ solitons)**.** *Let (N^7, φ, g_N) be connected and complete, with $\mathcal{T} = c g_N$ for a constant $c \neq 0$, and put $\psi = *\varphi$. On $\mathbb{R}_s \times N$ set*

$$\Phi = ds \wedge \varphi + \psi, \qquad g = ds^2 + g_N.$$

For every $a, b \in \mathbb{R}$ this is a complete gradient MHR$_{a,b}$ soliton with

$$\rho = \tfrac{1}{2}(a + 7b - 12)c^2, \qquad F(s) = \tfrac{1}{4}(12 - a)c^2 s^2 + \ell s + C, \qquad \ell, C \in \mathbb{R}. \tag{5.22}$$

These are all gradient soliton potentials for this fixed product structure. The soliton is shrinking, steady or expanding according as $a + 7b < 12$, $a + 7b = 12$ or $a + 7b > 12$.

Proof. The nearly parallel equations give

$$d\Phi = -4c\, ds \wedge \Phi.$$

Thus the Lee form is $T_8^1 = -4c\, ds$ and $T_{48}^5 = 0$. Moreover, $e^{4cs}\Phi$ is closed and defines a torsion-free $\mathrm{Spin}(7)$-structure locally. Harmonicity of locally conformally parallel $\mathrm{Spin}(7)$-structures [FSE24, arXiv:2301.12494v2, Corollary 3.4] gives $\operatorname{div} T = 0$. By (5.13),

$$|T|^2 = \tfrac{7}{32}|-4c\, ds|^2 = \tfrac{7}{2}c^2.$$

Since $\nabla_{\partial_s}\Phi = 0$, the defining identity for intrinsic torsion gives $T_{\partial_s} = 0$. For $X \in TN$, the covariant derivative identities [Kar09, Definition 2.26 and (2.28)] give

$$\nabla_X \Phi = c\big(ds \wedge (X \lrcorner \psi) - X^\flat \wedge \varphi\big).$$

This first jet is G_2-equivariant at each point. Intrinsic torsion is recovered equivariantly from that jet, so $T \star T$ restricts to a G_2-invariant symmetric bilinear form on the irreducible module TN. It is a scalar multiple of g_N; its vertical and mixed components vanish because $T_{\partial_s} = 0$. Taking the trace therefore gives

$$T \star T = \tfrac{1}{2}c^2 g_N, \qquad |T|^2 = \tfrac{7}{2}c^2, \qquad \operatorname{div} T = 0. \tag{5.23}$$

The nearly parallel Einstein identity [Bry06, (4.28) and (4.30)], with $\tau_0 = 4c$, gives $\mathrm{Ric} = 6c^2 g_N$. Hence the symmetric MHR velocity is

$$h_{a,b} = \tfrac{1}{2}c^2\big[7b\,ds^2 + (a + 7b - 12)g_N\big].$$

For the potential in (5.22),

$$\nabla^2 F = \tfrac{1}{2}(12 - a)c^2 ds^2, \qquad T_{\nabla F} = 0.$$

These identities give $h_{a,b} = \rho g + \nabla^2 F$ and $\operatorname{div} T = T_{\nabla F}$, proving both soliton equations. The product metric is complete, and ∇F is affine in s, so its vector field is complete.

Conversely, the skew equation for any gradient soliton potential gives $T_{\nabla F} = 0$. Positivity of $T \star T$ on TN implies that F depends only on s. The horizontal metric equation fixes ρ as above, and the vertical equation fixes $F'' = (12 - a)c^2/2$. Integration gives precisely (5.22). □

Corollary 5.18 (Circle products and the upper rigidity endpoint)**.** *If N is closed, the same product structure on $S^1_L \times N$ is an MHR$_{a,b}$ soliton, allowing an arbitrary vector field Y, exactly when $a = 12$. At that parameter,*

$$\rho = \tfrac{7}{2}bc^2, \qquad \mathcal{L}_Y \Phi = 0.$$

In particular, $(a, b) = (12, 0)$ gives a compact stationary structure with nonzero torsion and $\mu = a + 8b = 12$.

Proof. Write $Y = u(s, z)\partial_s + V(s, z)$ with V tangent to N. The vertical metric equation gives

$$\partial_s u = \tfrac{7}{2}bc^2 - \rho.$$

Integrating around the circle yields $\rho = 7bc^2/2$. The horizontal trace is

$$\operatorname{div}_N V = 7\big(\tfrac{1}{2}(a + 7b - 12)c^2 - \rho\big).$$

Integrating over N for each s gives $\rho = (a + 7b - 12)c^2/2$, so $a = 12$. At this value $h_{a,b} = \rho g$ and $\operatorname{div} T = 0$. The full soliton equation is therefore exactly $\mathcal{L}_Y \Phi = 0$, which also proves the converse. □

The stationary circle products show that the strict inequality $\mu > 12$ in the torsion-free part of Corollary 5.7 cannot be weakened to $\mu \geq 12$. They settle this explicit endpoint family without classifying all steady solitons at $\mu = 12$.

5.5 Contact Calabi–Yau solutions

Here the geometry supplies two independent length scales: one along the Reeb direction and one transverse to it. This reduces MHR to two scalar ordinary differential equations. The resulting formulas describe the flow; completeness and the existence of a descending soliton field are checked separately below.

Let $(M^7, \eta, \omega, \Upsilon)$ be a contact Calabi–Yau manifold, normalised by $d\eta = \omega$. The Reeb foliation has transverse Calabi–Yau metric $g_{\mathcal{D}}$, Kähler form ω, and parallel complex volume form Υ, with $\omega^3/6 = (\mathrm{i}/8)\Upsilon \wedge \overline{\Upsilon}$. Let $f, h > 0$ be spatially constant length scales in the Reeb and transverse directions, respectively. The compatible G_2 form and metric are

$$\begin{aligned} \varphi_{f,h} &= fh^2 \eta \wedge \omega + h^3 \operatorname{Re} \Upsilon, \\ g_{f,h} &= f^2 \eta^2 + h^2 g_{\mathcal{D}}. \end{aligned} \tag{5.24}$$

This ansatz was used for the Laplacian flow and coflow by Lotay–Sá Earp–Saavedra [LSES22, (3.2), (3.4)], for the modified coflow by Sá Earp–Saavedra–Suan [SESS25, Section 4, (78)–(81)], and for the Ricci–harmonic flow by Dwivedi–Singhal [DS26, Section 2.3, (2.15)–(2.19)]. Its torsion is divergence-free, and the metric equation preserves the two scales. The flow therefore reduces to an explicit system for f and h.

Proposition 5.19 (The contact Calabi–Yau family)**.** *Fix $c_0 > 0$ and $(a, b) \in \mathbb{R}^2$, and put*

$$\begin{aligned} x &= \tfrac{1}{2} + \tfrac{a}{6} + \tfrac{5b}{2}, & y &= -\tfrac{3}{2} + \tfrac{3a}{2} + \tfrac{5b}{2}, \\ \sigma &= 2x - y = \tfrac{5}{2} - \tfrac{7a}{6} + \tfrac{5b}{2}. \end{aligned} \tag{5.25}$$

If $\sigma \neq 0$, define

$$s(t) = 1 + 2\sigma c_0^2 t, \qquad h(t) = s(t)^{x/(2\sigma)}, \qquad f(t) = c_0 s(t)^{y/(2\sigma)}. \tag{5.26}$$

Then $\varphi_{f(t),h(t)}$ solves MHR$_{a,b}$ throughout $s(t) > 0$. If $\sigma = 0$, the corresponding solution is

$$h(t) = e^{xc_0^2 t}, \qquad f(t) = c_0 e^{yc_0^2 t}, \qquad t \in \mathbb{R}. \tag{5.27}$$

In both cases $h(0) = 1$, $f(0) = c_0$, and every time slice is co-closed.

Proof. Write $\theta = f\eta$, $\omega_h = h^2\omega$, $\Upsilon_h = h^3\Upsilon$, and $c = f/h^2$. Thus $d\theta = c\omega_h$ and

$$\varphi = \theta \wedge \omega_h + \operatorname{Re}\Upsilon_h, \qquad \psi = -\tfrac{1}{2}\omega_h^2 + \theta \wedge \operatorname{Im}\Upsilon_h.$$

The Calabi–Yau identities give $d\varphi = c\omega_h^2$ and $d\psi = 0$. Here and below the transverse objects are interpreted in local foliation charts; no regularity assumption on the Reeb quotient is needed.

To compute the torsion and Ricci tensor, let Z be dual to θ and let X, Y be horizontal basic fields. If $\omega_h(X, Y) = g(JX, Y)$, the Koszul formula gives

$$\begin{aligned} \nabla_X Y &= (\nabla^T_X Y)^{\mathcal{D}} - \tfrac{c}{2}\omega_h(X, Y)Z, \\ \nabla_X Z &= \nabla_Z X = \tfrac{c}{2}JX, & \nabla_Z Z &= 0. \end{aligned} \tag{5.28}$$

In a transverse parallel frame at the point, substitution in $\nabla\varphi$ gives $\nabla_X\varphi = -(c/2)X \lrcorner \psi$ for horizontal X and $\nabla_Z\varphi = (3c/2)Z \lrcorner \psi$. Consequently the full torsion, in horizontal–vertical orthonormal blocks, is

$$\begin{aligned} \mathcal{T} &= -c\ \operatorname{diag}(\tfrac{1}{2}, \dots, \tfrac{1}{2}, -\tfrac{3}{2}), \\ \operatorname{Ric} &= c^2\ \operatorname{diag}(-\tfrac{1}{2}, \dots, -\tfrac{1}{2}, \tfrac{3}{2}). \end{aligned} \tag{5.29}$$

For the Ricci formula, (5.28) gives the horizontal sectional-curvature correction $-3c^2\omega_h(X, Y)^2/4$ and the vertical sectional curvature $c^2/4$. Their traces, together with $\operatorname{Ric}^T = 0$, give the second formula. The mixed components vanish by O'Neill's formula, because the integrability tensor $A_X Y = -\frac{c}{2}\,\omega_h(X, Y)Z$ is transversally parallel ($\nabla^T\omega = 0$). The Ricci tensor and the quadratic torsion terms agree with [DS26, Proposition 2.5 and Lemma 2.6]. Their full torsion [DS26, Proposition 2.4] is $-\mathcal{T}$: they use the opposite orientation, $(X \lrcorner\varphi) \wedge (Y \lrcorner\varphi) \wedge \varphi = 6g(X, Y)\,\mathrm{vol}$ [DS26, (2.1)], whereas [Dwi26, (2.2)] and this paper use -6. Since $\mathcal{T} = -(c/2)g + 2c\theta^2$, while Z is divergence-free and geodesic, $\operatorname{div}\mathcal{T} = 0$.

By (2.12), $|T|^2 = 5c^2/2$ and $T \star T = c^2\operatorname{diag}(1/6, \dots, 1/6, 3/2)$. The MHR velocity is therefore $c^2\operatorname{diag}(x, \dots, x, y) \diamond \varphi$, with zero skew part. Comparing its two form components with (5.24) yields

$$\frac{h'}{h} = xc^2, \qquad \frac{f'}{f} = yc^2, \qquad (c^2)' = -2\sigma c^4. \tag{5.30}$$

For $\sigma \neq 0$ this gives $c^2 = c_0^2/s$ and then (5.26); for $\sigma = 0$, $c = c_0$ and (5.27) follows. All expressions remain smooth and positive on the stated intervals, completing the proof. □

Several earlier solutions occur at the following parameter values. The constant A in the modified coflow is the one in [SESS25, (2)].

Evolution	(a, b)	(x, y)	Source
Laplacian coflow	$(-3/2, 3/10)$	$(1, -3)$	[LSES22, (3.15)]
Laplacian flow	$(9/2, -1/2)$	$(0, 4)$	[LSES22, (4.4)]
Modified coflow, $A = 0$	$(3, -3/5)$	$(-1/2, 3/2)$	[SESS25, (94)–(95)]
Ricci–harmonic flow	$(9/2, -3/2)$	$(-5/2, 3/2)$	[DS26, Theorem 3.1 and (3.4)]

All four solutions use $d\eta = \omega$ and the same phase of Υ. Their initial radius is our c_0, and the functions (a_t, b_t) of [SESS25, (78)] are our (f, h). Substitution in (5.25) gives the displayed (x, y); integrating $h'/h = x(f/h^2)^2$ and $f'/f = y(f/h^2)^2$ gives the cited solutions without a change of time. These are agreements of trajectories within the contact ansatz; the unrestricted flow equations are different.

There are two contributions to the curvature scale: the contact torsion, measured by $c^2 = f^2/h^4$, and the transverse curvature, measured by $h^{-2}|\mathrm{Rm}^T|$. Their relative rates determine the singularity type.

Corollary 5.20 (Curvature rates of the contact family)**.** *Assume M is closed. If its transverse metric is flat, the solutions in Proposition 5.19 have a finite Type I singularity for $\sigma < 0$ and are immortal of Type III for $\sigma > 0$. If the transverse metric is not flat, these conclusions hold precisely when $a \geq 3/2$; for $a < 3/2$ the corresponding types are IIa and IIb. When $\sigma = 0$, the solution is immortal and $\sup_{t\geq 0} t\Lambda_{\mathrm{H}}(t) = \infty$, including the stationary case $x = y = 0$: the factor t makes $t\Lambda_{\mathrm{H}}(t)$ unbounded even when the positive torsion norm is constant.*

Proof. Put $K_T = \sup_M |\mathrm{Rm}^T|_{g_{\mathcal{D}}}$. The connection formula (5.28) expresses the horizontal curvature as $h^{-2}\mathrm{Rm}^T$ plus a universal algebraic term of size c^2; the other curvature components have size at most Cc^2. It also gives $|\nabla T| \leq Cc^2$. Since $|T|^2 = 5c^2/2$, the triangle inequality in both directions gives constants $C_1, C_2 > 0$, independent of time, such that

$$C_1(c^2 + h^{-2}K_T) \leq \Lambda_{\mathrm{H}}(t) \leq C_2(c^2 + h^{-2}K_T). \tag{5.31}$$

If $\sigma < 0$, the maximal forward time is $T_0 = -1/(2\sigma c_0^2)$ and $\Lambda_{\mathrm{H}}(t) \asymp s^{-1} + K_T s^{-x/\sigma}$. Thus $(T_0 - t)\Lambda_{\mathrm{H}}(t)$ is bounded precisely when $K_T = 0$ or $x/\sigma \leq 1$. If $\sigma > 0$, the solution is immortal and $t\Lambda_{\mathrm{H}}(t)$ is bounded precisely when $K_T = 0$ or $x/\sigma \geq 1$. Both nonflat conditions reduce to $x \geq \sigma$, equivalently $y - x = 4(a - 3/2)/3 \geq 0$. Finally, when $\sigma = 0$, the positive constant c_0^2 is a lower bound for Λ_{H} up to a fixed factor. This proves all assertions. □

A local formula for the evolving tensors can descend to a quotient even when the vector field giving its soliton representation does not. The following proposition keeps completeness, descent of the flow and descent of a soliton representation separate.

Proposition 5.21 (Completeness and descent)**.** *(i) (Cylinders.) The solitons of Proposition 5.17 are complete, with complete fields. The gradient field $\nabla F = (\frac{1}{2}(12 - a)c^2 s + \ell)\partial_s$ is invariant under $s \mapsto s + L$ if and only if $a = 12$. By Corollary 5.18, $S^1_L \times N$ carries no other soliton field, so the soliton property descends to $S^1_L \times N$ exactly when $a = 12$; then every soliton field is an automorphism field.*

(ii) (Contact family: completeness.) For every contact Calabi–Yau $(M, \eta, \omega, \Upsilon)$ and all $f, h > 0$, $\min\{f^2, h^2\}g_{1,1} \leq g_{f,h} \leq \max\{f^2, h^2\}g_{1,1}$. So every time slice of the explicit MHR$_{a,b}$ solution of Proposition 5.19 is complete if and only if $g_{1,1}$ is: always on closed M and on the Heisenberg group. The solution formula is local, so the flow descends to every quotient.

(iii) (Contact family: soliton descent.) Let M be a closed contact Calabi–Yau 7-manifold. The structure $\varphi_{f,h}$ is an MHR$_{a,b}$ soliton for some vector field Y if and only if $a = \frac{3}{2}$. Then $\rho = (\frac{3}{4} + \frac{5}{2}b)c^2 = \frac{5\mu+3}{14}c^2$ with $\mu = a + 7b$, and every soliton field is an infinitesimal automorphism. In particular, on a compact Heisenberg nilmanifold $\Gamma\backslash G$ the algebraic soliton of Remark 5.33 and Proposition 5.22, whose soliton field $c^2(y - x)V_D$ is a multiple of the field V_D generating $\exp(tD)$, $D = \mathrm{diag}(1, \dots, 1, 2)$, descends as a soliton only at $a = \frac{3}{2}$, where that multiple vanishes. For $\Gamma \neq 1$ the field V_D itself never descends.

Proof. (i) N is closed by Myers; product of complete metrics; ∇F is affine in s. (ii) $g_{f,h} = f^2\eta^2 + h^2 g_{\mathcal{D}}$ with η^2 and $g_{\mathcal{D}}$ orthogonal. (iii) Since $\nabla_Z Z = 0$ and $\nabla Z = \frac{c}{2}J$ by (5.28) on the horizontal space, which is skew, Z is Killing and $\mathrm{div}\, Z = 0$. For any Y, $(\frac{1}{2}\mathcal{L}_Y g)(Z, Z) = g(\nabla_Z Y, Z) = Z(g(Y, Z))$, whose integral vanishes. The velocity is $c^2\mathrm{diag}(x, \dots, x, y)$ with c constant and zero skew part. The (Z, Z)-component of (5.2), integrated, gives $\rho = yc^2$. The integrated trace gives $7\rho = (6x + y)c^2$. Hence $x = y$, i.e. $y - x = \frac{4}{3}(a - \frac{3}{2}) = 0$. Conversely, at $a = \frac{3}{2}$ the velocity is $xc^2 g \diamond \varphi$, so $(\varphi, 0, xc^2)$ is a soliton, and Proposition 5.4(b) gives the last assertion. If δ_t are the grading automorphisms, then $V_D(\gamma p) = dR_p V_D(\gamma) + dL_\gamma V_D(p)$, and $V_D(\gamma) \neq 0$ for $\gamma \neq e$. □

5.6 Heisenberg solitons and collapse

Earlier explicit evolutions on the same Heisenberg group were obtained for the Laplacian coflow and its modification by [BFF20, Corollary 5 and Theorem 7], and for the Ricci–harmonic flow by [DS26, Theorem 4.6]. The scalar-plus-derivation calculation below implements for MHR the algebraic-soliton construction of [Lau17, Theorem 3.8, (22), and Definition 3.9].

On the Heisenberg group the transverse geometry is flat and the contact Calabi–Yau ansatz is left-invariant. Grading automorphisms change the horizontal and central length scales at different rates. Combining these automorphisms with a homothety gives the soliton representation on the simply connected group.

Proposition 5.22 (Algebraic solitons on the Heisenberg group)**.** *Let G be the simply connected seven-dimensional Heisenberg group with left-invariant coframe satisfying $d\eta = \omega = e^{12} + e^{34} + e^{56}$ and $de^j = 0$ for $1 \le j \le 6$. For $c_0 > 0$, set*

$$\varphi_0 = c_0\eta \wedge \omega + \operatorname{Re}\Upsilon, \qquad g_0 = c_0^2\eta^2 + \sum_{j=1}^{6}(e^j)^2.$$

Let D be the grading derivation, equal to the identity on the horizontal space and twice the identity on the centre, and let V_D generate its one-parameter group of automorphisms. With x, y, σ as in (5.25),

$$(\varphi_0, Y, \rho), \qquad Y = c_0^2(y-x)V_D, \qquad \rho = \sigma c_0^2, \tag{5.32}$$

is a complete MHR$_{a,b}$ soliton. It admits no gradient soliton representation when $a \neq 3/2$. When $a = 3/2$, it is homothetic, with $Y = 0$, and is stationary precisely when $b = -3/10$.

Proof. The grading is a derivation because the only nonzero brackets pair two horizontal vectors to a central vector. Its automorphisms and the field V_D are defined for all real times. Since D is symmetric for g_0,

$$\begin{aligned} \tfrac{1}{2}\mathcal{L}_Y g_0 &= c_0^2(y-x)D, \\ \mathcal{L}_Y\varphi_0 &= c_0^2(y-x)D \diamond \varphi_0, \end{aligned}$$

where the first expression identifies the endomorphism with its associated symmetric tensor. Hence $\mathcal{K}_{\varphi_0}(Y) = 0$. The identity

$$c_0^2 \operatorname{diag}(x, \dots, x, y) = \sigma c_0^2 \operatorname{Id} + c_0^2(y-x)D$$

and (5.29) verify both soliton equations. The left-invariant metric on G is complete.

If the same structure admitted a gradient soliton potential u, then Proposition 5.9 and $\operatorname{div}\mathcal{T} = 0$ would give $\mathcal{T}^{\mathsf{t}}\nabla u = 0$. The torsion matrix in (5.29) is invertible, so $\nabla u = 0$. The symmetric soliton equation would force $x = y$, equivalent to $a = 3/2$. At this parameter $x = y = 3/4 + 5b/2$, proving the remaining claims. □

At the bridge parameters, $\rho = -13c_0^2/2$, giving a complete non-gradient shrinking MHR soliton. The explicit flow also descends to every compact Heisenberg quotient, although the automorphism field need not descend. The quotient geometry can behave differently from its universal cover: the following solution collapses even after rescaling by its Type I singularity rate.

Corollary 5.23 (A collapsing Type I solution)**.** *Let $M = \Gamma\backslash G$ be a compact quotient of the Heisenberg group in Proposition 5.22, with $c_0 = 1$. For $(a, b) = (0, -6/5)$, its MHR solution has a Type I singularity at $t = 1$, whereas the metrics $\widehat{g}(t) = (1-t)^{-1}g(t)$ satisfy*

$$\begin{aligned} \operatorname{diam}_{\widehat{g}(t)} M &\longrightarrow 0, \\ \operatorname{Vol}_{\widehat{g}(t)} M = (1-t)^{16} \operatorname{Vol}_{g_0} M &\longrightarrow 0. \end{aligned}$$

Consequently their injectivity radii tend to zero uniformly, and the volumes of balls of every fixed positive radius tend to zero. Thus a Type I bound for Λ_{H} alone does not imply noncollapse at the singularity scale for general MHR parameters.

Proof. Here $x = -5/2$, $y = -9/2$, $\sigma = -1/2$ and $s = 1 - t$, so

$$h = s^{5/2}, \qquad f = s^{9/2}, \qquad c^2 = s^{-1},$$

$$\widehat{g} = s^8\eta^2 + s^4\sum_{j=1}^{6}(e^j)^2.$$

The Type I assertion follows from Corollary 5.20. Let δ_r be the Heisenberg automorphism acting by r on the horizontal space and by r^2 on the centre. Then $\widehat{g} = \delta_{s^2}^* g_0$ on G, and hence $(M, \widehat{g})$ is isometric to $(\delta_{s^2}(\Gamma)\backslash G, g_0)$. On the fixed quotient the displayed metric satisfies $\widehat{g} \le s^4 g_0$ for $0 < s \le 1$. Hence $\operatorname{diam}_{\widehat{g}} M \le s^2 \operatorname{diam}_{g_0} M \to 0$. The volume formula follows directly from the six horizontal metric factors s^4 and the central factor s^8. The injectivity radius at any point is at most the diameter; moreover, every ball of a fixed positive radius is eventually the whole quotient. These observations prove the remaining assertions. The curvature of $\widehat{g}$ remains uniformly bounded, since it is locally isometric to g_0 throughout. □

The behaviour depends on the parameters. For the same seed with $c_0 = 1$ at the bridge pair $(a, b) = (9/2, -3/2)$, (5.25) gives $x = -5/2$, $y = 3/2$ and $\sigma = -13/2$. Writing $s = 1 - 13t$, the singularity-rescaled metric is

$$s^{-1}g(t) = s^{-16/13}\eta^2 + s^{-8/13}\sum_{j=1}^{6}(e^j)^2.$$

Here every coefficient grows as the singular time is approached, whereas the coefficients in Corollary 5.23 tend to zero.

5.7 Homogeneous solitons and bracket flow

Left invariance turns the flow into a finite-dimensional problem. One may either evolve the defining form on a fixed Lie algebra or keep the form fixed and evolve the bracket. We derive the equivalence, then use it to express the soliton condition as exact algebraic equations.

Throughout, $\mathfrak{g} = (\mathbb{R}^7, \mu)$ is a Lie algebra with inner product $\langle\cdot,\cdot\rangle$ and orthonormal basis $e_1, \dots, e_7$, and φ is a positive 3-form with $g_\varphi = \langle\cdot,\cdot\rangle$. Structure constants are $\mu_{ijk} = \langle\mu(e_i, e_j), e_k\rangle$, and $de^k = s\, e^{ij}$ means $\mu_{ijk} = -s$. As in Section 2, a 2-tensor A is identified with the endomorphism E_A given by $A(x, y) = \langle E_A x, y\rangle$ (so the matrix of E_A is A^{t}), and $A \diamond \varphi = \varphi(E_A\cdot,\cdot,\cdot) + \varphi(\cdot, E_A\cdot,\cdot) + \varphi(\cdot,\cdot, E_A\cdot)$. We write $\mathfrak{g}_2 = \mathfrak{g}_2(\varphi) \subset \mathfrak{so}(\mathfrak{g})$ for the stabiliser and $\mathfrak{m} = \mathfrak{g}_2^{\perp} = \{v \lrcorner \varphi\} \cong \Lambda^2_7$, with orthogonal projection

$$(\operatorname{pr}_{\mathfrak{m}} A)_{ab} = \tfrac{1}{6}\, A_{pq}\varphi_{pqm}\varphi_{mab} \qquad (A \in \mathfrak{so}(\mathfrak{g})), \tag{5.33}$$

which uses $\varphi_{mpq}\varphi_{npq} = 6\delta_{mn}$ and no orientation. For $E \in \mathfrak{gl}(\mathfrak{g})$ put

$$\pi(E)\mu = E\mu(\cdot,\cdot) - \mu(E\cdot,\cdot) - \mu(\cdot, E\cdot) = \tfrac{d}{ds}\big|_0\, e^{sE}\cdot\mu, \qquad h\cdot\mu := h\mu(h^{-1}\cdot, h^{-1}\cdot), \tag{5.34}$$

so $\pi(E)\mu = 0$ if and only if $E \in \operatorname{Der}(\mu)$, and $\pi(I)\mu = -\mu$. Lauret writes $\delta_\mu(E) = -\pi(E)\mu$ and $\theta(E)\varphi = -E \diamond \varphi$ [Lau17, (19) and §2].

The next formulas provide the velocity from the bracket alone. Their polynomial dependence is what makes both the bracket ODE and the exact catalogue computations possible.

Proposition 5.24 (MHR velocity of a left-invariant G_2-structure)**.** *Let $\nabla_x \in \mathfrak{so}(\mathfrak{g})$ be the Levi-Civita endomorphism, $\langle\nabla_x y, z\rangle = \frac{1}{2}\big(\langle\mu(x,y),z\rangle - \langle\mu(y,z),x\rangle + \langle\mu(z,x),y\rangle\big)$, let $H_\mu \in \mathfrak{g}$ be given by $\langle H_\mu, x\rangle = \operatorname{tr}\operatorname{ad}_\mu x$, and let B_μ be the Killing form regarded as an endomorphism. Then, for the intrinsic torsion of (2.1):*

(i) $T_x = -\operatorname{pr}_{\mathfrak{m}}\nabla_x$;

(ii) $\langle(T \star T)x, y\rangle = \langle T_x, T_y\rangle$ *and* $|T|^2 = \sum_i |T_{e_i}|^2$, *with* $\langle A, B\rangle = A_{pq}B_{pq}$;

(iii) $\operatorname{Ric}_\mu = M_\mu - \frac{1}{2}B_\mu - \frac{1}{2}(\operatorname{ad}_\mu H_\mu + (\operatorname{ad}_\mu H_\mu)^{\mathsf{t}})$, *where* $M_\mu = -\frac{1}{2}\sum_i (\operatorname{ad}_\mu e_i)^{\mathsf{t}}\operatorname{ad}_\mu e_i + \frac{1}{4}\sum_i \operatorname{ad}_\mu e_i (\operatorname{ad}_\mu e_i)^{\mathsf{t}}$;

(iv) $\operatorname{div} T = \sum_i [\nabla_{e_i}, T_{e_i}] - T_{H_\mu} = \sum_i [\operatorname{pr}_{\mathfrak{g}_2}\nabla_{e_i}, T_{e_i}] - T_{H_\mu} \in \mathfrak{m}$.

Consequently the MHR$_{a,b}$ velocity (2.4) is $\partial_t\varphi = \mathcal{Q}_\mu \diamond \varphi$ with

$$\mathcal{Q}_\mu = h_\mu + X_\mu, \qquad h_\mu = -\operatorname{Ric}_\mu + a\, T \star T + b|T|^2 I \in \operatorname{Sym}(\mathfrak{g}), \qquad X_\mu = \operatorname{div} T \in \mathfrak{m} \cong \Lambda^2_7. \tag{5.35}$$

Every entry is a polynomial in μ: ∇, T and H_μ are linear, and Ric, $T \star T$, $|T|^2$ *and* $\operatorname{div} T$ *are quadratic. On a unimodular algebra $H_\mu = 0$; on a nilpotent algebra also $B_\mu = 0$, so* $\operatorname{Ric}_\mu = M_\mu$ *and* $\operatorname{scal}_\mu = -\frac{1}{4}|\mu|^2$ *with* $|\mu|^2 = \sum_{ijk}\mu_{ijk}^2$. *In the* G_2 *convention of Proposition 2.2, $T \star T = \frac{2}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}}$, $|T|^2 = \frac{2}{3}|\mathcal{T}|^2$ and $X_\mu = -\frac{1}{3}(\operatorname{div}\mathcal{T})\lrcorner\varphi$, so that $X_\mu \diamond \varphi = (\operatorname{div}\mathcal{T})\lrcorner\psi$.*

Proof. For left-invariant fields y, z, w one has $(\nabla_x\varphi)(y,z,w) = -\varphi(\nabla_x y, z, w) - \dots$, that is, $\nabla_x\varphi = -\nabla_x \diamond \varphi$. Since $\mathfrak{g}_2 \diamond \varphi = 0$, this equals $(-\mathrm{pr}_{\mathfrak{m}}\nabla_x)\diamond\varphi$. The map $\mathfrak{m} \to \Lambda^3$, $A \mapsto A\diamond\varphi$, is injective, so (i) holds. (ii) is (2.2) read in the identification fixed at the start of this subsection. (iii) is the standard formula for left-invariant metrics [Bes87, 7.38]. For (iv), differentiate the left-invariant tensor T: $(\nabla_{e_p}T)_{e_p}(y,z) = -T_{\nabla_{e_p}e_p}(y,z) - T_{e_p}(\nabla_{e_p}y, z) - T_{e_p}(y, \nabla_{e_p}z)$. Use $\sum_p \nabla_{e_p}e_p = H_\mu$ (from $\langle\nabla_{e_p}e_p, e_m\rangle = -\mu_{pmp}$) and the skew-symmetry of ∇_{e_p}; this gives the first expression. Splitting $\nabla_{e_p} = \mathrm{pr}_{\mathfrak{g}_2}\nabla_{e_p} - T_{e_p}$ and $[T_{e_p}, T_{e_p}] = 0$ gives the second, which lies in $\mathfrak{m}$ because $[\mathfrak{g}_2, \mathfrak{m}] \subset \mathfrak{m}$. The G_2 identities follow from Proposition 2.2 in the orientation fixed in Section 2. □

Lemma 5.25 (Vanishing of the skew part)**.** *For every left-invariant G_2-structure, in the orientation convention of Section 2 (that of [Dwi26]), $\mathrm{div}(\mathcal{T}^{\mathsf{t}}) = \mathcal{T}\,V_{\mathcal{T}}$ with $(V_{\mathcal{T}})_k = \mathcal{T}_{ij}\varphi_{ijk}$; reversing the orientation ($\psi \mapsto -\psi$, $\mathcal{T} \mapsto -\mathcal{T}$) changes the sign of the right-hand side. Hence $X_\mu = 0$ whenever φ is co-closed (then $\mathcal{T}$ is symmetric) or closed (then $\mathcal{T} \in \Lambda^2_{14}$). For a closed structure on any manifold, $\mathrm{div}\,\mathcal{T} = 0$ pointwise.*

Proof. The first identity is the contracted G_2-Bianchi identity $\mathrm{div}\,\mathcal{T}^{\mathsf{t}} = \nabla\,\mathrm{tr}\,\mathcal{T} + \mathcal{T}V_{\mathcal{T}}$ ([DGK25, relation $(\mathrm{G}7_a)$ of the decomposed G_2-Bianchi identity]), with $\mathrm{tr}\,\mathcal{T}$ constant. V annihilates Sym^2 and Λ^2_{14}, so in both cases $\mathrm{div}\,\mathcal{T} = \pm\,\mathrm{div}\,\mathcal{T}^{\mathsf{t}} = 0$. In the closed case $\mathrm{tr}\,\mathcal{T} = 0$ pointwise, so the conclusion holds on any manifold. □

Proposition 5.26 (Closed structures: $\mathrm{MHR}_{3,-1/2}$ is the Laplacian flow)**.** *For a closed G_2-structure on any manifold, $\Delta_\varphi\varphi = \mathcal{F}_{3,-1/2}(\varphi)$; for left-invariant φ this is $\mathcal{Q}_{3,-1/2}\diamond\varphi$. Consequently, for left-invariant closed initial data on a Lie group and every b, the $\mathrm{MHR}_{3,b}$ solution stays closed and equals the Laplacian-flow solution up to a time-dependent scaling and reparametrisation. The (semi-)algebraic Laplacian solitons of [Lau17] are (semi-)algebraic $\mathrm{MHR}_{3,b}$ solitons for every b, with $\rho_{\mathrm{MHR}} = -c_{\mathrm{Lauret}} + (b+\frac{1}{2})|T|^2$.*

Proof. By [Lau17, Proposition 2.2], which rests on [LW17, Proposition 2.2 and (3.4)], $\Delta\varphi = \theta(Q^L)\varphi = -Q^L\diamond\varphi$ with $Q^L = \mathrm{Ric} - \frac{1}{12}(\mathrm{tr}\,\tau^2)I + \frac{1}{2}\tau^2$ and $\mathcal{T} = -\frac{1}{2}\tau$. So $\tau^2 = -4\mathcal{T}\mathcal{T}^{\mathsf{t}}$, $\mathrm{tr}\,\tau^2 = -4|\mathcal{T}|^2$ and $-Q^L = -\mathrm{Ric} + 2\mathcal{T}\mathcal{T}^{\mathsf{t}} - \frac{1}{3}|\mathcal{T}|^2 I$. This is h_μ for $(a,b) = (3, -\frac{1}{2})$ by Proposition 5.24, and $X_\mu = 0$ by Lemma 5.25. For left-invariant φ, $\mathcal{Q}_{3,b}\diamond\varphi = \Delta\varphi + 3(b+\frac{1}{2})|T|^2\varphi$ with $|T|^2$ constant. If φ_L solves the Laplacian flow, then $\varphi(t) = \lambda(t)^3\varphi_L(\sigma(t))$ with $\dot\sigma = \lambda^{-2}$ and $\dot\lambda = (b+\frac{1}{2})|T_{\varphi_L(\sigma)}|^2/\lambda$ solves $\mathrm{MHR}_{3,b}$; uniqueness of left-invariant solutions concludes. □

Theorem 5.27 (Bracket-flow reduction of MHR)**.** *Let G be a simply connected Lie group with Lie algebra $(\mathbb{R}^7, \mu_0)$. Fix a positive 3-form φ_0 on $\mathbb{R}^7$ with $g_{\varphi_0} = \langle\cdot,\cdot\rangle$, and write $\mathcal{Q}_\mu := \mathcal{Q}_{\mu,\varphi_0}$ as in (5.35).*

(a) Let $\varphi(t)$, $t \in (T_-, T_+)$, be the maximal left-invariant $\mathrm{MHR}_{a,b}$ solution with $\varphi(0) = \varphi_0$. This is the maximal solution of the real-analytic ODE $\dot\varphi = \mathcal{Q}_{\mu_0,\varphi}\diamond\varphi$ on the open cone of positive 3-forms on $\mathfrak{g}$.

(b) Let $\mu(t)$, $t \in (S_-, S_+)$, be the maximal solution of the bracket flow

$$\tfrac{d}{dt}\mu = \pi(\mathcal{Q}_\mu)\mu = \pi(h_\mu + X_\mu)\mu, \qquad \mu(0) = \mu_0, \tag{5.36}$$

a polynomial (cubic) ODE on $\Lambda^2(\mathbb{R}^7)^* \otimes \mathbb{R}^7$. In Lauret's sign it reads $\frac{d}{dt}\mu = -\pi(Q^L_\mu)\mu = \delta_\mu(Q^L_\mu)$ with $Q^L_\mu = -\mathcal{Q}_\mu$.

Then $(T_-, T_+) = (S_-, S_+)$. The solution $h(t) \in \mathrm{GL}(7)$ of $\dot h = \mathcal{Q}_{\mu(t)}h$, $h(0) = I$, equivalently of $\dot h = h\,\mathcal{Q}_{\mu_0,\varphi(t)}$, satisfies

$$\varphi(t) = h(t)^*\varphi_0, \qquad \mu(t) = h(t)\cdot\mu_0.$$

The Lie group isomorphisms $F_t\colon G \to G_{\mu(t)}$ with $dF_t|_e = h(t)$ therefore satisfy $F_t^*\varphi_0 = \varphi(t)$ and are isometries $(G, g_{\varphi(t)}) \to (G_{\mu(t)}, \langle\cdot,\cdot\rangle)$. In particular $\mu(t)$ stays in the Jacobi variety and in the $\mathrm{GL}(7)$-orbit of μ_0; it stays nilpotent or unimodular if μ_0 is. An arbitrary left-invariant initial structure $\bar\varphi = h_0^*\varphi_0$ is handled by replacing μ_0 with $h_0\cdot\mu_0$.

Proof. Naturality. For $h \in \mathrm{GL}(7)$ the map $h\colon (\mathfrak{g}, \mu, \varphi) \to (\mathfrak{g}, h\cdot\mu, (h^{-1})^*\varphi)$ is an isomorphism of Lie algebras with G_2-structures. Every ingredient of (5.35) is natural, since Ric, T and $\mathrm{div}\,T$ are natural under diffeomorphisms, in particular under Lie group isomorphisms, so $\mathcal{Q}_{h\cdot\mu,(h^{-1})^*\varphi} = h\,\mathcal{Q}_{\mu,\varphi}\,h^{-1}$ as endomorphisms. Also, for any curve $h(t)$, $\frac{d}{dt}h^*\varphi_0 = (h^{-1}\dot h)\diamond h^*\varphi_0$ and $\frac{d}{dt}(h\cdot\mu_0) = \pi(\dot h h^{-1})(h\cdot\mu_0)$. *From (a) to (b).* Solve the linear ODE $\dot h = h\mathcal{Q}_{\mu_0,\varphi(t)}$ on (T_-, T_+); h stays invertible. Then $\eta = h^*\varphi_0$ and $\eta = \varphi(t)$ both solve the linear ODE $\dot\eta = \mathcal{Q}_{\mu_0,\varphi(t)}\diamond\eta$, whose coefficient is a fixed function of t, with the same initial value. Hence

$h^*\varphi_0 = \varphi(t)$. By naturality, $\mu := h\cdot\mu_0$ satisfies $\dot\mu = \pi(h\mathcal{Q}_{\mu_0,\varphi(t)}h^{-1})\mu = \pi(\mathcal{Q}_{\mu,\varphi_0})\mu$, so $(T_-,T_+)\subset(S_-,S_+)$, and $\dot h = \mathcal{Q}_{\mu(t)}h$. *From (b) to (a).* Solve $\dot h = \mathcal{Q}_{\mu(t)}h$ on (S_-,S_+). Both $h\cdot\mu_0$ and $\mu(t)$ solve the linear ODE $\dot\nu = \pi(\mathcal{Q}_{\mu(t)})\nu$, so they coincide. Then $\varphi := h^*\varphi_0$ has $\dot\varphi = (h^{-1}\mathcal{Q}_{\mu(t)}h)\diamond\varphi = \mathcal{Q}_{\mu_0,\varphi}\diamond\varphi$ by naturality, so $(S_-,S_+)\subset(T_-,T_+)$. Finally, $h(t)$ integrates to F_t because G is simply connected. □

Remark 5.28 (Gauge and the skew part)**.** (i) *The $\mathfrak{g}_2$-part is gauge.* Let $Z\colon \Lambda^2(\mathbb{R}^7)^*\otimes\mathbb{R}^7\to\mathfrak{g}_2(\varphi_0)$ be smooth. If $\mu(t)$ solves (5.36) and $\dot k = Z(k\cdot\mu)\,k$, $k(0)=I$, then $k(t)\in\mathrm{G}_2$ and $\tilde\mu = k\cdot\mu$ solves $\dot{\tilde\mu} = \pi(\mathcal{Q}_{\tilde\mu}+Z(\tilde\mu))\tilde\mu$. The proof uses G_2-equivariance, $\mathcal{Q}_{k\cdot\mu} = k\mathcal{Q}_\mu k^{-1}$. Since k fixes φ_0, $(G_{\tilde\mu},\varphi_0)\cong(G_\mu,\varphi_0)$, and the velocity endomorphism matters only modulo $\mathfrak{g}_2$. (ii) *The skew part X_μ is not gauge.* $X_\mu\in\mathfrak{m}$ and $\mathfrak{m}\cap\mathfrak{g}_2 = 0$. As an element of $\mathfrak{so}(7)$ it preserves $\langle\cdot,\cdot\rangle$ and $|\mu|$, since $\langle\pi(X)\mu,\mu\rangle = 0$, but it rotates φ_0 relative to μ: $X_\mu\diamond\varphi_0 = (\operatorname{div}\mathcal{T})\lrcorner\psi$, which is nonzero whenever $X_\mu\neq 0$. Dropping it replaces MHR by the different flow $\partial_t\varphi = h\diamond\varphi$. Because h_μ depends on φ through T, the resulting metric evolutions can also differ. (iii) The condition $X=0$ along a solution needs its own justification. It holds for the Heisenberg solutions of Proposition 5.22 and for left-invariant closed data when $a=3$, by Proposition 5.26.

In the bracket description, self-similarity becomes scaling together with a change of frame preserving the fixed form. The distinction between algebraic and semi-algebraic solitons records which derivations can generate that change of frame.

Definition 5.29 (Algebraic MHR solitons)**.** (μ,φ_0) is a *semi-algebraic* $\mathrm{MHR}_{a,b}$ soliton if there are $\rho\in\mathbb{R}$ and $D\in\operatorname{Der}(\mu)$ with

$$\mathcal{Q}_\mu - \rho I - D\in\mathfrak{g}_2(\varphi_0),\quad \text{equivalently}\quad h_\mu = \rho I + \tfrac12(D+D^{\mathsf t}),\qquad X_\mu = \operatorname{pr}_{\mathfrak{m}}\tfrac12(D-D^{\mathsf t}). \tag{5.37}$$

It is *algebraic* if moreover $\mathcal{Q}_\mu = h_\mu + X_\mu = \rho I + D$. These are the notions of [Lau17, Definition 3.9 and Proposition 4.5], with $Q^L = -\mathcal{Q}$, $c=-\rho$ and $D_{\mathrm L} = -D$. For co-closed or closed φ_0 one has $X_\mu = 0$, and (5.37) reduces to $h_\mu = \rho I + \frac12(D+D^{\mathsf t})$ with $\frac12(D-D^{\mathsf t})\in\mathfrak{g}_2$.

Proposition 5.30 (Algebraic solitons are MHR solitons)**.** *Let G be simply connected and let (μ,φ_0) satisfy (5.37) with (ρ,D), and put $Z = \mathcal{Q}_\mu - \rho I - D$. Let $F_s\in\operatorname{Aut}(G)$ satisfy $dF_s|_e = e^{sD}$, and let Y_D be the complete vector field generating F_s. Then (φ_0,Y_D,ρ) is an MHR soliton in the sense of Definition 5.1:*

$$-\mathrm{Ric} + a(T\star T) + b|T|^2 g = \rho g + \tfrac12\mathcal{L}_{Y_D}g,\qquad \operatorname{div}T = \mathcal{K}_{\varphi_0}(Y_D),$$

where at e one has $\frac12\mathcal{L}_{Y_D}g = \frac12(D+D^{\mathsf t})$ and $\mathcal{K}_{\varphi_0}(Y_D) = \operatorname{pr}_{\mathfrak{m}}\frac12(D-D^{\mathsf t})$. The solution with $\varphi(0)=\varphi_0$ is

$$\varphi(t) = s(t)^{3/2}F^*_{\sigma(t)}\varphi_0,\qquad s = 1+2\rho t,\quad \sigma = \tfrac{1}{2\rho}\log s\ \ (\sigma = t \text{ if } \rho = 0),$$

which is (5.4), since $F_{\sigma(t)}$ is generated by $Y_D/s(t)$. The corresponding bracket flow is $\mu(t) = s(t)^{-1/2}\,e^{\sigma(t)Z}\cdot\mu$ with $e^{\sigma Z}\in\mathrm{G}_2$, and $\mu(t) = s(t)^{-1/2}\mu$ in the algebraic case. Conversely, if the bracket flow has the form $\mu(t) = c(t)k(t)\cdot\mu$ with $c>0$ and $k(t)\in\mathrm{G}_2$ (C^1, $c(0)=1$, $k(0)=I$), then μ is semi-algebraic with $\rho = -\dot c(0)$ and $Z = \dot k(0)$. For $\mu\neq 0$,

$$\rho = -\frac{\langle\pi(\mathcal{Q}_\mu)\mu,\mu\rangle}{|\mu|^2} = -\frac{\langle\pi(h_\mu)\mu,\mu\rangle}{|\mu|^2} \overset{\text{nilpotent}}{=} \frac{\operatorname{tr}(h_\mu\mathrm{Ric}_\mu)}{\mathrm{scal}_\mu},\qquad 7\rho + \operatorname{tr}D = -\mathrm{scal}_\mu + (a+7b)|T|^2. \tag{5.38}$$

Proof. $F_s^*\varphi_0$ is left-invariant because F_s is an automorphism, and its value at e is $\varphi_0(e^{sD}\cdot,e^{sD}\cdot,e^{sD}\cdot)$. Hence $\mathcal{L}_{Y_D}\varphi_0 = D\diamond\varphi_0$ everywhere. Since $Z\diamond\varphi_0 = 0$ and $I\diamond\varphi_0 = g\diamond\varphi_0$, $\mathcal{F}_{a,b}(\varphi_0) = \mathcal{Q}_\mu\diamond\varphi_0 = \rho\,g\diamond\varphi_0 + \mathcal{L}_{Y_D}\varphi_0$. Decompose with (5.1) and use the injectivity of $(S,C)\mapsto(S+C)\diamond\varphi_0$ on $\mathrm{Sym}^2\oplus\Lambda^2_7$ to get both equations. The formula for $\varphi(t)$ follows by differentiation, using $\mathcal{F}_{a,b}(s^{3/2}F^*\varphi_0) = s^{1/2}F^*\mathcal{F}_{a,b}(\varphi_0)$, $\dot s = 2\rho$ and $\dot\sigma = 1/s$. For the bracket flow, $\mathcal{Q}_{ck\cdot\mu} = c^2k\mathcal{Q}_\mu k^{-1}$, $\pi(D)\mu = 0$ and $\pi(I)\mu = -\mu$ reduce (5.36) to $\dot c = -\rho c^3$ and $\dot k k^{-1} = c^2kZk^{-1}$. The converse is obtained by differentiating at $t=0$. For (5.38), take the inner product of $\pi(\mathcal{Q}_\mu)\mu = -\rho\mu + \pi(Z)\mu$ with μ: $\pi(Z)$ and $\pi(X_\mu)$ are skew-adjoint, and $\langle\pi(A)\mu,\mu\rangle = 4\operatorname{tr}(A\mathrm{Ric}_\mu)$ for symmetric A and nilpotent μ (Lauret's moment-map identity). The trace identity is the trace of $h_\mu = \rho I + \frac12(D+D^{\mathsf t})$, in agreement with (5.9) because $\operatorname{div}Y_D = \operatorname{tr}D$ on every Lie group ($F_s^*\mathrm{vol} = e^{s\operatorname{tr}D}\mathrm{vol}$). □

Proposition 5.31 (Scale-invariant bracket flow)**.** *For $\mu \neq 0$, put $r_\mu = -\langle \pi(\mathcal{Q}_\mu)\mu, \mu\rangle/|\mu|^2$. The flow $\dot\mu = \pi(\mathcal{Q}_\mu - r_\mu I)\mu$ preserves $|\mu|$; on a nilpotent algebra this is* $\mathrm{scal} = -\frac{1}{4}|\mu|^2$*. Its solutions are $c(t)\mu(\tau(t))$, where μ solves (5.36), $\dot c = r\,c$ and $\dot\tau = c^2$; recall $(G_{c\mu}, \varphi_0) \cong (G_\mu, c^{-3}\varphi_0)$ and $\mathcal{Q}_{c\mu} = c^2\mathcal{Q}_\mu$. Its fixed points are exactly the algebraic solitons, with $\rho = r_\mu$. Its G_2-invariant orbits, i.e. the fixed points of the induced flow on G_2-orbits, are exactly the semi-algebraic solitons. For $\mu \neq 0$ the rate of a semi-algebraic soliton is unique, $\rho = r_\mu$, by (5.38); for $\mu = 0$ every endomorphism is a derivation and every ρ occurs, with $D = -\rho I$ (cf. Proposition 5.4(b)). This is uniqueness of the algebraic data only: a flat non-abelian (G_μ, φ_0) with parallel φ_0 is an MHR soliton for every ρ.*

Proof. Taking the inner product with μ, using $\pi(I)\mu = -\mu$, gives $\frac{d}{dt}|\mu|^2 = 0$. For $\nu = c\mu(\tau)$, $\mathcal{Q}_\nu = c^2\mathcal{Q}_\mu$; differentiation gives the stated rescaling, with $r = r_\nu$. At a fixed point, $\pi(\mathcal{Q}_\mu - r_\mu I)\mu = 0$, so $\mathcal{Q}_\mu - r_\mu I$ is a derivation. A stationary G_2-orbit allows in addition $\pi(Z)\mu$, $Z \in \mathfrak{g}_2$, giving precisely (5.37). Formula (5.38) proves uniqueness of the rate when $\mu \neq 0$. □

Proposition 5.32 (No nearly parallel left-invariant structures)**.** *No 7-dimensional Lie group carries a left-invariant nearly parallel G_2-structure ($\tau_0 \neq 0$). On a nilpotent Lie group, every left-invariant G_2-structure has* $\mathrm{scal} = -\frac{1}{4}|\mu|^2 < 0$ *unless the group is abelian, in which case it is torsion-free and flat.*

Proof. A nearly parallel structure is Einstein with $\mathrm{Ric} = \frac{3}{8}\tau_0^2 g > 0$ (Proposition 5.15). Left-invariant metrics are complete, so by Myers' theorem G is compact with finite fundamental group, and $\mathfrak{g}$ is compact semisimple. There is no semisimple Lie algebra of dimension 7: the real simple Lie algebras of dimension below 8 have dimension 3 or 6. For nilpotent μ, $\mathrm{Ric}_\mu = M_\mu$ and $\mathrm{scal}_\mu = \mathrm{tr}\, M_\mu = -\frac{1}{4}|\mu|^2$. If $\mu = 0$, then $\nabla = 0$ and $T = 0$. □

Remark 5.33 (Homogeneous examples)**.** On the Heisenberg algebra with the structure of Proposition 5.22, $\mathcal{Q}_\mu = c_0^2\,\mathrm{diag}(x, \dots, x, y)$ with $x = \frac{1}{2} + \frac{a}{6} + \frac{5b}{2}$ and $y = -\frac{3}{2} + \frac{3a}{2} + \frac{5b}{2}$, $X_\mu = 0$, and for every (a, b) the pair is an algebraic soliton with $D = \frac{4}{3}(a - \frac{3}{2})c_0^2\,\mathrm{diag}(1, \dots, 1, 2)$ and $\rho = (\frac{5}{2} - \frac{7a}{6} + \frac{5b}{2})c_0^2$, in agreement with Proposition 5.22; $D = 0$ exactly at $a = \frac{3}{2}$. On the algebra $37A = (0, 0, 0, 0, 12, 23, 24)$ with unit brackets, the co-closed structure of [BFF18, proof of Theorem 17] has $T \star T = \frac{1}{6}I$ and is an algebraic soliton for every (a, b), with $\rho = (a + 7b + 15)/6$ and $D = -\mathrm{diag}(2, 1, 2, 2, 3, 3, 3)$. The bounded catalogue over the nilpotent algebras carrying closed or co-closed structures, with the inherited solitons (co-closed at $a = 0$ with nilsoliton metric, closed at $a = 3$) reported separately, is Remark 5.34.

We now apply these equations to a fixed list of structures in the literature. The unit of the computation is a specified structure, not an isomorphism class of Lie algebras with all its possible forms. The counts and the limitations of negative results below use this distinction.

Remark 5.34 (A finite catalogue of homogeneous solitons)**.** The catalogue uses Theorem 5.27 and Proposition 5.30. Its source list consists of the left-invariant closed or co-closed G_2-structures given in [CF11, Theorem 4 and Lemmas 5–6] (the twelve algebras of its Theorem 7), in [BFF18, Theorems 16–18 and their proofs] and, on the algebras of [BFF18, Theorem 14], by the half-flat frames of [Con11, Table 1], together with samples of the closed families of [Nic18] (its Laplacian solitons and a generic member) and [Lau17, Example 5.28]. Each structure was tested by solving (5.37) together with $D\mu(u, v) = \mu(Du, v) + \mu(u, Dv)$, using exact source coefficients. The calculation comprises 40 nilpotent algebras, 78 rows, of which 70 are valid. On 43 rows (21 algebras) the structure is an exactly certified algebraic or semi-algebraic MHR soliton for some (a, b). Most are inherited: at $a = 0$ whenever $X = 0$ and the metric is a nilsoliton (34 rows), at $a = 3$ for closed Laplacian solitons (10 rows, Proposition 5.26), along the affine span of those two on the closed algebra $(0, 0, 0, 0, 12, 13, 0)$ (the velocity is affine in a), and on $37A$, where $T \star T$ is scalar. Beyond these and the Heisenberg family, 13 rows in 8 invariant classes on 7 algebras have the following MHR soliton parameters:

Algebra (co-closed, source)	a	$\rho(a, b)$ at the source scale
$\mathfrak{n}_2$ (BFF Thm 18)	all	$\frac{3}{2} + \frac{a}{6} + \frac{b}{2}$
$\mathfrak{n}_3$ (BFF Thm 18)	all	$2 + \frac{a}{3} + \frac{2b}{3}$
$\mathfrak{n}_4$ (BFF Thm 18)	all	$2 + \frac{2b}{3}$
$\mathfrak{n}_6$ (BFF Thm 18)	all	$3 - \frac{2a}{3} + \frac{4b}{3}$
$\mathfrak{n}_{13}$ (Conti's frame, BFF Thm 14)	all	$2 + \frac{2a}{3} + 2b$
$37B1$ (BFF Thm 17, repaired)	all	$\frac{7}{20} + \frac{a}{60} + \frac{19b}{60}$
$37D1$ (BFF Thm 17, uniform reading; same class as Thm 16)	all	$\frac{1}{3} - \frac{a}{36} + \frac{2b}{9}$
$37D1$ (BFF's printed multipliers)	$\frac{3}{2}$	$\frac{11}{24} + \frac{b}{3}$

The first seven are co-closed structures with nilsoliton metric, and are algebraic MHR$_{a,b}$ solitons for every (a,b), equivalently $T \star T \in \mathbb{R}I + \mathrm{Der}(\mu)$; they are also semi-algebraic Laplacian coflow solitons on $\mathfrak{n}_2, \mathfrak{n}_3, \mathfrak{n}_4, \mathfrak{n}_6, \mathfrak{n}_{13}$, while the structures on $37B1$ and $37D1$ are not semi-algebraic coflow solitons. The last structure, from the multipliers as printed in the source, is co-closed but not a nilsoliton, and is a (semi-)algebraic MHR soliton only at $a = \frac{3}{2}$; it is neither a Ricci soliton nor a semi-algebraic coflow soliton. All structures and metrics are in the cited literature; the table records their MHR parameters without a priority claim. Eight printed structures fail their sources' own claims (misprints in the TEX sources); seven repaired readings, each the only one tried that meets the source's claims, are hypotheses. This is a bounded search over a fixed list: an empty set of parameters means that the structure is not a (semi-)algebraic soliton; it refers to that structure only, not to arbitrary solitons on the algebra or to other positive forms.

The homogeneous calculations also suggest a useful distinction in the nonhomogeneous setting. The next result tests the velocity at a closed or co-closed structure. Preservation of that class along a solution is a further question, except in the cases already proved above.

Corollary 5.35 (Closed and co-closed G_2-structures on any manifold)**.** *(i) If $d\varphi = 0$, then $\mathcal{F}_{3,b}(\varphi) = \Delta_\varphi\varphi + 3(b+\frac{1}{2})|T|^2\varphi$. Hence every closed Laplacian soliton $\Delta_\varphi\varphi = \lambda\varphi + \mathcal{L}_X\varphi$ is an MHR$_{3,-1/2}$ soliton $(\varphi, X, \lambda/3)$, and an MHR$_{3,b}$ soliton $(\varphi, X, \lambda/3 + (b+\frac{1}{2})|T|^2)$ whenever $|T|^2$ is constant. At $t = 0$, $d\,\partial_t\varphi = 3(b+\frac{1}{2})\,d|T|^2 \wedge \varphi$, so the MHR$_{3,b}$ velocity of closed data is closed if and only if $b = -\frac{1}{2}$ or $|T|^2$ is locally constant. No preservation of closedness is asserted beyond Proposition 5.26.*

(ii) If $d{*}\varphi = 0$, then, in the convention (2.10) (Dwivedi's orientation, $\nabla\varphi = \mathcal{T}\psi$), $(\mathrm{div}\, T) \diamond \varphi = \frac{7}{4}(\nabla\tau_0)\lrcorner\psi$. So the skew MHR velocity of a co-closed structure vanishes if and only if τ_0 is locally constant. This is a statement about the velocity at the given structure.

Proof. (i) The first sentence of Proposition 5.26 (Lotay–Wei's formula, valid pointwise) gives $\Delta\varphi = \mathcal{Q}_{3,-1/2} \diamond \varphi$, and $\mathcal{Q}_{3,b} = \mathcal{Q}_{3,-1/2} + (b+\frac{1}{2})|T|^2 I$ with $I \diamond \varphi = 3\varphi$. For closed φ, $\Delta\varphi = dd^*\varphi$ is exact; $\alpha \mapsto \alpha \wedge \varphi$ is injective on 1-forms, since ${*}(\alpha \wedge \varphi) = \pm\alpha\lrcorner\psi$. (ii) Co-closed means $\tau_1 = \tau_2 = 0$, so $\mathcal{T}$ is symmetric and $V_\mathcal{T} = 0$, where $(V_\mathcal{T})_k = \mathcal{T}_{ij}\varphi_{ijk}$. The general contracted G_2-Bianchi identity, valid on any manifold, reads $\mathrm{div}\,\mathcal{T}^{\mathsf{t}} = \nabla\,\mathrm{tr}\,\mathcal{T} + \mathcal{T}V_\mathcal{T}$ in this convention; it is the contraction of the G_2-Bianchi identity [Kar09, Theorem 4.2 and Corollary 4.4], as used in [DGK25]. Hence $\mathrm{div}\,\mathcal{T} = \nabla\,\mathrm{tr}\,\mathcal{T}$, with $\mathrm{tr}\,\mathcal{T} = \frac{7}{4}\tau_0$ (as $\mathcal{T} = cg$, $\tau_0 = 4c$ in the nearly parallel case). Apply (2.13). □

Lemma 5.36 (Averaging on compact quotients)**.** *Let G be a unimodular Lie group with a lattice Γ, and ξ a left-invariant* H*-structure on $M = \Gamma\backslash G$. If (ξ, Y, ρ) is an MHR$_{a,b}$ soliton on M for some smooth Y, then so is $(\xi, \bar{Y}, \rho)$, where $\bar{Y} = \sum_i (\frac{1}{\mathrm{Vol}}\int_M y^i)e_i$ for $Y = \sum_i y^i e_i$ in a left-invariant frame. For left-invariant X, $\mathcal{L}_X\xi = (-\mathrm{ad}_X) \diamond \xi$, so this means $\mathcal{Q} - \rho I + \mathrm{ad}_X \in \mathfrak{h}$ for some $X \in \mathfrak{g}$.*

Proof. $\mathcal{L}_{ye_i}\xi = y\mathcal{L}_{e_i}\xi + E_{dy,e_i} \diamond \xi$, with $E_{dy,e_i}v = dy(v)e_i$. In the left-invariant frame the coefficients of $\mathcal{S}(\xi, Y, \rho)$ are constants plus constants times y^i plus constants times $e_j(y^i)$. Integrate with respect to dV: $\int_M e_j(h)\,dV = -\int_M h\,\mathrm{div}\,e_j\,dV = 0$, because $\mathrm{div}\,e_j = -\,\mathrm{tr}\,\mathrm{ad}_{e_j} = 0$. Then $(\mathcal{L}_X\xi)(Z_1, \dots) = -\sum_s \xi(\dots, [X, Z_s], \dots)$ for left-invariant Z_s. □

5.8 The $\mathrm{SU}(3)$ dictionary and product comparisons

For an $\mathrm{SU}(3)$-structure the defining forms have different scaling weights. We first express the intrinsic torsion and MHR in this tuple, then compare the resulting equation with the Type IIA flow and with G_2-MHR on a product. These comparisons require the skew velocity as well as the metric equation.

Let (g, J, Ψ) be an $\mathrm{SU}(3)$-structure on M^6, $\omega = g(J\cdot, \cdot)$ and $\Psi = \Psi_+ + i\Psi_- \in \Omega^{3,0}$, modelled on $\omega_0 = e^{12} + e^{34} + e^{56}$ and $\Psi_0 = (e^1 + ie^2) \wedge (e^3 + ie^4) \wedge (e^5 + ie^6)$; so $Je_1 = e_2$, $|\Psi_\pm|^2 = 4$ in the form norm, ${*}\Psi_+ = \Psi_-$ and $X\lrcorner\Psi_+ = (JX)\lrcorner\Psi_-$. As in Papers I–II, J acts on forms by pull-back,

$$(J\alpha)(X_1, \dots, X_k) = \alpha(JX_1, \dots, JX_k),$$

so $J\Psi_+ = \Psi_-$; on 1-forms this is the diamond action. For an arbitrary endomorphism, the diamond action is

$$(A \diamond \alpha)(X_1, \dots, X_k) = \sum_s \alpha(\dots, AX_s, \dots),$$

acting componentwise on the tuple $\xi = (\omega, \Psi_+, \Psi_-)$; 2-tensors and endomorphisms are identified by $A_{ij} = g(Ae_i, e_j)$, and the intrinsic torsion is defined by $\nabla_X \xi = T_X \diamond \xi$, $T_{x,ij} = g(T_{e_x} e_i, e_j)$, $T \star T$, $|T|^2$ and $\operatorname{div} T$ as in (2.2) (tensor norms).

The torsion forms are those of [BV07], in the form of [FFL+26, Appendix A]:

$$\begin{aligned} d\omega &= -\tfrac{3}{2}\sigma_0 \Psi_+ + \tfrac{3}{2}\pi_0 \Psi_- + \nu_1 \wedge \omega + \nu_3, \\ d\Psi_+ &= \pi_0\, \omega^2 + \pi_1 \wedge \Psi_+ - \pi_2 \wedge \omega, \qquad d\Psi_- = \sigma_0\, \omega^2 + J\pi_1 \wedge \Psi_+ - \sigma_2 \wedge \omega, \end{aligned} \tag{5.39}$$

with $\pi_0, \sigma_0 \in \Omega^0$, $\pi_1, \nu_1 \in \Omega^1$, $\pi_2, \sigma_2 \in \Omega^2_8$ and $\nu_3 \in \Omega^3_{12}$, and with norms of forms (divided by $k!$). The following correspondences identify these forms with the customary Chiossi–Salamon torsion classes: $W_1 \leftrightarrow (\pi_0, \sigma_0)$, $W_2 \leftrightarrow (\pi_2, \sigma_2)$, $W_3 \leftrightarrow \nu_3$, $W_4 \leftrightarrow \nu_1$ and $W_5 \leftrightarrow \pi_1$. We write $d^* = -\operatorname{div}$ on 1-forms, $\Sigma^2 = \mathbb{R}g \oplus \Sigma^2_+ \oplus \Sigma^2_-$ for symmetric tensors commuting (Σ^2_+, trace-free) and anticommuting (Σ^2_-) with J, and π_8, π_{12} for the projections onto $\Lambda^2_8, \Lambda^3_{12}$. For $v \in \mathbb{R}^6$, $E_v \in \mathfrak{so}(6)$ denotes the endomorphism of the 2-form $v \lrcorner \Psi_+$.

The dictionary below has four parts: the stabiliser complement, the components of intrinsic torsion, their expression in exterior torsion forms, and the norm conversion. The last two are needed to carry the metric and divergence equations into the same convention.

Proposition 5.37 (The $\mathrm{SU}(3)$ torsion dictionary)**.** *(i)* $\mathfrak{m} = \mathfrak{su}(3)^\perp = \mathfrak{m}_1 \oplus \mathfrak{m}_2$, *where* $\mathfrak{m}_1 = \mathbb{R}J$ *is trivial and* $\mathfrak{m}_2 = \mathfrak{u}(3)^\perp = \{A \in \mathfrak{so}(6) : AJ = -JA\} = \{E_v : v \in \mathbb{R}^6\} \cong \Lambda^2_6 \cong \mathbb{R}^6$ *is the standard real representation; thus* $\mathfrak{m} \cong \mathbf{1} \oplus \mathbf{6}$, *with* $|J|^2 = 6$ *and* $|E_v|^2 = 4|v|^2$. *Accordingly* $\Lambda^1 \otimes \mathfrak{m} = (\Lambda^1 \otimes \mathfrak{m}_1) \oplus (\mathbb{R}^6 \otimes \mathbb{R}^6) = \mathbf{6} \oplus (\mathbf{1} \oplus \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{12} \oplus \mathbf{6})$, 42 *dimensions.*

(ii) Every $T \in \Lambda^1 \otimes \mathfrak{m}$ *is uniquely*

$$T_{x,ij} = \theta_x\, \omega_{ij} + \mathcal{T}_{xp}\, (\Psi_+)_{pij}, \qquad \theta_x = \tfrac{1}{6} T_{x,ij}\omega_{ij}, \quad \mathcal{T}_{xp} = \tfrac{1}{4} T_{x,ij}(\Psi_+)_{pij}, \tag{5.40}$$

with $\theta \in \Lambda^1$ *and* $\mathcal{T} \in \mathbb{R}^6 \otimes \mathbb{R}^6$ *(no symmetry); this is* $T_{l,mn} = -\frac{1}{24}\eta_l J_{mn} + \mathcal{T}_l{}^p (\Psi_+)_{pmn}$ *of* [FFL+26, §2.3, example on the torsion of an $\mathrm{SU}(3)$-structure] *with* $\eta = -24\theta$.

(iii) In the torsion forms (5.39)*,*

$$\begin{aligned} \theta &= \tfrac{1}{3} J(\pi_1 - \nu_1), \\ \mathcal{T}_{xp} &= \tfrac{1}{4}\pi_0\, \delta_{xp} - \tfrac{1}{4}\sigma_0\, \omega_{xp} + \tfrac{1}{4}(\sigma_2)_{xp} - \tfrac{1}{4}\pi_2(e_x, Je_p) + \tfrac{1}{8}(\nu_3)_{xkl}(\Psi_-)_{pkl} - \tfrac{1}{4}(\nu_1)_m (\Psi_+)_{mxp}. \end{aligned} \tag{5.41}$$

The six pieces of $\mathcal{T}$ *lie in* $\mathbb{R}g$, $\mathbb{R}\omega$, Λ^2_8, Σ^2_+, Σ^2_- *and* Λ^2_6 *respectively:* W_1 *is the* $(\mathbb{R}g \oplus \mathbb{R}\omega)$*-part of* $\mathcal{T}$, W_2 *its* $(\Sigma^2_+ \oplus \Lambda^2_8)$*-part,* W_3 *its* Σ^2_-*-part, while the two isotypic* $\mathbf{6}$*'s mix:* ν_1 *enters both the* Λ^2_6*-part of* $\mathcal{T}$ *and the* $\mathfrak{m}_1$*-part* θ.

(iv) Consequently

$$\begin{aligned} |T|^2 &= 6|\theta|^2 + 4|\mathcal{T}|^2 \\ &= \tfrac{3}{2}(\pi_0^2 + \sigma_0^2) + \tfrac{1}{2}\big(|\pi_2|^2 + |\sigma_2|^2 + |\nu_3|^2\big) + \tfrac{5}{3}|\nu_1|^2 + \tfrac{2}{3}|\pi_1|^2 - \tfrac{4}{3}\langle \pi_1, \nu_1 \rangle, \end{aligned} \tag{5.42}$$

where $|T_{\mathfrak{m}_1}|^2 = 6|\theta|^2 = \frac{2}{3}|\pi_1 - \nu_1|^2$ *and* $|T_{\mathfrak{m}_2}|^2 = 4|\mathcal{T}|^2$.

Proof. The orthogonal splitting $\mathfrak{so}(6) = \mathfrak{su}(3) \oplus \mathbb{R}J \oplus \mathfrak{u}(3)^\perp$ and the $\mathrm{SU}(3)$-equivariant injection $v \mapsto v \lrcorner \Psi_+$ give (i). In the displayed standard frame,

$$\omega_{ij}\omega_{ij} = 6, \qquad (\Psi_+)_{pij}(\Psi_+)_{qij} = 4\delta_{pq}, \qquad \omega_{ij}(\Psi_+)_{pij} = 0.$$

These contractions prove (ii). Contracting the intrinsic-torsion formula of [FFL+26, Appendix A, proposition expressing $T(X)$ in torsion forms] with $\omega/6$ and $\Psi_+/4$ gives (iii). The six displayed summands of $\mathcal{T}$ lie in orthogonal irreducible modules. Their squared tensor norms are, respectively,

$$\tfrac{3}{8}\pi_0^2, \quad \tfrac{3}{8}\sigma_0^2, \quad \tfrac{1}{8}|\sigma_2|^2, \quad \tfrac{1}{8}|\pi_2|^2, \quad \tfrac{1}{8}|\nu_3|^2, \quad \tfrac{1}{4}|\nu_1|^2.$$

Adding $6|\theta|^2 = \frac{2}{3}|\pi_1 - \nu_1|^2$ to four times this sum proves (iv), including the mixed term. □

The dictionary gives the quadratic metric terms immediately. For the skew equation we also need its differentiated form; the next result isolates that divergence calculation.

Proposition 5.38 (Quadratic terms and divergence)**.** *In the notation above,*

$$T \star T = 6\theta \otimes \theta + 4\mathcal{T}\mathcal{T}^{\mathsf{t}}, \tag{5.43}$$

$$\operatorname{div} T = (\operatorname{div}\theta)\omega + (\operatorname{div}\mathcal{T})\lrcorner\Psi_+ - (\mathcal{T}^{\mathsf{t}}\theta)\lrcorner\Psi_-, \tag{5.44}$$

where $(\operatorname{div}\mathcal{T})_p = \nabla_x \mathcal{T}_{xp}$ *and* $(\mathcal{T}^{\mathsf{t}}\theta)_p = \theta_x \mathcal{T}_{xp}$*. Moreover,* $\operatorname{div}\theta = -\frac{1}{3}d^*(J\pi_1)$.

Proof. The same contractions give (5.43). Differentiate (5.40) in a normal orthonormal frame. Apart from the two differentiated coefficients, the terms are

$$\theta_x \nabla_x \omega = 2(\mathcal{T}^{\mathsf{t}}\theta)\lrcorner\Psi_-, \qquad \mathcal{T}_{xp}(\nabla_x \Psi_+)_{pij} = -3\big((\mathcal{T}^{\mathsf{t}}\theta)\lrcorner\Psi_-\big)_{ij}.$$

For the second identity use $J \diamond \Psi_+ = -3\Psi_-$; the term quadratic in each row $v = \mathcal{T}(x)$ vanishes because $v\lrcorner(E_v \diamond \Psi_+) = 0$. This proves the divergence formula. Finally $*J\nu_1$ is a constant multiple of $\nu_1 \wedge \omega^2 = d\omega \wedge \omega$. Its exterior derivative vanishes since $d\omega$ has odd degree. Thus $d^*(J\nu_1) = 0$, and the formula for θ concludes. □

The scalar curvature is [BV07, (3.11)]

$$\begin{aligned} R = {} & \tfrac{15}{2}(\pi_0^2 + \sigma_0^2) + 2d^*(\pi_1 + \nu_1) - |\nu_1|^2 + 4\langle \pi_1, \nu_1 \rangle \\ & - \tfrac{1}{2}\big(|\pi_2|^2 + |\sigma_2|^2 + |\nu_3|^2\big). \end{aligned} \tag{5.45}$$

For the trace-free Ricci components we use [FFL+26, Appendix A, Propositions A.6 and A.8] (arXiv v1 numbering): Proposition A.6 gives the Λ^2_8 component and Proposition A.8 the Λ^3_{12} component. In the latter formula (A.3), the term $-2\pi_2 \wedge \pi_1$ must read $-2 * (\pi_2 \wedge \pi_1)$.

Proposition 5.39 (Trace identity and the $\mathrm{SU}(3)$ thresholds)**.** *For MHR$_{a,b}$ on* $\mathrm{SU}(3)$*-structures, with* $\mu = a + 6b$*,*

$$\operatorname{tr}_g\big(-\mathrm{Ric} + aT \star T + b|T|^2 g\big) = -R + \mu|T|^2 = -2d^*(\pi_1 + \nu_1) + q_\mu(T), \tag{5.46}$$

$$\begin{aligned} q_\mu = {} & \tfrac{3}{2}(\mu - 5)(\pi_0^2 + \sigma_0^2) + \tfrac{1}{2}(\mu + 1)(|\pi_2|^2 + |\sigma_2|^2 + |\nu_3|^2) \\ & + \big(1 + \tfrac{5}{3}\mu\big)|\nu_1|^2 + \tfrac{2}{3}\mu|\pi_1|^2 - \big(4 + \tfrac{4}{3}\mu\big)\langle \pi_1, \nu_1 \rangle. \end{aligned} \tag{5.47}$$

The (π_1, ν_1)*-block has determinant* $\frac{2}{3}(\mu^2 - 3\mu - 6)$*, so* q_μ *is positive definite if and only if* $\mu > 5$*, and negative definite if and only if* $\mu < \frac{1}{2}(3 - \sqrt{33})$*. Hence, on a closed manifold, the trace argument yields torsion-freeness of stationary points exactly for these* μ*, as in* [FFL+26, Section 3.3.2, $\mathrm{SU}(3)$ case]*. At* $\mu = 5$*,* $q_5 \geq 0$ *with kernel exactly* W_1*; at* $\mu = \frac{1}{2}(3 - \sqrt{33})$*,* $q_\mu \leq 0$ *with kernel the line* $\pi_1 = \frac{1}{4}(1 - \sqrt{33})\,\nu_1$ *inside* $W_4 \oplus W_5$.

Proof. Subtract (5.45) from μ times (5.42). On each pair of corresponding components of (π_1, ν_1) the matrix is

$$\begin{pmatrix} 2\mu/3 & -2 - 2\mu/3 \\ -2 - 2\mu/3 & 1 + 5\mu/3 \end{pmatrix}.$$

Its determinant is $\frac{2}{3}(\mu^2 - 3\mu - 6)$. Combining its definiteness with the W_1 and $W_2 \oplus W_3$ coefficients gives exactly the stated ranges. At $\mu = 5$ the matrix is positive definite and only the W_1 coefficient vanishes. At $\mu_* = (3 - \sqrt{33})/2$ it is negative semidefinite with the stated kernel, and the other coefficients are negative. On a closed manifold the divergence integrates to zero, so stationarity and definiteness force $T = 0$. □

Corollary 5.40 (The critical value $\mu = 5$)**.** *Let* $a + 6b = 5$ *and* M *be closed and connected. An* $\mathrm{SU}(3)$*-structure is a stationary point of MHR$_{a,b}$ if and only if it is torsion-free or nearly Kähler (torsion in* W_1 *with* (π_0, σ_0) *constant and not both zero). For every* (a, b)*, a nearly Kähler structure has MHR velocity* $\frac{\mu-5}{6}|T|^2\, g$*, and*

$$g(t) = c(t)^2 g_0, \quad \omega(t) = c(t)^2 \omega_0, \quad \Psi_\pm(t) = c(t)^3 \Psi_{\pm,0}, \qquad c(t)^2 = 1 + \tfrac{\mu-5}{3}|T_0|^2\, t, \tag{5.48}$$

is the exact solution; it is a homothetic shrinker for $\mu < 5$ *and an expander for* $\mu > 5$.

Proof. At $\mu = 5$, the integrated trace forces all torsion forms except π_0, σ_0 to vanish. The identities $d^2\Psi_\pm = 0$ then reduce to $d\pi_0 \wedge \omega^2 = d\sigma_0 \wedge \omega^2 = 0$. Wedge multiplication by ω^2 is injective on one-forms, so both functions are constant. Connectedness gives the stated alternatives. Conversely, for constant W_1 torsion the dictionary gives

$$\operatorname{div} T = 0, \qquad T \star T = \tfrac{1}{6}|T|^2 g, \qquad \operatorname{Ric} = \tfrac{5}{6}|T|^2 g.$$

The last identity also follows by setting the other torsion forms to zero in the Ricci formula cited above. Hence the velocity endomorphism is $(\mu - 5)|T|^2 I/6$. Under metric scaling by c^2, $|T|^2$ scales by c^{-2}; applying the weights two and three to the tuple gives (5.48). □

Example 5.41 (A stationary point with torsion at $\mu = -1$)**.** *On $\mathfrak{e}(1,1) \oplus \mathfrak{e}(1,1)$ take the coframe f with $df^1 = -f^{15}$, $df^2 = f^{25}$, $df^3 = f^{46}$, $df^4 = f^{36}$, $df^5 = df^6 = 0$, and the standard ω, $\Psi_\pm$ in this coframe (the symplectic half-flat structure of* [TV08]*). Then $d\omega = d\Psi_+ = 0$, $\sigma_2 = -2(f^{13} + f^{24})$ is the only torsion form, $|T|^2 = 4$, $\operatorname{Ric} = -2(f^5 \otimes f^5 + f^6 \otimes f^6)$ and $T \star T = \operatorname{diag}(1,1,1,1,0,0)$, $\operatorname{div} T = 0$. The MHR$_{2,-1/2}$ velocity vanishes identically, while every other pair with $\mu = -1$ moves ω by the displayed tensors and* (5.43)*. The structure descends to compact quotients of $E(1,1) \times E(1,1)$* [TV08, Theorem 3.5]*; it is the case $C = 0$ of* [FPPZ21, §9.3.2]*, and Ricci-Hermitian in the sense of* [Raf23]*. Thus no condition on μ alone forcing torsion-freeness can include $\mu = -1$.*

We can now write the flow on the defining forms themselves. The scalar part acts with different weights on the two- and three-forms, while the remaining terms separate into their symmetric and skew actions. This form of the equation is the basis for the two comparisons that follow.

Theorem 5.42 (MHR$_{a,b}$ for SU(3)-structures)**.** *Write $A = -\operatorname{Ric} + aT \star T + b|T|^2 g + \operatorname{div} T = f\, g + h_+ + h_- + xJ + E_\alpha$ with $f = \frac{1}{6}(-R + \mu|T|^2)$, $h_\pm = (-\operatorname{Ric} + aT \star T)_\pm$, $x = -\frac{1}{3} d^*(J\pi_1)$ and $\operatorname{div} T = x\,\omega + \alpha^\sharp \lrcorner \Psi_+$. Then MHR$_{a,b}$* (2.4) *for the tuple (ω, Ψ_+, Ψ_-) is*

$$\begin{aligned} \partial_t \omega &= \tfrac{1}{3}\big(-R + \mu|T|^2\big)\omega + h_+ \diamond \omega + 2\,\alpha^\sharp \lrcorner \Psi_-, \\ \partial_t \Psi_+ &= \tfrac{1}{2}\big(-R + \mu|T|^2\big)\Psi_+ + h_- \diamond \Psi_+ + d^*(J\pi_1)\,\Psi_- - 2\,J\alpha \wedge \omega, \\ \partial_t \Psi_- &= \tfrac{1}{2}\big(-R + \mu|T|^2\big)\Psi_- - J(h_- \diamond \Psi_+) - d^*(J\pi_1)\,\Psi_+ - 2\,\alpha \wedge \omega, \end{aligned} \tag{5.49}$$

and for the tuple (g, J, Ψ) of [FFL$^+$26]*: $\partial_t g = -2\operatorname{Ric} + 2aT \star T + 2b|T|^2 g$, $\partial_t J = [J, A] = 2J(h_- + E_\alpha)$, $\partial_t \Psi = A \diamond \Psi$. Here $h_\pm$ are obtained from* (5.43) *and the Ricci tensor, and α from* (5.44)*. In the full-torsion form,*

$$\begin{aligned} \partial_t \xi = &\Big[-\operatorname{Ric} + a(6\theta \otimes \theta + 4\mathcal{T}\mathcal{T}^{\mathsf{t}}) + b(6|\theta|^2 + 4|\mathcal{T}|^2) g\Big] \diamond \xi \\ &+ \Big[(\operatorname{div}\theta)\omega + (\operatorname{div}\mathcal{T}) \lrcorner \Psi_+ - (\mathcal{T}^{\mathsf{t}}\theta) \lrcorner \Psi_-\Big] \diamond \xi. \end{aligned} \tag{5.50}$$

This is the SU(3) *analogue of* (2.14)*; Paper II's $(\lambda, \hat{\lambda}) = (\mu, b)$.*

Proof. The scalar endomorphism acts with weights two and three. In the standard frame, the remaining actions are

$$\begin{aligned} &J \diamond \omega = 0, && J \diamond \Psi_+ = -3\Psi_-, && J \diamond \Psi_- = 3\Psi_+, \\ &E_\alpha \diamond \omega = 2\alpha^\sharp \lrcorner \Psi_-, && E_\alpha \diamond \Psi_+ = -2J\alpha \wedge \omega, && E_\alpha \diamond \Psi_- = -2\alpha \wedge \omega. \end{aligned}$$

The Σ^2_+ component acts trivially on $\Psi_\pm$, and Σ^2_- acts trivially on ω; moreover $h_- \diamond \Psi_- = -J(h_- \diamond \Psi_+)$. Substitution proves the three form equations. Differentiating the endomorphism J gives $\partial_t J = [J, A]$, and (5.50) follows from (5.43)–(5.44). □

Since the stabiliser of $(\omega_0, \Psi_{+,0}, \Psi_{-,0})$ in $\mathrm{GL}(6,\mathbb{R})$ is $\mathrm{SU}(3) \subset \mathrm{SO}(6)$, (5.50) is one fixed equation of the class (2.3). The retained-regularity, continuation and compactness results of Section 4 therefore apply to it independently of the comparisons below.

Comparison with the Type IIA flow. There are two levels of comparison here. The first identifies the normalised velocities on symplectic half-flat data. The second identifies left-invariant solutions after a time change. The final part of the proposition explains why this does not give a general identification of the two flows on closed manifolds.

Proposition 5.43 ($\mathrm{MHR}_{2,-1/2}$ and the Type IIA flow)**.** *Let (ω, Ψ_+) be symplectic half-flat, so its only torsion form is σ_2.*

(i) $\operatorname{div} T = 0$, $(T\star T)_- = 0$, $\mathrm{Ric}_+ \diamond \omega = \frac{1}{2}\pi_8 * (\sigma_2 \wedge \sigma_2)$, $(T\star T)_+ \diamond \omega = \frac{1}{4}\pi_8 * (\sigma_2 \wedge \sigma_2)$, $\mathrm{Ric}_- \diamond \Psi_+ = \pi_{12} d\sigma_2$ *and* $d\sigma_2 = -\frac{1}{4}|\sigma_2|^2 \Psi_+ + \pi_{12} d\sigma_2$*, so that*

$$\partial_t \omega = \tfrac{1+\mu}{6}|\sigma_2|^2 \omega + \big(\tfrac{a}{4} - \tfrac{1}{2}\big)\pi_8 * (\sigma_2 \wedge \sigma_2), \qquad \partial_t \Psi_+ = \tfrac{1+\mu}{4}|\sigma_2|^2 \Psi_+ - \pi_{12} d\sigma_2, \tag{5.51}$$

and $\frac{d}{dt}\mathrm{Vol} = \frac{1+\mu}{2}\int |\sigma_2|^2$.

(ii) For $(a,b) = (2, -\frac{1}{2})$, that is $(\lambda, \hat{\lambda}) = (-1, -\frac{1}{2})$, this is $\partial_t \omega = 0$, $\partial_t \Psi_+ = -\pi_{12} d\sigma_2 = (d\Lambda d \Psi_-)^{\perp \Psi_+}$, which is $|\varphi|^{-2}$ times the velocity of the normalised $\mathrm{SU}(3)$-structure of the source-free Type IIA flow $\partial_t \varphi = d\Lambda d(|\varphi|^2 * \varphi)$, $\varphi = \Psi_+$, of [FPPZ21, (1.4)] *with source $\rho_A = 0$, normalised as $(\omega, 2\varphi/|\varphi|)$. Ricci-Hermitian symplectic half-flat structures are stationary points of $\mathrm{MHR}_{2,-1/2}$.*

(iii) On a Lie group with left-invariant symplectic half-flat initial data, the left-invariant $\mathrm{MHR}_{2,-1/2}$ solution stays symplectic half-flat. More precisely, if $\varphi(t)$ is the source-free Type IIA solution, its normalised tuple, parametrised by $\tau(t) = \int_0^t |\varphi(s)|^2 ds$, solves $\mathrm{MHR}_{2,-1/2}$; the norm is spatially constant.

(iv) On a closed manifold with $\sigma_2 \neq 0$ and $\mu \neq -1$, $\mathrm{MHR}_{a,b}$ changes the volume while the Type IIA flow fixes ω, so no time reparametrisation and diffeomorphism identifies them. For $(a,b) = (2, -\frac{1}{2})$ and data with $\Delta_d |\sigma_2|^2 \not\equiv 0$ (on a closed manifold: $|\sigma_2|$ not constant), $\partial_t \omega(0) = 0$ but $\frac{d}{dt}\langle \partial_t \omega, \omega\rangle|_{t=0} = \frac{1}{2}\Delta_d |\sigma_2|^2$, while the Type IIA flow keeps ω fixed: without diffeomorphism gauge the agreement is first order only.

Proof. With only σ_2 nonzero, the dictionary gives $\theta = 0$ and $\mathcal{T} = \sigma_2/4$. Substitution in the quadratic and Ricci formulae gives (i). The decomposition of $d\sigma_2$ can also be read directly: $d^2\Psi_- = 0$ removes its Λ^3_6 part, and differentiating $\sigma_2 \wedge \Psi_+ = 0$ removes its Ψ_- component. Finally

$$d\sigma_2 \wedge \Psi_- = \sigma_2 \wedge \sigma_2 \wedge \omega = -|\sigma_2|^2 dV$$

gives the coefficient $-|\sigma_2|^2/4$ of Ψ_+. For (ii), $d\Psi_- = -\sigma_2 \wedge \omega$ and the primitive identity $\Lambda(\omega \wedge \sigma_2) = \sigma_2$ give $d\Lambda d\Psi_- = -d\sigma_2$. At $\varphi = \Psi_+$ the norm $|\varphi| = 2$ is constant. Normalising the Type IIA velocity therefore removes precisely its Ψ_+ component and leaves $-|\varphi|^2 \pi_{12} d\sigma_2$. The Ricci-Hermitian case has $\pi_{12} d\sigma_2 = 0$.

For (iii), the Type IIA flow fixes ω and preserves $d\varphi = 0$. Left invariance makes $|\varphi|$ spatially constant, so its normalised tuple remains symplectic half-flat. The preceding velocity calculation and the chain rule give the time change, and uniqueness of the left-invariant ODE identifies the two solutions on $\tau([0, T_{\mathrm{IIA}}))$. This is the whole forward MHR interval. Write $\varphi(t) = \frac{1}{2}c(t)\tilde{\Psi}_+(\tau(t))$, with $\tilde{\Psi}_+$ the normalised tuple, so $c = |\varphi|$. The $\tilde{\Psi}_+$-component of the Type IIA velocity gives $dc/d\tau = \frac{1}{4}|\tilde{\sigma}_2|^2 c$. If $\tau(T_{\mathrm{IIA}})$ were smaller than the maximal MHR time, c would stay bounded and positive, and $\varphi(t)$ would converge to a closed positive form as $t \to T_{\mathrm{IIA}}$, contradicting maximality.

For (iv), the volume formula in (i) excludes an identification when $\mu \neq -1$. At $(a,b) = (2, -1/2)$, put $u = |\sigma_2|^2$. Since $\dot{\omega}(0) = 0$, differentiating the torsion decomposition of $d\omega$ gives $\dot{\nu}_1(0) = 0$. On the other hand,

$$d\dot{\Psi}_+(0) = -\tfrac{1}{4} du \wedge \Psi_+, \qquad \dot{\pi}_1(0) = -\tfrac{1}{4} du.$$

The quadratic form q_{-1} vanishes on W_2 and has zero first variation at these data. Differentiating (5.46) gives $\partial_t \operatorname{tr} A_{\mathrm{sym}}(0) = \frac{1}{2}\Delta_d u$. Since $\langle \dot{\omega}, \omega\rangle = \operatorname{tr} A_{\mathrm{sym}}$, this is the claimed second-order discrepancy. On a closed manifold, $\Delta_d u = 0$ forces u to be constant. □

Products and the two tensor weights. A product with a circle relates the defining forms in six and seven dimensions, but does not automatically identify their MHR velocities. The next calculation displays the correction terms and the precise conditions under which they vanish.

Proposition 5.44 (Products $N^6 \times S^1$ and G_2-MHR)**.** *Let $\varphi = \omega \wedge e^7 + \Psi_+$ on $N^6 \times S^1$ with e^7 the unit parallel circle form. With the convention of* [FLMSE24, (1.10)] *and* [Dwi26, (2.2)], $\mathrm{vol}_\varphi = -e^{1\cdots 7}$ *and $\psi = -\frac{1}{2}\omega^2 - \Psi_- \wedge e^7$; no seven-dimensional Hodge star enters below. Then $T^{\mathrm{G}_2}_X = \theta(X)\,\omega + \frac{2}{3}\,\mathcal{T}(X)\lrcorner\varphi$ for $X \in TN$, $T^{\mathrm{G}_2}_{e_7} = 0$, so*

$$|T^{\mathrm{G}_2}|^2 = 6|\theta|^2 + \tfrac{8}{3}|\mathcal{T}|^2, \qquad T^{\mathrm{G}_2} \star T^{\mathrm{G}_2} = \big(6\theta\otimes\theta + \tfrac{8}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}}\big) \oplus 0,$$

and $(\operatorname{div} T^{\mathrm{G}_2}) \diamond \varphi = X \diamond \varphi$ with $X \in \mathfrak{so}(6)$, $X \equiv (\operatorname{div}\theta)\omega + (\operatorname{div}\mathcal{T})\lrcorner\Psi_+ = \operatorname{div} T^{\mathrm{SU}(3)} + (\mathcal{T}^{\mathsf{t}}\theta)\lrcorner\Psi_-$ modulo $\mathfrak{su}(3)$. Hence G_2-MHR$_{a,b}$ moves g_{77} by $2b|T^{\mathrm{G}_2}|^2$ and keeps the product form only if $b|T^{\mathrm{G}_2}|^2$ is constant on N; for $b = 0$ it induces on N

$$\mathrm{SU}(3)\text{-MHR}_{2a/3,\,0} \;+\; \big[2a\,\theta\otimes\theta + (\mathcal{T}^{\mathsf{t}}\theta)\lrcorner\Psi_-\big] \diamond \xi.$$

For $a \neq 0$, the two velocities agree exactly when $\theta = 0$ ($\pi_1 = \nu_1$), whereas for $a = b = 0$ they agree exactly when $\mathcal{T}^{\mathsf{t}}\theta = 0$; otherwise the two correction terms are the precise obstruction. Dwivedi's parameters $(\frac{9}{2}, -\frac{3}{2})$ (Proposition 2.2) have $b \neq 0$.

Proof. The Λ^2_7-projection sends ω to itself and $v\lrcorner\Psi_+$ to $\frac{2}{3}v\lrcorner\varphi$: the squared tensor norms are $4|v|^2$ and $6|v|^2$, respectively. Project the $\mathrm{SU}(3)$ torsion to obtain T^{G_2} and its two quadratic contractions. Differentiating that projection gives

$$\operatorname{div} T^{\mathrm{G}_2} = (\operatorname{div}\theta)\omega + \tfrac{2}{3}(\operatorname{div}\mathcal{T})\lrcorner\varphi.$$

Indeed, with $v = \mathcal{T}^{\mathsf{t}}\theta$, the two derivatives of the defining forms cancel:

$$\theta_x \nabla_x \omega = 2v\lrcorner\Psi_-, \qquad \tfrac{2}{3}\mathcal{T}_{xp} e_p \lrcorner \nabla_x\varphi = -2v\lrcorner\Psi_-.$$

These follow from the standard-frame actions above and (2.10); the term quadratic in each row of $\mathcal{T}$ contracts a symmetric product against an alternating form and vanishes. Modulo $\mathfrak{g}_2$, $\frac{2}{3}v\lrcorner\varphi \equiv v\lrcorner\Psi_+$. Together with $\mathrm{Ric}_{N\times S^1} = \mathrm{Ric}_N \oplus 0$, this proves the velocity formula. Its two correction terms lie in the symmetric and $\mathfrak{m}$ summands, whose actions are independent. Thus for $a \neq 0$ equality forces $\theta = 0$; for $a = 0$ it forces only $\mathcal{T}^{\mathsf{t}}\theta = 0$. These are pointwise velocity comparisons; any identification along trajectories also requires the relevant condition to persist. □

The two scaling weights also matter when a norm is placed on the defining tuple. The next proposition concerns that choice of pairing; it does not change the intrinsic torsion or the MHR equation.

Proposition 5.45 (Two weights for the tuple)**.** *Give the tuple the $\mathrm{SO}(6)$-invariant norm $w_\omega|\omega|^2 + w_+|\Psi_+|^2 + w_-|\Psi_-|^2$ (form norms) and $\mathfrak{m}$ the tensor norm of (2.2). Then for $A, B \in \mathfrak{m}$*

$$\langle A\diamond\xi, B\diamond\xi\rangle = \lambda_1\langle A_{\mathfrak{m}_1}, B_{\mathfrak{m}_1}\rangle + \lambda_2\langle A_{\mathfrak{m}_2}, B_{\mathfrak{m}_2}\rangle, \qquad \lambda_1 = 6(w_+ + w_-), \quad \lambda_2 = 2(w_\omega + w_+ + w_-),$$

so $|\nabla\xi|^2 = 6\lambda_1|\theta|^2 + 4\lambda_2|\mathcal{T}|^2$. The single-weight identity $\langle A\diamond\xi, B\diamond\xi\rangle = c_\diamond\langle A, B\rangle$ holds iff $w_\omega = 2(w_+ + w_-)$ (e.g. $4|\omega|^2 + |\Psi_+|^2 + |\Psi_-|^2$, $c_\diamond = 12$); for equal weights $(\lambda_1, \lambda_2) = (12, 6)$, and for Paper I's (J, Υ) $(\lambda_1, \lambda_2) = (12, 8)$ [FLMSE24, Example 1.16]*, restored by $2|J|^2 + |\Upsilon|^2$. None of T, $T\star T$, $|T|^2$, $\operatorname{div} T$ or (2.4) depends on this choice.*

Proof. The two irreducible summands of $\mathfrak{m}$ are inequivalent, so the pairing is diagonal and scalar on each. The standard-frame actions give $|J\diamond\Psi_\pm|^2 = 36$ and $|E_v\diamond\omega|^2 = |E_v\diamond\Psi_\pm|^2 = 8|v|^2$. Divide by $|J|^2 = 6$ and $|E_v|^2 = 4|v|^2$, respectively. The two scalars coincide exactly under the stated weight condition. □

Remark 5.46 (The projected Taylor term for the tuple)**.** The $\mathrm{Spin}(7)$ identity in Corollary B.3 does not extend to this tuple. Give it positive weights with $w_+ = w_-$ and set $\mathcal{B}(T) = \frac{1}{2}\sum_p T_p \diamond (T_p \diamond \xi)$. There are no constants α, β such that its tangent projection equals $(\alpha T\star T + \beta|T|^2 g + C)\diamond\xi$ for all T, even allowing a skew term C.

To see this, take only $T_1 = J$ nonzero. Then $\mathcal{B} = (0, -\frac{9}{2}\Psi_+, -\frac{9}{2}\Psi_-)$ is $\mathrm{SU}(3)$-invariant, so its symmetric tangent component is scalar. But $T \star T = 6e^1 \otimes e^1$, forcing $\alpha = 0$. Next take only $T_1 = E_{e_1}$ nonzero, where $E_{e_1} = e^{35} - e^{46}$. For $H = \mathrm{diag}(-2, -2, 1, 1, 1, 1)$ one has

$$\mathcal{B}_\omega = -2(e^{34} + e^{56}), \quad H \diamond \omega = -4e^{12} + 2e^{34} + 2e^{56}, \quad H \diamond \Psi_\pm = 0.$$

Hence $\langle \mathcal{B}, H \diamond \xi \rangle = -8w_\omega \neq 0$. The direction $H \diamond \xi$ is orthogonal to the scalar and skew tangent summands, contradicting $\alpha = 0$.

Compact endpoints and sharpness. We finish the soliton analysis by returning to the trace restrictions. At an endpoint, vanishing of the integrated quadratic form leaves specific torsion components, which must still satisfy the full soliton equation. The explicit families then determine which exclusions are sharp.

Proposition 5.47 (Endpoint calculations)**.** *Let M be closed and put $\mu_* = (3 - \sqrt{33})/2$, with $\mu = a + 6b$ for $\mathrm{SU}(3)$ and $\mu = a + 8b$ for $\mathrm{Spin}(7)$.*

(i) ($\mathrm{SU}(3)$.) There are no expanders for $\mu \leq \mu_*$ and no shrinkers for $\mu \geq 5$, and steady solitons are torsion-free for $\mu < \mu_*$ or $\mu > 5$. At $\mu = 5$ a steady soliton $(\xi, Y, 0)$ has $\mathcal{L}_Y \xi = 0$ and is, on each connected component of M, torsion-free or nearly Kähler; conversely, every such structure gives a steady soliton. At $\mu = \mu_*$ a steady soliton has $W_1 = W_2 = W_3 = 0$ and $\pi_1 = \frac{1}{4}(1 - \sqrt{33})\nu_1$.

(ii) ($\mathrm{Spin}(7)$, $\mu = 12$.) A steady soliton has $T^5_{48} = 0$, closed Lee form ($dT^1_8 = 0$, so it is locally conformally parallel), $\operatorname{div} T = 0$ and $\mathcal{K}_\Phi(Y) = 0$.

Proof. (i) Apply Proposition 5.6 with (5.46)–(5.47): q_μ is negative semidefinite for $\mu \leq \mu_*$, positive semidefinite for $\mu \geq 5$, and definite (hence coercive) in the open ranges. At $\mu = 5$, $\int q_5 = 0$ forces $T \in W_1$; the proof of Corollary 5.40 gives (π_0, σ_0) constant on each connected component; the velocity vanishes at $\mu = 5$; and injectivity on $\Sigma^2 \oplus \mathfrak{m}$ (the $\mathfrak{gl}(6)$-stabiliser of the tuple is $\mathfrak{su}(3)$) turns (5.6) into $\mathcal{L}_Y \xi = 0$. At μ_*, $q_{\mu_*} \leq 0$ with $\int q_{\mu_*} = 0$ puts T in the kernel described in Proposition 5.39. (ii) $\int q_{12} = \frac{7}{2} \int |T^5_{48}|^2 = 0$. Then $d\Phi = T^1_8 \wedge \Phi$ and $0 = d^2\Phi = dT^1_8 \wedge \Phi$; the map $\beta \mapsto \beta \wedge \Phi$ is injective on Λ^2 (eigenvalues 3 and -1 of $*(\cdot \wedge \Phi)$ on $\Lambda^2_7 \oplus \Lambda^2_{21}$). Locally conformally parallel structures are harmonic [FSE24, arXiv:2301.12494v2, Corollary 3.4] (pointwise; exactness of the Lee form is irrelevant), and the skew equation gives $\mathcal{K}_\Phi(Y) = 0$. □

Corollary 5.48 (Sharpness on closed manifolds)**.** *(i) The closed contact Calabi–Yau family gives the solitons*

$$(\varphi_{f,h}, 0, \rho) \quad \text{of } MHR_{3/2,b}, \qquad \mu = \tfrac{3}{2} + 7b, \qquad \rho = \frac{5\mu + 3}{14} c^2.$$

It is a shrinker, stationary or an expander according as $\mu < -\frac{3}{5}$, $\mu = -\frac{3}{5}$ or $\mu > -\frac{3}{5}$. It is stationary with nonzero torsion at $(a, b) = (\frac{3}{2}, -\frac{3}{10})$.

(ii) On compact quotients of $E(1,1) \times E(1,1)$ the Tomassini–Vezzoni structure of Example 5.41 is a homothetic $MHR_{2,b}$ soliton for every b, with $\rho = 2 + 4b = \frac{2}{3}(\mu + 1)$ at the scale of that example; it is stationary exactly at $\mu = -1$, that is $b = -\frac{1}{2}$. It is a soliton for some Y only when $a = 2$.

(iii) Consequently: for G_2, no condition on μ alone forcing steady solitons to be torsion-free can include $\mu = -\frac{3}{5}$ or $\mu = 9$, and closed expanders exist for every $\mu > -\frac{3}{5}$. For $\mathrm{Spin}(7)$, closed shrinkers exist for every $\mu < 12$ and expanders for every $\mu > 12$. For $\mathrm{SU}(3)$, closed shrinkers exist for every $\mu < 5$ and expanders for every $\mu > -1$. The shrinker exclusions $\mu \geq 9, 12, 5$ are therefore sharp.

Proof. (i) Proposition 5.21(iii). (ii) Exact computation: $\mathrm{Ric} = -2(f^5 \otimes f^5 + f^6 \otimes f^6)$, $T \star T = \mathrm{diag}(1, 1, 1, 1, 0, 0)$, $|T|^2 = 4$, $\operatorname{div} T = 0$, so $\mathcal{Q}_{a,b} = \mathrm{diag}(a, a, a, a, 2, 2) + 4bI$. For $a = 2$ this is $(2 + 4b)I$; for $a \neq 2$, average a hypothetical field using Lemma 5.36. For every left-invariant X, $(\mathrm{ad}_X)_{55} = (\mathrm{ad}_X)_{66} = 0$ and $(\mathrm{ad}_X)_{11} + (\mathrm{ad}_X)_{22} = 0$ in this coframe. The fifth diagonal soliton equation gives $\rho = 2 + 4b$; the sum of the first two gives $2(a + 4b) = 2\rho$, hence $a = 2$. (iii) Combine (i), (ii), Proposition 5.15, Corollary 5.18 and Corollary 5.40. □

Remark 5.49 (Completeness and soliton type)**.** The homothetic nearly parallel and nearly Kähler seeds above are taken closed. The cylindrical solitons are complete by Proposition 5.17, and their fields are affine in the line coordinate. Every left-invariant metric on a connected Lie group is complete; the fields in Proposition 5.30 are complete because they generate automorphisms. This covers the Heisenberg group, $37A$ and the catalogue. Their soliton type is the sign of the stated ρ. The Heisenberg soliton has no gradient representation when $a \neq 3/2$, by Proposition 5.22; no such assertion is made for the other catalogue rows. Descent to compact quotients is addressed separately in Proposition 5.21 and Lemma 5.36.

6 Variational obstructions for MHR

The Dirichlet energy of Weiß–Witt has a well-posed negative gradient flow [WW12, Definition 4.1 and Theorem 5.1], and the variations of the irreducible torsion energies, including the total torsion-energy flow, are computed in [DGK25, Corollary 5.36 and Remark 6.84]. Gianniotis–Zacharopoulos construct the G_2-Hilbert functional and its product-metric gradient [GZ26, (4.2) and Proposition 4.2]. These variational flows provide the comparison for the question considered here: whether MHR itself has such a representation.

We ask whether MHR is the negative gradient of a scalar-curvature and quadratic-torsion functional, up to a diffeomorphism term. For the natural constant-coefficient choices described below, the answer is negative. For G_2, three pointwise torsion tests expose an incompatibility between the trace-free metric equation and the torsion-divergence equation. For $\mathrm{Spin}(7)$, exact variations on nilmanifolds give the obstruction directly. In both cases, evaluation at a constant weight gives the fixed-measure weighted obstruction.

6.1 The G_2 obstruction

For the G_2 argument below, M is closed and admits a G_2-structure. We use the full torsion convention $\nabla_i \varphi_{jkl} = \mathcal{T}_{im}\psi_{mjkl}$, where $\psi = *_\varphi\varphi$. Its irreducible decomposition is

$$\mathcal{T} = \mathcal{T}_1 + \mathcal{T}_7 + \mathcal{T}_{14} + \mathcal{T}_{27},$$
$$\mathcal{T}_1 = \frac{\operatorname{tr}\mathcal{T}}{7} g, \qquad \mathcal{T}_{27} = \frac{\mathcal{T} + \mathcal{T}^{\mathsf{t}}}{2} - \mathcal{T}_1.$$

The skew part splits orthogonally into $\mathcal{T}_7 \in \Lambda^2_7$ and $\mathcal{T}_{14} \in \Lambda^2_{14}$, where

$$\Lambda^2_7 = \{v \lrcorner \varphi : v \in TM\}, \qquad \Lambda^2_{14} = \{A \in \Lambda^2 : A_{ij}\varphi_{ijk} = 0\}.$$

Thus $\mathcal{T}_1$ is the scalar part, $\mathcal{T}_{27}$ is symmetric and trace-free, and the remaining two components are skew. We write $A_{\mathrm{sym}} = (A + A^{\mathsf{t}})/2$, use subscript 0 for the trace-free part of a symmetric tensor, and form matrix products using g. Set

$$(V_{\mathcal{T}})_k = \mathcal{T}_{ij}\varphi_{ijk}, \qquad \mathsf{P}\mathcal{T} = -4\mathcal{T}_7 + 2\mathcal{T}_{14},$$

as in [DGK25, (2.59)–(2.64)].

A variation is written

$$\dot\varphi = h \diamond \varphi + X \lrcorner \psi, \qquad \dot g = 2h,$$

with h symmetric and X a vector field. In these variables the MHR vector field $\mathcal{F}_{a,b}$ has components

$$h_{a,b} = -\mathrm{Ric} + \frac{2a}{3}\mathcal{T}\mathcal{T}^{\mathsf{t}} + \frac{2b}{3}|\mathcal{T}|^2 g, \qquad X_{a,b} = \operatorname{div}\mathcal{T}.$$

We test gradient representations with respect to the zeroth-order bilinear forms

$$\mathsf{G}_{p,q,r}\big((h,X),(k,Z)\big) = \int_M \big(p\langle h,k\rangle + q(\operatorname{tr} h)(\operatorname{tr} k) + r\langle X,Z\rangle\big)\, dV. \tag{6.1}$$

Here p, q, r are constants. The eigenvalues on the trace-free, scalar, and vector parts are respectively p, $p + 7q$, and r. Consequently, the form is nondegenerate when these three numbers are nonzero, and positive definite when they are positive. In particular, q is a trace correction. This is the general constant-coefficient fibrewise G_2-invariant bilinear form on the variation space: the summands $\mathbb{R}g$, Sym^2_0, and Λ^1 are pairwise inequivalent

irreducible real representations. Likewise, the vector representation occurs once in the torsion module, so a natural vector field linear in torsion is $cV_{\mathcal{T}}$.

For the product pairing $p = r = 1$, $q = 0$, the obstruction already follows from [GZ26, Proposition 3.3]. Indeed, [DGK25, Corollary 5.18 and (5.30)] identifies the principal part of $\mathcal{F}_{a,b} - \mathcal{L}_{cV_{\mathcal{T}}}\varphi$ as

$$\left(-\mathrm{Ric} - \tfrac{c}{2}\mathcal{L}_{V_{\mathcal{T}}}g\right) \diamond \varphi + \left((1 - \tfrac{c}{2})\,\mathrm{div}\,\mathcal{T} + \tfrac{c}{2}\nabla\,\mathrm{tr}\,\mathcal{T}\right) \lrcorner \psi.$$

This is their special Ricci-like principal part, with no scalar-curvature trace term, which Proposition 3.3 excludes as a gradient. Their statement permits a wider functional class but fixes the pairing. The following result allows every nondegenerate pairing (6.1), for the specified quadratic functionals; the exact lower-order terms are needed to obtain this conclusion.

Theorem 6.1 (Obstruction for quadratic torsion functionals)**.** *Fix $a, b \in \mathbb{R}$. There are no constants $\alpha, \beta_1, \beta_7, \beta_{14}, \beta_{27}, c, p, q, r$, with $\mathsf{G}_{p,q,r}$ nondegenerate, for which*

$$\mathcal{F}_{a,b}(\varphi) = -\,\mathrm{grad}_{\mathsf{G}_{p,q,r}}\,\mathcal{W}(\varphi) + \mathcal{L}_{cV_{\mathcal{T}}}\varphi \tag{6.2}$$

holds for every smooth G_2-structure, where

$$\mathcal{W}(\varphi) = \int_M \left(\alpha R + \sum_{k\in\{1,7,14,27\}} \beta_k |\mathcal{T}_k|^2\right) dV.$$

More strongly, the variational identity obtained from (6.2) by pairing with arbitrary variations forces $r = 0$, without any assumption on p or q.

Proof. We first compare the independent terms containing second derivatives of the structure. Their coefficients restrict the functional. We then test the remaining quadratic torsion identity on the two skew summands and on symmetric torsion.

Write the first variation as

$$D\mathcal{W}_\varphi(h, X) = \int_M \big(\langle \ell, h\rangle + \langle Z, X\rangle\big)\, dV.$$

The required identity is equivalent to the pointwise equations

$$p h_{a,b} + q(\mathrm{tr}\, h_{a,b})g - \frac{pc}{2}\mathcal{L}_{V_{\mathcal{T}}}g - qc(\mathrm{div}\, V_{\mathcal{T}})g + \ell = 0, \tag{6.3}$$

$$r\,\mathrm{div}\,\mathcal{T} - rc\mathcal{K}(V_{\mathcal{T}}) + Z = 0. \tag{6.4}$$

Here $\mathcal{K}(V_{\mathcal{T}})$ is the vector component of $\mathcal{L}_{V_{\mathcal{T}}}\varphi$; explicitly, [DGK25, (5.30)] gives

$$\mathcal{K}(V_{\mathcal{T}}) = \frac{1}{2}\left(\mathrm{div}\,\mathcal{T} - \mathrm{div}\,\mathcal{T}^{\mathsf{t}} + \mathcal{T}^{\mathsf{t}}V_{\mathcal{T}} + \mathcal{T}V_{\mathcal{T}}\right).$$

The fundamental lemma applies because all smooth variations of a positive 3-form are allowed.

The six leading differential invariants $Rg, \mathrm{Ric}, F, \mathcal{L}_{V_{\mathcal{T}}}g, \mathrm{div}\,\mathcal{T}, \mathrm{div}\,\mathcal{T}^{\mathsf{t}}$ are independent modulo torsion-quadratic terms [DGK25, Theorems 5.66 and 6.2]. We use the independent trace-free symmetric tensors Ric_0, F_0 and $(\mathcal{L}_{V_{\mathcal{T}}}g)_0$, together with the two vector divergences. Here the φ-Ricci tensor is $F_{pq} = R_{ijkl}\varphi_{ijp}\varphi_{klq}$ in an orthonormal frame, with the curvature convention of [DGK25, Definition 2.117 and (2.118)]. For $E_k = \int_M |\mathcal{T}_k|^2\, dV$, the coefficients needed below in the symmetric and vector first variations are [DGK25, Corollary 5.36]

	E_1	E_7	E_{14}	E_{27}
F	0	0	$-\frac{1}{4}$	$\frac{1}{4}$
$\mathrm{div}\,\mathcal{T}$	0	$-\frac{1}{3}$	$-\frac{2}{3}$	-1
$\mathrm{div}\,\mathcal{T}^{\mathsf{t}}$	$-\frac{2}{7}$	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{5}{7}$

.

The Einstein–Hilbert first variation contributes only $\alpha(Rg - 2\mathrm{Ric})$ to ℓ. The coefficient of F_0 in the trace-free part of (6.3) therefore forces $\beta_{27} = \beta_{14} = t$. Define

$$B = \frac{5t + \beta_7}{3}, \qquad C = -\frac{2(\beta_1 + 6t)}{7}.$$

The two independent divergence coefficients in (6.4) then give

$$r\left(1-\frac{c}{2}\right)-B=0, \qquad \frac{rc}{2}+B+C=0. \tag{6.5}$$

All leading coefficients in the trace-free symmetric equation must vanish. Substituting $\beta_{27}=\beta_{14}=t$ into the quadratic terms of [DGK25, Corollary 5.36], its remaining trace-free part is

$$\left[\left(\frac{2pa}{3}-B\right)\mathcal{T}\mathcal{T}^{\mathsf{t}}+B(\mathcal{T}^2)_{\mathrm{sym}}-B\big(\mathcal{T}(\mathsf{P}\mathcal{T})\big)_{\mathrm{sym}}+C(\operatorname{tr}\mathcal{T})\mathcal{T}_{\mathrm{sym}}\right]_0=0. \tag{6.6}$$

The other nonscalar quadratic terms in the source cancel in pairs when $\beta_{27}=\beta_{14}$. Every term omitted from (6.6) is a scalar multiple of g.

We may now prescribe $\mathcal{T}$ arbitrarily at a point. To see this, keep a background metric fixed and act on the G_2-form by a local map into $\mathrm{SO}(7)$ whose value at the point is the identity. In each coordinate direction, its derivative can be prescribed independently in $\mathfrak{so}(7)$. The orbit differential identifies $\mathfrak{so}(7)/\mathfrak{g}_2$ with Λ^3_7, so these derivatives prescribe the seven rows of $\mathcal{T}$. Multiplying a local Lie-algebra parameter by a cut-off before exponentiating realises the prescribed jet in a global structure. Hence (6.6) holds for every torsion matrix.

First choose a nonzero $\mathcal{T}\in\Lambda^2_{14}$. Then $\mathcal{T}^{\mathsf{t}}=-\mathcal{T}$ and $\mathsf{P}\mathcal{T}=2\mathcal{T}$. Thus

$$\mathcal{T}\in\Lambda^2_{14}\setminus\{0\}: \qquad -\frac{2pa}{3}(\mathcal{T}^2)_0=0 \qquad\Longrightarrow\qquad pa=0.$$

For the implication, a nonzero skew matrix in odd dimension has a kernel and cannot have scalar square; hence $(\mathcal{T}^2)_0\neq 0$.

Next choose nonzero $\mathcal{T}\in\Lambda^2_7$. Now $\mathsf{P}\mathcal{T}=-4\mathcal{T}$. Using $pa=0$ and the same skew-matrix observation gives

$$\mathcal{T}\in\Lambda^2_7\setminus\{0\}: \qquad 6B(\mathcal{T}^2)_0=0 \qquad\Longrightarrow\qquad B=0.$$

Finally take symmetric $\mathcal{T}$ with $\operatorname{tr}\mathcal{T}\neq 0$ and $\mathcal{T}_0\neq 0$. Since $\mathsf{P}\mathcal{T}=0$, we obtain

$$\mathcal{T}=\mathcal{T}^{\mathsf{t}}: \qquad C(\operatorname{tr}\mathcal{T})\mathcal{T}_0=0 \qquad\Longrightarrow\qquad C=0.$$

Equations (6.5) now imply $rc=0$ and $r=0$, proving the stronger assertion and contradicting nondegeneracy of the vector block. □

Only the trace-free symmetric and vector equations were used. This also yields a weighted obstruction: at a constant weight, the extra terms affect only the trace equation.

Corollary 6.2 (Fixed-measure weighted obstruction)**.** *For a pair (φ,f) set $dm=e^{-f}dV$ and take variations preserving dm, so that $\dot f=\operatorname{tr}h$. Replace dV by dm in (6.1). There are no constants with $r\neq 0$ (and arbitrary p,q) for which the first-variation identity associated with*

$$\mathcal{F}_{a,b}(\varphi)=-\operatorname{grad}_{\mathsf{G}_{p,q,r}}\mathcal{W}_f(\varphi,f)+\mathcal{L}_{cV_{\mathcal{T}}+d\nabla f}\varphi$$

holds for every smooth pair, where

$$\mathcal{W}_f=\int_M\left(\alpha R+\sum_{k\in\{1,7,14,27\}}\beta_k|\mathcal{T}_k|^2+\gamma|\nabla f|^2+\kappa\langle V_{\mathcal{T}},\nabla f\rangle\right)dm. \tag{6.7}$$

For a nondegenerate pairing, this is the usual gradient on the fixed-measure variations, identified with (h,X). For a degenerate symmetric block the assertion means that $\mathsf{G}_{p,q,r}\big(\mathcal{F}_{a,b}(\varphi)-\mathcal{L}_{cV_{\mathcal{T}}+d\nabla f}\varphi,\cdot\big)$ cannot equal $-d\mathcal{W}_f$ on all fixed-measure variations.

Proof. Evaluate the proposed universal identity at a locally constant function f. This also tests an identity on any prescribed positive measure: Moser's volume-form theorem [Mos65, Theorem, pp. 286–287], applied on each connected component, identifies that measure with a constant multiple of dV_φ. Naturality pulls the identity back to this constant-weight pair. For the first five terms of (6.7), the fixed-measure variation differs from the unweighted variation only by removing the volume term $L(\operatorname{tr}h)$, where $L=\alpha R+\sum_k\beta_k|\mathcal{T}_k|^2$. This changes

only the trace of the symmetric Euler–Lagrange tensor. The first variation of $|\nabla f|^2$ vanishes. The coupling term contributes

$$\kappa \int_M \langle V_{\mathcal{T}}, \nabla \dot{f} \rangle \, dm = -\kappa \int_M (\operatorname{div} V_{\mathcal{T}})(\operatorname{tr} h) \, dm,$$

again a pure-trace term. Also $d\nabla f = 0$ at this pair. Consequently, the trace-free symmetric equation and the vector equation used in the preceding proof are unchanged. They again force $r = 0$, a contradiction. □

On a closed manifold, adding constant multiples of Δf or $\operatorname{div} V_{\mathcal{T}}$ does not enlarge this weighted class, since

$$\int_M \Delta f \, dm = \int_M |\nabla f|^2 \, dm, \qquad \int_M \operatorname{div} V_{\mathcal{T}} \, dm = \int_M \langle V_{\mathcal{T}}, \nabla f \rangle \, dm.$$

The conclusions concern universal gradient identities for the displayed functionals, metrics, and diffeomorphism generators. They do not exclude a functional evaluated only at a minimising weight, or a monotonicity formula of another form.

6.2 The $\mathrm{Spin}(7)$ obstruction

For $\mathrm{Spin}(7)$, five variations on one compact nilmanifold rule out a nondegenerate pairing. Additional tests, using a second nilmanifold, determine the degenerate exception. This argument computes only the first variations needed for the tests. Dwivedi's torsion-energy gradient flow [Dwi25, Definition 1.2, Lemma 3.3 and (3.5)] provides the relevant comparison; Duthie describes its additional torsion-induced Lie derivative in [Dut26, Remark 3.4 and (3.21)].

Write $\dot{\Phi} = (h + X) \diamond \Phi$, with h symmetric and $X \in \Lambda^2_7$, and use full index sums for the norms of X and T. The general invariant pairing again has the form (6.1); its three eigenvalues are now p, $p + 8q$ and r. Set $V_k = T_{i,ik}$ and write $T = T_8 + T_{48}$ for the irreducible torsion decomposition.

The proof first obtains the obstruction to a nondegenerate pairing. Only after that conclusion does it use further tests to determine the remaining degenerate possibility. Thus the classification of that exception is separate from the nonexistence assertion.

Proposition 6.3 ($\mathrm{Spin}(7)$ obstruction and the degenerate exception)**.** *Let*

$$\mathcal{W}(\Phi) = \int_M (\alpha R + \beta_8 |T_8|^2 + \beta_{48} |T_{48}|^2) \, dV.$$

If the universal variational identity

$$\mathsf{G}_{p,q,r}(\mathcal{F}_{a,b} - \mathcal{L}_{cV}\Phi, \dot{\Phi}) = -D\mathcal{W}_\Phi(\dot{\Phi}) \tag{6.8}$$

holds for every smooth structure and variation, then $r = 0$. If $\mathsf{G}_{p,q,r} \neq 0$, it holds precisely when

$$a = b = c = 0, \qquad p \neq 0, \qquad q = -\tfrac{p}{2}, \qquad r = 0, \qquad \mathcal{W} = -\tfrac{p}{2} \int_M R \, dV,$$

up to adding a multiple of the null functional $\int_M (R - 12|T_8|^2 + 2|T_{48}|^2) \, dV$. In particular, no $\mathrm{MHR}_{a,b}$ has such a gradient representation for a nondegenerate invariant pairing.

Proof. The identity is local after integration by parts, so any universal identity must hold on compact quotients of left-invariant test structures. Equivalently, their local jets can be inserted in an orbit chart on any fixed closed manifold admitting a $\mathrm{Spin}(7)$-structure. We give the necessary equations explicitly.

In an oriented orthonormal coframe use

$$\Phi_0 = \tfrac{1}{2}\omega^2 + \operatorname{Re}\Omega, \qquad \omega = e^{12} + e^{34} + e^{56} + e^{78}, \qquad \Omega = \bigwedge_{j=1}^{4} (e^{2j-1} + \mathrm{i}e^{2j}).$$

Our first Lie algebra has the following nonzero brackets and their skew counterparts:

$$\begin{gathered} [e_1, e_2] = z_1, \quad [e_1, e_3] = z_2, \quad [e_1, e_6] = -z_2, \quad [e_2, e_3] = z_3, \quad [e_2, e_6] = -z_3, \\ z_1 = e_4 + e_7, \qquad z_2 = e_5 + e_8, \qquad z_3 = e_3 + e_6 - e_7. \end{gathered} \tag{6.9}$$

The z_i are central. The second algebra has $[e_1,e_2]=[e_3,e_4]=[e_5,e_6]=e_7$ and no other nonzero brackets. Both are two-step nilpotent with integral structure constants and admit compact quotients. Equip these quotients with the indicated left-invariant Cayley forms.

Here is a direct way to verify every coefficient used below. Put $c_{ijk}=\langle[e_i,e_j],e_k\rangle$ and define

$$\Gamma_{ijk}=\tfrac12(c_{ijk}-c_{ikj}-c_{jki}),\qquad K(A)_{ij}=\tfrac12(\Phi_0)_{ijkl}A_{kl},\qquad \pi_7(A)=\tfrac14(A+K(A))$$

for skew A. The Cayley contractions give $K^2=2K+3\,\mathrm{Id}$, $T_i=-\pi_7(\Gamma_i)$ and $|T_8|^2=\frac87|V|^2$; thus $|T_{48}|^2=|T|^2-\frac87|V|^2$. On these unimodular algebras,

$$\begin{aligned}R&=-\tfrac14\sum_{i,j,k}c_{ijk}^2-\tfrac12\sum_{i,j,k}c_{ijk}c_{ikj},\\ \mathrm{Ric}_{jl}&=-\tfrac12\sum_{i,k}c_{jik}c_{lik}+\tfrac14\sum_{i,k}c_{ikj}c_{ikl}-\tfrac12\sum_{i,k}c_{jik}c_{lki}.\end{aligned}$$

For $Q(t)=\mathrm{Id}+tA$, the moving orthonormal frame has brackets

$$c(t)_{abk}=(Q^{-1})_{ai}(Q^{-1})_{bj}c_{ijm}Q_{mk}.$$

Consequently, with $d_A=\partial_t|_0$,

$$d_Ac_{abk}=-A_{ai}c_{ibk}-A_{bj}c_{ajk}+c_{abm}A_{mk},\qquad d_A(dV)=(\operatorname{tr}A)dV. \tag{6.10}$$

Differentiating the displayed quadratic densities therefore computes $D\mathcal{W}$ exactly, per unit volume. For the other side of (6.8), use

$$\nabla_iV_j=-\Gamma_{ijm}V_m,\qquad h_V=(\nabla V)_{\mathrm{sym}},\qquad X_V=\pi_7\big((\nabla V)_{\mathrm{skew}}+V^iT_i\big),$$

and compute $\operatorname{div}T$ by covariantly differentiating its three indices. These formulae reduce the tests to finite sums of rational numbers.

Let E_{ij} denote an elementary matrix. For $i<j$ put $H_{ij}=E_{ij}+E_{ji}$ and $M_{ij}=(\mathrm{Id}+K)(E_{ij}-E_{ji})\in\Lambda^2_7$. Write

$$A_0=pa,\qquad C_0=pc,\qquad B_0=pb+q(a+8b),\qquad S=\beta_8+6\beta_{48}.$$

On the first algebra, the symmetric variations H_{12},H_{14},H_{18} and skew variations M_{12},M_{14} give, respectively,

$$\begin{aligned}-\tfrac34A_0+\tfrac{3}{28}S&=0,\\ -\tfrac14A_0+\tfrac18C_0-\tfrac{1}{28}S&=0,\\ \tfrac14A_0-\tfrac18C_0-\tfrac{1}{14}S&=0,\\ -\tfrac12r+\tfrac17S&=0,\\ -\tfrac12r+\tfrac18rc+\tfrac17S&=0.\end{aligned} \tag{6.11}$$

The first three equations force $A_0=C_0=S=0$. The last two then give $r=rc=0$. This already proves the nondegenerate obstruction.

To classify the possible degenerate exception, test H_{13} and E_{11} on the first algebra, and E_{11} on the second. They give

$$\begin{aligned}p+2\alpha-\tfrac{1}{28}\beta_8-\tfrac{17}{14}\beta_{48}&=0,\\ 3p+\tfrac12A_0+\tfrac72B_0+6q+\tfrac{1}{28}\beta_8+\tfrac{3}{14}\beta_{48}&=0,\\ \tfrac12p+\tfrac18A_0+\tfrac{15}{8}B_0+\tfrac32q-\tfrac12\alpha+\tfrac{3}{56}\beta_8+\tfrac47\beta_{48}&=0.\end{aligned}$$

Substitution yields

$$\beta_{48}=p+2\alpha,\qquad \beta_8=-6(p+2\alpha),\qquad B_0=0,\qquad q=-\tfrac{p}{2}.$$

If the pairing is nonzero, $r=0$ and $q=-p/2$ imply $p\neq0$. Thus $pa=pc=0$ give $a=c=0$, and $B_0=pb+q(a+8b)=-3pb=0$ gives $b=0$. The scalar identity, in the conventions of [FFL+26, Section 3.3.2, pp. 28–29],

$$R=12|T_8|^2-2|T_{48}|^2-8\operatorname{div}V$$

identifies the functional modulo the stated divergence. Conversely, the Einstein–Hilbert first variation $D\int R(h,X)=\int\langle Rg-2\mathrm{Ric},h\rangle\,dV$ verifies (6.8) for these coefficients. □

Corollary 6.4 ($\mathrm{Spin}(7)$ fixed-measure obstruction)**.** *Replace dV by $dm = e^{-f}dV$ in the pairing and consider*

$$\mathcal{W}_f = \int_M \big(\alpha R + \beta_8|T_8|^2 + \beta_{48}|T_{48}|^2 + \gamma|\nabla f|^2 + \kappa\langle V, \nabla f\rangle\big)\, dm.$$

If a gradient identity modulo $\mathcal{L}_{cV+d\nabla f}\Phi$ holds for every smooth pair (Φ, f) with fixed-measure variations, its variational form forces $r = 0$. Hence such an identity is impossible for every nondegenerate invariant pairing and every (a, b).

Proof. At constant f, fixing the measure removes only the volume variation of the quadratic densities. The $|\nabla f|^2$ variation vanishes, and the coupling contributes only $-\kappa(\operatorname{div} V)(\operatorname{tr} h)$ after integration by parts. All five variations in (6.11) have zero trace, so their equations are unchanged and again imply $r = 0$. □

The same five tests apply when the coefficients in the displayed degree-two Lagrangian depend smoothly on f: at constant f their additional variations are multiples of $\dot{f} = \operatorname{tr} h$. This does not decide all possible identities with a degenerate skew pairing, nor a construction required to hold only at a minimising weight.

Afterword

The results give a common analytic framework for the fixed quadratic Ricci–harmonic class, while keeping coefficient-dependent MHR identities separate. Extension to other fixed reasonable operators requires verification of the energy inequalities, coefficient bounds and scaling properties actually used in each proof. The initial-boundary compactness theorem concerns sequences of smooth solutions with finitely many uniform initial bounds. Constructing solutions from rough initial structures, finding optimal trace topologies and treating noncompact initial-value problems require additional arguments. These conclusions do not follow from the principal symbol alone.

The weighted metric-error condition supplies one sufficient route from relative noncollapse to complete ancient limits. The collapsing Type I example shows that no corresponding unrestricted assertion holds for every MHR parameter pair. It remains to determine which parameter ranges or geometric subclasses admit a stronger noncollapse mechanism. For general H, convergence of selected rescaled time slices to torsion-free limits is not established here; the Ricci-flat and volume-ratio steps that the G_2 and $\mathrm{Spin}(7)$ arguments derive from torsion-freeness would need another justification. Identifying an ancient limit as a shrinking soliton requires more than a Type I rate and smooth compactness: a flow-specific monotonicity formula, its equality case and control of the limiting defect would provide such a mechanism. The variational obstructions proved here concern specified functionals, pairings and diffeomorphism terms; they leave open other monotone quantities, including constructions involving an optimising weight.

The explicit solutions exhibit shrinking, steady and expanding behaviour. The finite homogeneous catalogue concerns the specified structures, not all positive forms on those algebras. The product comparisons likewise require their stated torsion and coefficient conditions. A general local moduli theory for solitons would need a slice and integrability beyond the principal-symbol calculation. Other questions include the general locally conformally parallel and balanced $\mathrm{Spin}(7)$ classes, warped or cohomogeneity-one solutions, and compact steady or expanding solitons in the intermediate parameter ranges left by the trace inequalities. Universal classification and gap theorems for complete noncompact models remain open. Any such extension must verify the skew equation, completeness and descent as well as the metric equation.

Finally, the stability theorem uses the full-holonomy torsion-free moduli space and its verified linearised kernel. Backgrounds with smaller holonomy can have additional neutral directions, whose integrability and geometric meaning must be established. Outside the coercive ranges $\mu < -3/2$ for G_2 and $\mu < -2$ for $\mathrm{Spin}(7)$, stationarity of immortal limits and convergence of the whole flow would need another Lyapunov quantity. The perturbation estimates are finite-time statements in a fixed parabolic gauge, with $p > n + 2$ in the maximal-L^p setting. Critical-exponent estimates, scale-critical residuals and long-time shadowing are further questions. Each would need its own analytic estimates and reconstruction bounds; none is implicit in the present stability or compactness results.

A Parabolic perturbation estimates

An approximate solution has two errors: its residual in the flow equation and its discrepancy from the prescribed initial data. We bound the distance to an exact solution by these errors, first in Hölder spaces and then in maximal-L^p spaces. Both arguments invert the linearisation and control the nonlinear remainder by a contraction. For MHR, the estimates hold on a fixed finite interval in a chosen DeTurck gauge.

A.1 Hölder estimates

In the Hölder setting, the linearised initial-value problem has a Schauder inverse and the nonlinear error is quadratic. We state the analytic estimate in an orbit chart, then explain how to read it as an estimate for the defining tensors.

Let M be closed. Choose a smooth bundle chart for the admissible tensor orbit, with model vector bundle $E \to M$, and write the gauged equation as

$$\partial_t u = \mathcal{P}(u), \qquad \mathcal{P}(u)(x,t) = P(x,t,u,\nabla u,\nabla^2 u).$$

Use a background connection and norms throughout. We assume that P is smooth on an open set of second jets, as for a smooth quasilinear gauged operator. The orbit chart keeps the unknown admissible even when the tensor orbit is not open in the ambient tensor bundle. For a path $\widetilde{u}$, define its residual by

$$\mathcal{R}_{\mathcal{P}}(\widetilde{u}) := \partial_t \widetilde{u} - \mathcal{P}(\widetilde{u}).$$

Theorem A.1 (Hölder estimate for approximate solutions)**.** *Let $0 < \alpha < 1$, $0 < \tau < \infty$, and set*

$$\mathcal{X} = C^{2+\alpha,1+\alpha/2}(M \times [0,\tau]; E), \qquad \mathcal{Z} = C^{2,\alpha}(M; E),$$
$$\mathcal{Y} = C^{\alpha,\alpha/2}(M \times [0,\tau]; E).$$

Suppose $\widetilde{u} \in \mathcal{X}$ has second-jet image compactly contained in the domain of P, and $L_t = D\mathcal{P}(\widetilde{u}(t))$ is uniformly strongly elliptic: writing A^{ij} for its principal coefficients, assume

$$\left\langle A^{ij}(x,t)\zeta_i\zeta_j v, v\right\rangle \geq \lambda|\zeta|^2|v|^2 \qquad (\lambda > 0)$$

in local bundle coordinates. There are $\varepsilon, C > 0$ such that, if $u_0 \in \mathcal{Z}$ and

$$\delta := \|u_0 - \widetilde{u}(0)\|_{\mathcal{Z}} + \|\mathcal{R}_{\mathcal{P}}(\widetilde{u})\|_{\mathcal{Y}} \leq \varepsilon,$$

there is a solution $u \in \mathcal{X}$ on $[0,\tau]$ with $u(0) = u_0$, unique among solutions in a closed $\mathcal{X}$-ball about $\widetilde{u}$ whose radius is independent of u_0 and δ, and

$$\|u - \widetilde{u}\|_{\mathcal{X}} \leq C\delta. \tag{A.1}$$

The constants depend on the background geometry, α, τ, λ, $\|\widetilde{u}\|_{\mathcal{X}}$, its second-jet distance from the chart and operator boundaries, and bounds for the derivatives of P on a neighbourhood of that image.

Proof. The linearised initial-value problem has a bounded inverse. The nonlinear remainder is quadratic, so this inverse turns the equation for the correction into a contraction.

The linear estimate. The coefficients of L_t belong to $\mathcal{Y}$. By linear Schauder theory [Hua15, Theorems 2.3–2.4],

$$\mathscr{B} : \mathcal{X} \longrightarrow \mathcal{Y} \times \mathcal{Z}, \qquad v \longmapsto (\partial_t v - L_t v, v(0))$$

is an isomorphism. Give the product the sum norm and choose $A \geq \|\mathscr{B}^{-1}\|$. Since M has no boundary, there are no boundary compatibility conditions; the initial trace is $\mathcal{Z}$.

The nonlinear remainder. Choose $\rho_0 > 0$ such that the closed $\mathcal{X}$-ball of radius ρ_0 about $\widetilde{u}$ stays in the operator domain. Smooth composition and multiplication give a finite K such that

$$\|D^2\mathcal{P}(w)[h,k]\|_{\mathcal{Y}} \leq K\|h\|_{\mathcal{X}}\|k\|_{\mathcal{X}} \quad (\|w - \widetilde{u}\|_{\mathcal{X}} \leq \rho_0).$$

Indeed, each term is a derivative of P evaluated at $j^2 w$, multiplied by components of $j^2 h$ and $j^2 k$. These components belong to the Banach algebra $\mathcal{Y}$, and the coefficient norms are uniformly bounded on this ball.

Put $r = \mathcal{R}_{\mathcal{P}}(\widetilde{u})$, $h_0 = u_0 - \widetilde{u}(0)$, and

$$N(v) = \mathcal{P}(\widetilde{u} + v) - \mathcal{P}(\widetilde{u}) - L_t v.$$

Taylor's formula gives, whenever $\|v\|_{\mathcal{X}}, \|w\|_{\mathcal{X}} \le \rho \le \rho_0$,

$$\|N(v)\|_{\mathcal{Y}} \le \tfrac{1}{2} K \|v\|_{\mathcal{X}}^2,$$
$$\|N(v) - N(w)\|_{\mathcal{Y}} \le K\rho \|v - w\|_{\mathcal{X}}.$$

The contraction. Choose $\rho \le \rho_0$ with $AK\rho \le \frac{1}{2}$, and take $\varepsilon = \rho/(2A)$. The map

$$\mathscr{T}(v) = \mathscr{B}^{-1}\big(-r + N(v), h_0\big)$$

has Lipschitz constant at most $\frac{1}{2}$ on the closed radius-ρ ball, and maps that ball into itself because

$$\|\mathscr{T}(v)\|_{\mathcal{X}} \le A\delta + \tfrac{1}{2} AK\rho^2 \le \tfrac{3}{4}\rho.$$

Its unique fixed point v gives the required solution $u = \widetilde{u} + v$. Finally,

$$\|v\|_{\mathcal{X}} \le A\delta + \tfrac{1}{2} AK\rho \|v\|_{\mathcal{X}} \le A\delta + \tfrac{1}{4}\|v\|_{\mathcal{X}},$$

which implies (A.1) with $C = 2A$. □

A smooth orbit parametrisation and its inverse on the admissible image are locally Lipschitz on these spaces. The residual transforms by the inverse differential of the parametrisation. Thus (A.1) also holds for ambient tensor discrepancies, with a changed constant, provided the initial structure is admissible and lies in the chart. This gives the following geometric application.

Corollary A.2 (Application to the DeTurck MHR equation)**.** *Fix (a, b) and the DeTurck modification in* [FFL+26, Theorem 4.2]. *Every $C^{2+\alpha,1+\alpha/2}$ approximate gauged MHR path on a closed manifold, contained in a smooth admissible-orbit chart with its chart second-jet image compactly contained in the operator domain, satisfies the conclusion of Theorem A.1. The initial discrepancy and residual may be measured in the ambient tensor bundle.*

Proof. The gauged MHR operator has smooth coefficients and scalar heat symbol by [FFL+26, proof of Theorem 4.2, pp. 33–34]. A change of orbit coordinates conjugates the principal symbol, preserving its scalar form. Compactness of the reference image gives uniform ellipticity relative to the background metric. Apply Theorem A.1 and the bounded chart changes above. For MHR the operator domain is the whole chart, so the second-jet condition reduces to the image of the path lying in a compact subset of the chart. □

The estimates hold in the chosen gauge. Reconstruction of an ungauged solution uses the DeTurck argument; transferring the estimate in the same tensor norm also requires the corresponding regularity bounds for the pullbacks.

A.2 Maximal-L^p estimates

At maximal-L^p regularity, the principal coefficients are still Hölder continuous, but the lower-order coefficients of the linearisation contain second derivatives of the approximate path and belong only to L^p. The main additional step is a linear estimate that controls these rough coefficients without requiring higher derivatives of the path.

Fix a smooth background metric and connection on a vector bundle $E \to M$, where M is closed of dimension n. For $p > n + 2$ and $0 < T \le \tau < \infty$ put

$$E_0 = L^p(M; E), \qquad E_1 = W^{2,p}(M; E), \qquad \mathcal{Z}_p = B_{pp}^{2-2/p}(M; E),$$
$$\mathcal{X}_p(T) = W^{1,p}(0, T; E_0) \cap L^p(0, T; E_1), \qquad \mathcal{Y}_p(T) = L^p(0, T; E_0).$$

Use the norm $\|v\|_{\mathcal{X}_p(T)}^p = \int_0^T (\|\partial_t v\|_{E_0}^p + \|v\|_{E_1}^p)\, dt$, and write ${}_0\mathcal{X}_p(T)$ for the subspace with zero initial trace. The space $\mathcal{Z}_p$ is equivalently the Slobodeckii space $W^{2-2/p,p}$ used in [Ama17, Section 1]; the identification on a closed manifold is made in a finite atlas [Tri92, Chapter 7].

Theorem A.3 (Maximal-L^p estimate for approximate solutions)**.** *Let $\mathcal{O}$ be an open set of first jets over $M \times [0,\tau]$, and let*

$$\mathcal{P}(u) = A(x,t,j^1u)[\nabla^2 u] + F(x,t,j^1u), \qquad j^1u = (u, \nabla u), \tag{A.2}$$

where A and F are C^2 on $\mathcal{O}$. Suppose $\widetilde{u} \in \mathcal{X}_p(\tau)$, its first-jet image lies in $\mathcal{O}$, and the frozen principal part is uniformly normally elliptic: for some $\lambda > 0$,

$$\operatorname{spec}\big(A(x,t,j^1\widetilde{u})[\zeta \otimes \zeta \otimes \cdot\,]\big) \subset \{\operatorname{Re} z \geq \lambda|\zeta|^2\}.$$

There are $\varepsilon, \rho, C > 0$ such that, if $u_0 \in \mathcal{Z}_p$ and

$$\delta = \|u_0 - \widetilde{u}(0)\|_{\mathcal{Z}_p} + \|\partial_t\widetilde{u} - \mathcal{P}(\widetilde{u})\|_{\mathcal{Y}_p(\tau)} \leq \varepsilon,$$

there is a solution $u \in \mathcal{X}_p(\tau)$ of $\partial_t u = \mathcal{P}(u)$, $u(0) = u_0$, unique in the closed radius-ρ ball about $\widetilde{u}$, with

$$\|u - \widetilde{u}\|_{\mathcal{X}_p(\tau)} + \sup_{0\leq t\leq\tau} \|u(t) - \widetilde{u}(t)\|_{\mathcal{Z}_p} \leq C\delta. \tag{A.3}$$

The constants depend only on the background data, p, τ, λ, a bound for $\|\widetilde{u}\|_{\mathcal{X}_p(\tau)}$, the first-jet distance to the boundary of $\mathcal{O}$, and C^2 bounds for A, F on a tube about that image. They do not require higher derivatives of $\widetilde{u}$.

Proof. We establish the bounded inverse of the linearisation and the quadratic remainder estimate; the contraction is then the one in the proof of Theorem A.1.

Trace and jet estimates. Choose

$$\beta = \tfrac{1}{2}\big(1 - (n+2)/p\big) > 0, \qquad \theta = \frac{p\beta}{(n+p)(1+\beta)}, \qquad \kappa = \theta(1-1/p) > 0.$$

The parabolic trace and Besov embeddings give

$$\mathcal{X}_p(\tau) \hookrightarrow C([0,\tau];\mathcal{Z}_p) \hookrightarrow C([0,\tau];C^{1,\beta}), \qquad [v]_{C^\kappa([0,\tau];C^1)} \leq C\|v\|_{\mathcal{X}_p(\tau)}. \tag{A.4}$$

Here is a verification that also supplies the uniform estimate on short intervals used below. Apply [Ama17, Theorem 1.23(i), Example 1.6 and Remark 1.17] to the smooth Bochner heat operator on $[0,\tau/2]$. Its initial trace is a bounded surjection onto $\mathcal{Z}_p$. Every time slice of a path on $[0,\tau]$ or $[0,2\tau]$ is the initial trace of a forward or reversed subinterval of length $\tau/2$. This gives a uniform trace bound; density of smooth paths then gives continuity into $\mathcal{Z}_p$. Thus no separate assertion of trace surjectivity from an abstract embedding theorem is needed. Since $2 - 2/p - n/p = 1 + 2\beta$, the Besov embedding into $C^{1,\beta}$ follows from [Tri83, Sections 2.7.1 and 2.5.7].

For $v \in {}_0\mathcal{X}_p(T)$, reflect it about T and extend by zero after $2T$ to obtain a path on $[0,2\tau]$. The pieces agree because $v(0) = 0$, and the pth power of the extended norm is twice the original. Consequently

$$\sup_{0\leq s\leq T} \|v(s)\|_{C^1} \leq c_0\|v\|_{\mathcal{X}_p(T)} \qquad (0 < T \leq \tau), \tag{A.5}$$

with c_0 independent of T. Finally, $\|v(t) - v(s)\|_{L^p} \leq |t-s|^{1-1/p}\|v\|_{\mathcal{X}_p(\tau)}$. The elementary interpolation inequality

$$\|f\|_{C^1} \leq C\|f\|_{L^p}^{\theta}\|f\|_{C^{1,\beta}}^{1-\theta}$$

proves the temporal estimate in (A.4). One obtains this inequality in charts by combining $\|f\|_\infty \leq C\|f\|_{L^p}^{p/(n+p)}\|f\|_{C^{1,\beta}}^{n/(n+p)}$ with $\|f\|_{C^1} \leq C\|f\|_\infty^{\beta/(1+\beta)}\|f\|_{C^{1,\beta}}^{1/(1+\beta)}$; these follow by averaging over a ball and by a first-order difference quotient, respectively.

The principal linear problem. Put $a = A(x,t,j^1\widetilde{u})$. The jet estimates and the C^2 bounds give uniformly bounded C^β spatial and C^κ temporal seminorms of a. To see that the bounds depend only on the stated data, compare nearby base points in one chart. A segment between their jet values stays in the prescribed tube when their base distance is small enough, by (A.4); apply the derivative bound there. For more distant base points use $2\|A\|_\infty$ divided by the fixed positive lower bound for that base distance. Do this separately in space and time.

For any $0 < \bar{s} < \min\{\beta, 2\kappa\}$, these bounds imply that a belongs to Amann's little anisotropic Hölder coefficient class $bc^{\bar{s}/(1,2)}$: spatial interpolation gives uniform continuity in $bc^{\bar{s}}$, and κ-Hölder continuity in time

implies little $\bar{s}/2$-Hölder continuity. Hence [Ama17, Theorem 1.23(i) and Remark 1.17], with $s = 0$, $r = 2$ and $q = p$, gives an isomorphism

$$\mathscr{L}_a : v \longmapsto (\partial_t v - a\nabla^2 v, v(0)) : \mathcal{X}_p(T) \longrightarrow \mathcal{Y}_p(T) \times \mathcal{Z}_p.$$

The symbol of $-a\nabla^2$ is $a[\zeta \otimes \zeta \otimes \cdot\,]$ in the convention of [Ama17, (1.7)].

The inverse on $[0, \tau]$ has a bound M_0 uniform over this bounded Hölder coefficient class with fixed ellipticity constant. Indeed, the class is compact in C^0 by Arzelà–Ascoli; its spatial and temporal Hölder bounds and its spectral condition are closed under uniform limits. Moreover, $\|\mathscr{L}_a - \mathscr{L}_{a'}\| \le \|a - a'\|_\infty$. Invertibility is stable in this operator norm, so a finite covering by Neumann-series neighbourhoods bounds all inverse norms. If $v(0) = 0$ on $[0, T]$, extend its forcing by zero to $[0, \tau]$ and restrict the resulting principal solution. Uniqueness on $[0, T]$ gives

$$\|v\|_{\mathcal{X}_p(T)} \le M_0 \|\partial_t v - a\nabla^2 v\|_{\mathcal{Y}_p(T)} \qquad (0 < T \le \tau). \tag{A.6}$$

Rough lower-order coefficients. Let $\mathscr{R}v = b[\nabla v] + c[v]$ satisfy

$$|\mathscr{R}v| \le m(|v| + |\nabla v|), \qquad 0 \le m \in L^p(M \times (0, \tau)).$$

This pointwise bound, rather than $|b| + |c| \le m$, is the hypothesis needed here. Write $B = \|m\|_{\mathcal{Y}_p(\tau)}$ and $\eta(s) = \|m(s)\|^p_{L^p(M)}$. For a zero-trace v, set $g = \partial_t v - a\nabla^2 v - \mathscr{R}v$ and $\phi(t) = \|v\|^p_{\mathcal{X}_p(t)}$. From (A.5)–(A.6),

$$\phi(t) \le 2^{p-1} M_0^p \|g\|^p_{\mathcal{Y}_p(\tau)} + 2^{p-1} M_0^p c_0^p \int_0^t \eta(s)\phi(s)\, ds.$$

Gronwall's inequality therefore gives

$$\|v\|_{\mathcal{X}_p(\tau)} \le 2^{1-1/p} M_0 \exp\!\big(2^{p-1} M_0^p c_0^p B^p/p\big) \|g\|_{\mathcal{Y}_p(\tau)}. \tag{A.7}$$

The estimate is uniform with $\mathscr{R}$ replaced by $\sigma\mathscr{R}$, $0 \le \sigma \le 1$. These operators are bounded from ${}_0\mathcal{X}_p(\tau)$ to $\mathcal{Y}_p(\tau)$. The method of continuity makes them onto: surjectivity is open by a Neumann series; for a convergent sequence of parameters, the uniform estimate makes the corresponding solutions Cauchy. Injectivity is already in (A.7). For nonzero initial data, first solve the principal homogeneous problem with that trace, subtract this lift, and use the zero-trace inverse. Thus

$$v \longmapsto (\partial_t v - a\nabla^2 v - \mathscr{R}v, v(0))$$

is an isomorphism, with norm bounded by the stated data and B. No time or spatial continuity of b, c is used. An equal-mass time subdivision gives an alternative proof with the same qualitative dependence on B; the Gronwall argument avoids that partition.

Linearisation and nonlinear remainder. Choose $\rho_0 > 0$ so that the closed radius-ρ_0 ball about $\widetilde{u}$ in $\mathcal{X}_p(\tau)$ stays inside the first-jet tube. This is possible by (A.4). Pointwise Taylor expansion gives

$$D\mathcal{P}(w)v = A(j^1 w)[\nabla^2 v] + D_\zeta A(j^1 w)[j^1 v][\nabla^2 w] + D_\zeta F(j^1 w)[j^1 v].$$

At $w = \widetilde{u}$ the latter two terms satisfy the preceding pointwise bound with $m = C\Lambda(1 + |\nabla^2 \widetilde{u}|)$, where Λ bounds the first two derivatives of A, F on the tube. Thus the full linearised initial-value operator has a bounded inverse $\mathscr{B}^{-1}$.

The map $\mathcal{P} : \mathcal{X}_p \to \mathcal{Y}_p$ is C^1 on this ball and its derivative is Lipschitz there. Indeed, the difference of the displayed derivatives consists of four terms: the difference of A multiplying $\nabla^2 v$; $D_\zeta A[j^1 v]$ multiplying $\nabla^2(w - w')$; the difference of $D_\zeta A$ multiplying $j^1 v$ and $\nabla^2 w'$; and the difference of $D_\zeta F$ multiplying $j^1 v$. Bounded first jets, the C^2 coefficient bounds and $\|w'\|_{\mathcal{X}_p} \le \|\widetilde{u}\|_{\mathcal{X}_p} + \rho_0$ bound their $\mathcal{Y}_p$ norms by $K\|w - w'\|_{\mathcal{X}_p}\|v\|_{\mathcal{X}_p}$. The same Taylor expansion gives the Fréchet remainder $O(\|v\|^2_{\mathcal{X}_p})$. Consequently, with $N(v) = \mathcal{P}(\widetilde{u}+v) - \mathcal{P}(\widetilde{u}) - D\mathcal{P}(\widetilde{u})v$,

$$\|N(v)\|_{\mathcal{Y}_p} \le \tfrac{1}{2} K \|v\|^2_{\mathcal{X}_p}, \qquad \|N(v) - N(w)\|_{\mathcal{Y}_p} \le K\rho \|v - w\|_{\mathcal{X}_p}$$

on each radius-$\rho \le \rho_0$ ball. The contraction from the proof of Theorem A.1, now with $(\mathcal{X}, \mathcal{Y}, \mathcal{Z}) = (\mathcal{X}_p, \mathcal{Y}_p, \mathcal{Z}_p)$, gives a unique correction v, with $\|v\|_{\mathcal{X}_p} \le 2\|\mathscr{B}^{-1}\|\delta$. The trace bound in (A.4) proves (A.3). Choosing the ball small enough keeps the corrected first jet in $\mathcal{O}$. □

Corollary A.4 (Maximal-L^p shadowing for fixed MHR)**.** *Fix (a, b) and a smooth background metric $\bar{g}$. Add $\mathcal{L}_{W(g_\xi,\bar{g})}\xi$ to the MHR equation, with the standard DeTurck field (3.2). In a smooth admissible-orbit chart on a closed manifold, every approximate gauged MHR path in $\mathcal{X}_p(\tau)$ whose image lies in a compact subset of the chart satisfies Theorem A.3. In particular, $p > 9$ for G_2 and $p > 10$ for $\mathrm{Spin}(7)$ suffice.*

Proof. The metric is algebraic in the defining structure and its torsion is linear in its first derivative. Ricci curvature and $\operatorname{div} T$ are therefore affine in the second jet, with coefficients smooth in the first jet. The same holds for the Lie derivative along the fixed first-order DeTurck vector field. The torsion-quadratic terms have only first-jet dependence. A change of orbit coordinates preserves this quasilinear form. In the convention $\dot{g} = 2h$, the field $W/2$ has linearised symbol $\zeta_i h_{ik} - \frac{1}{2}\zeta_k \operatorname{tr} h$, so $W = 2X_{DT}$ realises the gauge of [FFL+26, Proposition 4.1(1) and proof of Theorem 4.2]. That calculation gives the scalar heat symbol $|\zeta|^2_{g_\xi}\,\mathrm{Id}$ at every admissible structure. Compactness of the reference image therefore supplies uniform normal ellipticity and the required tube. Apply Theorem A.3. □

The residual can also be measured in the ambient tensor bundle: the chart differential and its inverse on the admissible tangent space are uniformly bounded on the relevant compact image. Smooth chart changes are locally Lipschitz on $\mathcal{X}_p$ by the same first-jet product estimates, and on $\mathcal{Z}_p$ by applying this fact to a bounded Bochner heat lift and taking traces. Initial tensor data must remain admissible and in the chart. The estimate is in the fixed gauge; no estimate in the identical ungauged tensor norm is asserted without the corresponding pullback regularity bounds.

B The Taylor-selected $\mathrm{Spin}(7)$ expression

The space of Cayley forms is a proper submanifold of the bundle of 4-forms. Thus a proposed evolution of $\mathrm{Spin}(7)$-structures must be tangent to this orbit, even if every term is tensorially natural. We examine the quadratic Taylor term in Dwivedi's and Duthie's Ricci–harmonic constructions before comparing it with MHR.

The literal expression in Dwivedi's $\mathrm{Spin}(7)$ definition is [Dwi26, (7.5) and Definition 7.2 (SRHF)]

$$\partial_t \Phi = \left[-\mathrm{Ric} + \tfrac{1}{2}\operatorname{div} T\right] \diamond \Phi + \mathcal{B}_\Phi(T), \qquad \mathcal{B}_\Phi(T) := \tfrac{1}{2}\sum_p T_p \diamond (T_p \diamond \Phi), \tag{B.1}$$

where the sum is taken in an orthonormal frame. Duthie retains this quadratic 4-form and changes the relative divergence coefficient from $1/2$ to 1 [Dut26, Definition 2.19 and (2.24)]. This change is not an overall time rescaling. Both sources write the quadratic term as the diamond action of a 2-tensor. Before comparing its coefficients with MHR, that representation requires a tangency check.

Proposition B.1 (Tangency obstruction for the literal Taylor term)**.** *The 4-form $\mathcal{B}_\Phi(T)$ in (B.1) need not belong to the tangent space $\mathfrak{gl}(8,\mathbb{R}) \diamond \Phi$ of the Cayley orbit, even when T is the intrinsic torsion of a smooth $\mathrm{Spin}(7)$-structure on a closed manifold. Thus the displayed expression does not define an orbit-preserving flow for arbitrary initial structures. Moreover,*

$$\langle \mathcal{B}_\Phi(T), \Phi\rangle = -\tfrac{1}{2}\sum_p |T_p \diamond \Phi|^2, \tag{B.2}$$

so $\mathcal{B}_\Phi(T) = 0$ if and only if $T = 0$.

Proof. On $\mathbb{R}^8 = \mathbb{C}^4$, choose the standard forms

$$\omega = e^{12} + e^{34} + e^{56} + e^{78}, \qquad \Omega = (e^1 + \mathrm{i}e^2) \wedge (e^3 + \mathrm{i}e^4) \wedge (e^5 + \mathrm{i}e^6) \wedge (e^7 + \mathrm{i}e^8).$$

Set $\Phi_0 = \frac{1}{2}\omega^2 + \operatorname{Re}\Omega$. Put $S = \omega^2/2$, $R = \operatorname{Re}\Omega$, and $I = \operatorname{Im}\Omega$. Let J be the standard complex rotation, choosing its sign so that $J \diamond \Omega = 4\mathrm{i}\Omega$. Then $J \diamond S = 0$ and

$$\tfrac{1}{2} J \diamond (J \diamond \Phi_0) = -8R. \tag{B.3}$$

The real centraliser of the standard $\mathrm{SU}(4)$ action on $\mathbb{C}^4$ is $\mathbb{R}\,\mathrm{Id} \oplus \mathbb{R}J$. Its skew part is $\mathbb{R}J$. The orthogonal projection of J to $\mathfrak{spin}(7)$ is $\mathrm{SU}(4)$-invariant, since $\mathrm{SU}(4)$ fixes both Φ_0 and J. Since $J \diamond \Phi_0 \neq 0$, that projection vanishes. Thus $J \in \mathfrak{m} = \mathfrak{spin}(7)^\perp$.

Suppose $A \diamond \Phi_0 = R$ for some $A \in \mathfrak{gl}(8,\mathbb{R})$. Average A over $\mathrm{SU}(4)$ with its normalised Haar measure. Both Φ_0 and R are fixed by this group, so the averaged endomorphism still maps Φ_0 to R. It has the form $a\,\mathrm{Id} + bJ$, whose diamond action is $4a(S+R) - 4bI$. The forms S, R, I are linearly independent: the S coefficient forces $a = 0$, contradicting the required R coefficient. Consequently (B.3) is not tangent.

For a geometric realisation, equip $\mathbb{R}^8/(2\pi\mathbb{Z})^8$ with its flat metric and set

$$f(x) = \sin x^1, \qquad \Phi(x) = \Phi_0 \cdot \exp(f(x)J).$$

The action here is the tensor action with infinitesimal action $\diamond$. It preserves the flat metric. Since J commutes with its exponential, it remains in the orthogonal complement of the moving stabiliser. Therefore $T_p = (\partial_p f)J$ and $\operatorname{div} T = (\Delta f)J$. At $x^1 = 0$, $T_1 = J$, all other T_p vanish, and $\mathrm{Ric} = \operatorname{div} T = 0$. The right-hand side of either source expression is then $-8R$, which is not tangent to the orbit. No smooth family of Cayley forms can have that initial velocity.

Finally, the infinitesimal action of each skew endomorphism T_p is skew-adjoint on forms. This proves (B.2). The diamond action is injective on $\mathfrak{m}$, so its right-hand side vanishes exactly when $T = 0$. □

Remark B.2 (Scope of the obstruction)**.** This concerns the literal quadratic 4-form in the cited displays. A tangent projection or a different quadratic tensor would define a different expression and must be specified before any comparison with MHR. Adding a Lie derivative cannot remove the transverse component, since Lie derivatives of a defining tensor are tangent to its orbit. The general theorems for flows satisfying Duthie's reasonable-flow hypotheses, and the MHR verification in Proposition 2.4, do not rely on this particular Taylor-selected expression. The index expression in [Dut26, (2.24)] is $\frac{1}{2}\sum_p T_{p;is}(T_p \diamond \Phi)_{sjkl}$ summed over the four slots; the obstruction therefore applies to both sources as written.

Orthogonal tangent projection. Orthogonal projection gives a precise correction: remove the normal component of the quadratic 4-form. The tangential component is a scalar multiple of the Cayley form, so its coefficient can be compared directly with MHR.

Corollary B.3 (Orthogonal tangent correction of the Taylor term)**.** *Let* $\Pi_{\tan}$ *denote the orthogonal projection of* 4*-forms onto* $\mathfrak{gl}(8,\mathbb{R}) \diamond \Phi$*, using the metric induced by* Φ*. In the torsion norm convention* (2.2)*,*

$$\Pi_{\tan}\mathcal{B}_\Phi(T) = -\tfrac{4}{7}|T|^2\Phi = \left[-\tfrac{1}{7}|T|^2 g\right] \diamond \Phi. \tag{B.4}$$

Consequently the explicitly corrected equation

$$\partial_t \Phi = (-\mathrm{Ric} + \operatorname{div} T) \diamond \Phi + \Pi_{\tan}\mathcal{B}_\Phi(T) \tag{B.5}$$

is exactly $\mathrm{MHR}_{0,-1/7}$*. This is a new projected equation; it is not the literal quadratic expression in the cited sources.*

Proof. The tangent space splits orthogonally as $\Lambda^4_1 \oplus \Lambda^4_7 \oplus \Lambda^4_{35}$, the images of $\mathbb{R}g$, $\mathfrak{m}$, and trace-free symmetric endomorphisms, respectively [Dut26, Corollary 2.10]. Each T_p is skew, so its infinitesimal action commutes with the Hodge star. Since Φ is self-dual, $\mathcal{B}_\Phi(T)$ is self-dual too. The trace-free symmetric image is anti-self-dual: diagonalise such an endomorphism and note that the sums of its eigenvalues over complementary four-indices are opposite. Thus $\mathcal{B}_\Phi(T)$ has no component in Λ^4_{35}.

For $A, C \in \mathfrak{m}$, the $\mathrm{Spin}(7)$ single-weight identity [FLMSE24, Example 1.14, (1.24)] reads

$$\langle A \diamond \Phi, C \diamond \Phi\rangle = 16\langle A, C\rangle.$$

Skew-adjointness and the right-action commutator identity give

$$\begin{aligned}\langle A \diamond (A \diamond \Phi), C \diamond \Phi\rangle &= -\langle A \diamond \Phi, A \diamond (C \diamond \Phi)\rangle \\ &= \langle A \diamond \Phi, [A, C] \diamond \Phi\rangle = 16\langle A, [A, C]\rangle = 0.\end{aligned}$$

Here the term with $C \diamond (A \diamond \Phi)$ vanishes by skew-adjointness, and $[A, C] \in \mathfrak{spin}(7)$ already acts trivially on Φ. Equivalently, the final equality follows from invariance of the inner product on skew endomorphisms. This is also the calculation in [FLMSE24, Lemma 1.10(iii), (1.20)–(1.21)]. Applying it to each $A = T_p$ shows that the Λ^4_7

component vanishes. Only the scalar component remains. Since $|\Phi|^2 = 14$, (B.2) and the single-weight identity give

$$\Pi_{\tan}\mathcal{B}_\Phi(T) = \frac{-\frac{1}{2}\sum_p |T_p \diamond \Phi|^2}{14}\Phi = -\tfrac{4}{7}|T|^2\Phi.$$

Finally, $g \diamond \Phi = 4\Phi$ proves (B.4) and the MHR identification. □

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